Anytime-Valid Certification of Robustness Margins for Safe Control via E-Values

Supplementary Ablation Report

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Notation, methods and metrics follow the arXiv version of the paper [1]. References with the prefix A., e.g., (A.46c) or Appendix A.F-A, refer to it.

Contents

1 Simulation Details **4**
- 1.1 How the Checks Are Done . . . 4
- 1.2 Secondary Metrics . . . 7
- 1.3 Parameters, Nominal Input and Lipschitz Constants . . . 8
- 1.4 Temporal Variation . . . 10

2 Simulation Figures and Tables **10**
- 2.1 Compared Parameters and Table Key . . . 10
- 2.2 Default and Selected Settings . . . 12
- 2.3 Margin Insufficiency and CBF-Violation Probability . . . 41
- 2.4 Error Confidence . . . 51
- 2.5 Test Period . . . 57
- 2.6 Budget Normalizer . . . 66
- 2.7 Bound Factor and Distribution-Drift Bound . . . 72
- 2.8 Certification, Deployment, and Assumption Checks . . . 90
- 2.9 Variation in Robustness Margin and Estimated Exceedance Risk . . . 95

List of Figures

1 Corridor, default setting: trajectories . . . 20
2 Corridor, S3: trajectories . . . 20
3 Corridor, S1: trajectories . . . 21
4 Corridor, S2: trajectories . . . 21
5 Corridor, S4: trajectories . . . 22
6 Corridor, S5: trajectories . . . 22
7 Maze, default setting: trajectories . . . 23
8 Maze, S3: trajectories . . . 23
9 Maze, S1: trajectories . . . 24
10 Maze, S2: trajectories . . . 24
11 Maze, S4: trajectories . . . 25
12 Maze, S5: trajectories . . . 25

13 Default: per-trajectory distributions . . . 26
14 S3: per-trajectory distributions . . . 26
15 S1: per-trajectory distributions . . . 27
16 S2: per-trajectory distributions . . . 27
17 S4: per-trajectory distributions . . . 28
18 S5: per-trajectory distributions . . . 28
19 Corridor, default setting: sweeps . . . 29
20 Maze, default setting: sweeps . . . 30
21 Corridor, S3: sweeps . . . 31
22 Maze, S3: sweeps . . . 32
23 Corridor, S1: sweeps . . . 33
24 Maze, S1: sweeps . . . 34
25 Corridor, S2: sweeps . . . 35
26 Maze, S2: sweeps . . . 36
27 Corridor, S4: sweeps . . . 37
28 Maze, S4: sweeps . . . 38
29 Corridor, S5: sweeps . . . 39
30 Maze, S5: sweeps . . . 40
31 Corridor, $\widehat{q}^{+}_{\mathrm{CBF}}$ heatmap at $\alpha_0 = 0.15$ . . . 41
32 Maze, $\widehat{q}^{+}_{\mathrm{CBF}}$ heatmap at $\alpha_0 = 0.15$ . . . 42
33 Corridor, $\widehat{m}_c/c_{\max}$ heatmap at $\alpha_0 = 0.15$ . . . 43
34 Maze, $\widehat{m}_c/c_{\max}$ heatmap at $\alpha_0 = 0.15$ . . . 43
35 Corridor, $\widehat{m}_c/c_{\max}$ heatmap at $\alpha_0 = 0.30$ . . . 44
36 Corridor, $\widehat{q}^{+}_{\mathrm{CBF}}$ heatmap at $\alpha_0 = 0.30$ . . . 45
37 Corridor, $\widehat{q}^{+}_{\mathrm{unsafe}}$ heatmap at $\alpha_0 = 0.30$ . . . 46
38 Maze, $\widehat{m}_c/c_{\max}$ heatmap at $\alpha_0 = 0.30$ . . . 47
39 Maze, $\widehat{q}^{+}_{\mathrm{CBF}}$ heatmap at $\alpha_0 = 0.30$ . . . 48
40 Maze, $\widehat{q}^{+}_{\mathrm{unsafe}}$ heatmap at $\alpha_0 = 0.30$ . . . 49
41 Corridor, $\widehat{q}^{+}_{\mathrm{unsafe}}$ heatmap at $\alpha_0 = 0.25$ . . . 50
42 Corridor, event frequencies at S3 . . . 58
43 Maze, event frequencies at S3 . . . 58
44 Corridor and Maze, event frequencies at the default setting . . . 58

List of Tables

1 Parameters of both scenarios . . . 9
2 Obstacle coordinates . . . 9
3 Compared parameter values . . . 12
4 Corridor, default setting, all nine methods . . . 13
5 Maze, default setting, all nine methods . . . 14
6 Corridor, S1–S5, all four variants . . . 15
7 Maze, S1–S5, all four variants . . . 17
8 Maze, S*, all four variants . . . 19
9 Corridor, error confidence . . . 51
10 Maze, error confidence . . . 54
11 Corridor, test period . . . 59
12 Maze, test period . . . 63
13 Corridor, budget normalizer . . . 67
14 Maze, budget normalizer . . . 70
15 Corridor, bound factor and distribution-drift bound . . . 73
16 Maze, bound factor and distribution-drift bound . . . 80
17 Corridor, bound factor: events and bound excesses . . . 87
18 Maze, bound factor: events and bound excesses . . . 89

19 Corridor, detailed checks at $L_\mu = 0.0005$ 92
20 Corridor, detailed checks at $L_\mu = 0.005$ 93
21 Maze, detailed checks at $L_\mu = 0.0005$ 94
22 Maze, detailed checks at $L_\mu = 0.005$ 95
23 Corridor, variation in robustness margin and estimated exceedance risk 96
24 Maze, variation in robustness margin and estimated exceedance risk 97

1 Simulation Details

1.1 How the Checks Are Done

The tables of this report give the outcomes of the checks of Appendix A.E. The checks evaluate the recorded trajectories after the simulation, with the full simulated disturbance distribution. The controller never uses these evaluations. Each step below names the equation it evaluates.

Angular risk At each recorded decision time t, the state, the applied input and the observed history are held fixed, and z_t is the predicted nose point of Appendix A.D-B. In the simulation, $\ell(x_t, u_t, w) = \psi(z_t) - \psi(z_t + w)$ by (A.52), and $w_t = A_t e(\Phi_t)$, where the magnitude A_t has the distribution μ_t^A defined in (A.88) of Appendix A.F-A and, given A_t, the direction Φ_t is uniform on $[0, 2\pi)$. Since $\{Y_t > c\} = \{\ell(x_t, u_t, w_t) > c\}$ for $c \geq 0$ (Appendix A.D), $p_t(c) = \int_0^{c_{\max}} p_t(c \mid A)\, \mu_t^A(dA)$ with the *angular risk*

$$p_t(c \mid A) := \frac{1}{2\pi} \int_0^{2\pi} \mathbf{1}_{\{\psi(z_t + Ae(\Phi)) < \psi(z_t) - c\}}\, d\Phi. \tag{1.1}$$

The clearance $\psi = \min_i \psi_i$ and its features i are those of Appendix A.D-B. Since ψ is the minimum of the ψ_i, the directions in the angular risk (1.1) form the union

$$\begin{aligned} \{\Phi : \psi(z_t + Ae(\Phi)) < \psi(z_t) - c\} &= \textstyle\bigcup_i \mathcal{A}_{t,i}(c, A), \\ \mathcal{A}_{t,i}(c, A) &:= \{\Phi : \psi_i(z_t + Ae(\Phi)) < \psi(z_t) - c\}. \end{aligned} \tag{1.2}$$

Since ψ_i is 1-Lipschitz, $\psi_i(z_t + Ae(\Phi)) \geq \psi_i(z_t) - A$. Hence $\mathcal{A}_{t,i}(c, A)$ is empty unless A exceeds the *entry magnitude*

$$A_{t,i}(c) := \psi_i(z_t) - \psi(z_t) + c. \tag{1.3}$$

For a circular feature with center o and effective radius R, $\Phi \in \mathcal{A}_{t,i}(c, A)$ if and only if $\|z_t + Ae(\Phi) - o\| < R + \psi(z_t) - c$. The set is empty when $R + \psi(z_t) - c \leq 0$. For a positive right-hand side and $A > 0$, expanding the square shows that membership is equivalent to the angle between $e(\Phi)$ and $o - z_t$ having cosine larger than

$$\frac{A^2 + \|o - z_t\|^2 - (R + \psi(z_t) - c)^2}{2A\|o - z_t\|},$$

so $\mathcal{A}_{t,i}(c, A)$ is an arc centered at the direction of $o - z_t$. For an affine feature with unit normal n, $\psi_i(z_t + Ae(\Phi)) = \psi_i(z_t) + An^\top e(\Phi)$, and (1.3) shows that $\Phi \in \mathcal{A}_{t,i}(c, A)$ if and only if the angle between $e(\Phi)$ and $-n$ has cosine larger than $A_{t,i}(c)/A$. For a rectangle, $R = \rho_{\text{body}}$, and $\mathcal{A}_{t,i}(c, A)$ is a finite union of arcs. Their end points lie among the intersections of the circle $\{z_t + Ae(\Phi) : \Phi \in [0, 2\pi)\}$ with the side lines and their shifts by the signed offset $R + \psi(z_t) - c$ (outward when positive and inward when negative). Circles about the corners with the offset as radius are included only when the offset is positive. The condition in (1.2) at the midpoint between two consecutive end points decides whether the arc between them belongs to $\mathcal{A}_{t,i}(c, A)$. With $|\cdot|$ the total length of a union of arcs, computed by sorting and merging the arcs,

$$p_t(c \mid A) = \frac{1}{2\pi} \Big| \bigcup_{i:\, A_{t,i}(c) < A} \mathcal{A}_{t,i}(c, A) \Big|. \tag{1.4}$$

This evaluates (1.1) exactly up to floating-point rounding.

Quadrature rule In the magnitude distribution μ_t^A defined in (A.88) of Appendix A.F-A, let $\beta_{t,1} := 1 - \beta_t$ and $\beta_{t,2} := \beta_t$ be the weights of U_1 and U_2, with the supports $\mathcal{I}_1 := [0, c_{\max}/2]$ and $\mathcal{I}_2 := [c_{\max}/2, c_{\max}]$ and the density $2/c_{\max}$ on them. Since ψ is 1-Lipschitz (A.51), $p_t(c \mid A) = 0$ for $A \leq c$. Hence, with $\mathcal{I}_j(c) := \mathcal{I}_j \cap [c, c_{\max}]$ the

part of the support above c, the weights and supports of U_1 and U_2 give

$$p_t(c) = \frac{2}{c_{\max}} \sum_{j=1}^{2} \beta_{t,j} \int_{\mathcal{I}_j(c)} p_t(c \mid A)\, dA. \tag{1.5}$$

The entry magnitudes (1.3) in the interior of $\mathcal{I}_j(c)$ split it into the set $\mathcal{M}_{t,j}(c)$ of magnitude intervals. On the interior of each $\mathcal{J} = [\underline{A}, \overline{A}] \in \mathcal{M}_{t,j}(c)$, the set of features in the union (1.4) does not change. At a positive, nondegenerate entry magnitude $\underline{A} = A_{t,i}(c)$, the cosine bound that defines $\mathcal{A}_{t,i}(c, A)$ equals one and has a nonzero derivative. Since $\arccos(1-x) = \sqrt{2x} + O(x^{3/2})$, the length of the arc grows like $(A - \underline{A})^{1/2}$, and $p_t(c \mid A)$ can have an unbounded derivative at $\underline{A}$. The substitution $A = \underline{A} + (\overline{A} - \underline{A})\vartheta^2$, $\vartheta \in [0,1]$, has $dA = 2(\overline{A} - \underline{A})\vartheta\, d\vartheta$ and $(A - \underline{A})^{1/2} = (\overline{A} - \underline{A})^{1/2}\vartheta$. It gives

$$\int_{\mathcal{J}} p_t(c \mid A)\, dA = \int_0^1 2(\overline{A} - \underline{A})\vartheta\, p_t\big(c \mid \underline{A} + (\overline{A} - \underline{A})\vartheta^2\big)\, d\vartheta, \tag{1.6}$$

in which a term of order $(A - \underline{A})^{1/2}$ in $p_t(c \mid A)$ becomes a term of order ϑ^2, smooth at $\vartheta = 0$. The 24-point Gauss–Legendre rule on $[0,1]$, with nodes ϑ_m and weights λ_m, integrates such smooth integrands accurately. Applying it to the right-hand side of (1.6) gives, for $\mathcal{J} \in \mathcal{M}_{t,j}(c)$ and $m = 1, \ldots, 24$, the nodes and weights

$$A_{\mathcal{J},m} := \underline{A} + (\overline{A} - \underline{A})\vartheta_m^2, \qquad \varpi_{\mathcal{J},m} := \frac{4\beta_{t,j}}{c_{\max}} (\overline{A} - \underline{A})\vartheta_m \lambda_m.$$

Substituting this rule into (1.6) and (1.5) yields the estimated exceedance risk of Appendix A.E-A,

$$\widehat{p}_t(c) := \sum_n \varpi_{t,n}\, p_t(c \mid A_{t,n}). \tag{1.8}$$

Its nodes $A_{t,n}$ and weights $\varpi_{t,n}$ are the $A_{\mathcal{J},m}$ and $\varpi_{\mathcal{J},m}$ over $j \in \{1,2\}$, $\mathcal{J} \in \mathcal{M}_{t,j}(c)$ and m. They depend on c through the intervals, and the weights are positive since $\vartheta_m, \lambda_m > 0$.

Numerical risk bounds The trajectory events use the numerical risk bounds of Appendix A.E-A instead of $\widehat{p}_t(c)$. The bounds use a partition $\mathcal{B}_{t,j}(c)$ of $\mathcal{I}_j(c)$ into magnitude intervals, initially $\{\mathcal{I}_j(c)\}$. An interval $\mathcal{J} \in \mathcal{B}_{t,j}(c)$ with midpoint $A_{\mathcal{J}}^{\mathrm{mid}}$ and half-width $\varepsilon_{\mathcal{J}}$ has the probability $\pi_{\mathcal{J}} := 4\beta_{t,j}\varepsilon_{\mathcal{J}}/c_{\max}$ under μ_t^A. Since ψ is 1-Lipschitz, every $A \in \mathcal{J}$ satisfies

$$p_t(c + \varepsilon_{\mathcal{J}} \mid A_{\mathcal{J}}^{\mathrm{mid}}) \le p_t(c \mid A) \le p_t(c - \varepsilon_{\mathcal{J}} \mid A_{\mathcal{J}}^{\mathrm{mid}}), \tag{1.9}$$

with $p_t(c' \mid A_{\mathcal{J}}^{\mathrm{mid}}) := 1$ for $c' < 0$, since $Y_t \ge 0$. Integrating the two bounds of (1.9) over each $\mathcal{J}$ and summing with (1.5) gives

$$\begin{aligned} p_t^-(c) &= \textstyle\sum_{j=1}^{2} \sum_{\mathcal{J} \in \mathcal{B}_{t,j}(c)} \pi_{\mathcal{J}}\, p_t(c + \varepsilon_{\mathcal{J}} \mid A_{\mathcal{J}}^{\mathrm{mid}}) \le p_t(c), \\ p_t^+(c) &= \textstyle\sum_{j=1}^{2} \sum_{\mathcal{J} \in \mathcal{B}_{t,j}(c)} \pi_{\mathcal{J}}\, p_t(c - \varepsilon_{\mathcal{J}} \mid A_{\mathcal{J}}^{\mathrm{mid}}) \ge p_t(c), \end{aligned} \tag{1.10}$$

for every such partition, with each angular risk evaluated by (1.4). The width $p_t^+(c) - p_t^-(c)$ is the sum over the intervals of $\pi_{\mathcal{J}}$ times the difference of the two angular risks in (1.10). Each bisection step replaces the interval with the largest such contribution by its two halves. The steps stop when the margin insufficiency α or the CBF-violation probability α_0 of the event lies outside $[p_t^-(c), p_t^+(c)]$, or at a refinement limit, $p_t^+(c) - p_t^-(c) \le 10^{-5}$ or 128 magnitude intervals. A failed risk evaluation gives $[p_t^-(c), p_t^+(c)] = [0,1]$.

The rectangle condition of Appendix A.D-B is checked in coordinates. For a rectangle $[\underline{z}_1, \bar{z}_1] \times [\underline{z}_2, \bar{z}_2]$, the disk of radius $c_{\max}$ about z lies in the exterior face region to the right if $z_1 - \bar{z}_1 \ge c_{\max}$ and $\min\{z_2 - \underline{z}_2, \bar{z}_2 - z_2\} \ge c_{\max}$, and in the corner region to the upper right if $z_1 - \bar{z}_1 \ge c_{\max}$ and $z_2 - \bar{z}_2 \ge c_{\max}$. The other faces and corners follow by symmetry, and the feature is affine on a face region and circular, with $R = \rho_{\mathrm{body}}$, on a corner region. If (A.58) and the rectangle condition of Appendix A.D-B hold at z_t and the nearest feature is affine, then $\psi(z_t + w) = \psi_{i_z}(z_t + w)$

for $\|w\| \le c_{\max}$, and $p_t(c \mid A) = P_\infty(c, A)$ in (A.54). For $0 < c < A$,

$$\frac{d}{dA}\Big[A \arccos\frac{c}{A} - c\,\mathrm{arcosh}\,\frac{A}{c}\Big] = \arccos\frac{c}{A} + \frac{c}{\sqrt{A^2 - c^2}} - \frac{c}{\sqrt{A^2 - c^2}} = \arccos\frac{c}{A}.$$

Hence, for a magnitude interval $[\underline{A}, \overline{A}]$ with $c \le \underline{A}$,

$$\int_{\underline{A}}^{\overline{A}} P_\infty(c, A)\, dA = F_c(\overline{A}) - F_c(\underline{A}), \qquad F_c(A) := \frac{1}{\pi}\Big[A \arccos\frac{c}{A} - c\,\mathrm{arcosh}\,\frac{A}{c}\Big]. \tag{1.11}$$

At $c = 0$, the continuous extension is $F_0(A) = A/2$. Substituting (1.11) with $[\underline{A}, \overline{A}] = \mathcal{I}_j(c)$ into (1.5) gives $p_t(c)$ without quadrature or bisection.

In (1.10), each angular risk is evaluated at $c \pm (\varepsilon_{\mathcal{J}} + \varepsilon_{\mathrm{len}})$ instead of $c \pm \varepsilon_{\mathcal{J}}$, and (1.11) at $c \pm \varepsilon_{\mathrm{len}}$. The length allowance $\varepsilon_{\mathrm{len}}$ is 256 machine epsilons times the largest of 1, $c_{\max}$, the absolute coordinates of z_t and the absolute clearances $\psi_i(z_t)$. Each angular risk in (1.10) and each of the two bounds is moved outward by 10^{-10}. These allowances are intended to account for floating-point error. Rigorous roundoff coverage is not established, and the bounds do not use outward-rounded interval arithmetic.

Events and counts Let $\mathcal{A}^-_{\mathrm{cert},k}$ and $\mathcal{A}^-_{\mathrm{dep},k}$ be the events (A.76) and (A.77) with every $p_{k,t}(c)$ replaced by $p^-_{k,t}(c)$, and $\mathcal{A}^+_{\mathrm{cert},k}$ and $\mathcal{A}^+_{\mathrm{dep},k}$ the events with $p^+_{k,t}(c)$. Both events are increasing in the exceedance risks. Since $p^-_{k,t}(c) \le p_{k,t}(c) \le p^+_{k,t}(c)$ by (1.10),

$$\mathcal{A}^-_{\mathrm{cert},k} \subseteq \mathcal{A}_{\mathrm{cert},k} \subseteq \mathcal{A}^+_{\mathrm{cert},k}, \qquad \mathcal{A}^-_{\mathrm{dep},k} \subseteq \mathcal{A}_{\mathrm{dep},k} \subseteq \mathcal{A}^+_{\mathrm{dep},k}. \tag{1.12}$$

An event is confirmed if its event with the lower bounds occurs and possible if its event with the upper bounds occurs. Thus the deployment event is confirmed if $p^-_{k,t}(c_{k,t}) > \alpha_0$ at some $t < N_k$ and possible if $p^+_{k,t}(c_{k,t}) > \alpha_0$ at some $t < N_k$. The false-certification event is confirmed if a certificate (c, r) has $p^-_{k,s}(c) \ge \alpha$ at every s of its monitoring period, and possible if a certificate has $p^+_{k,s}(c) \ge \alpha$ at every such s. A trajectory that stops at contact, at an infeasible safety filter or at a numerical failure counts as a possible event of both types. Summing the indicators in (1.12) over the N_{MC} trajectories shows that the confirmed count is at most the count of the event, which is at most the possible count. The tables give the possible count, or the range from the confirmed to the possible count, except the confirmed false-certification counts for S* in Table 8. Each reported count has the 95% upper confidence bound (A.79) of the possible count in percent.

For a binomial count $n \sim \mathrm{Bin}(N_{\mathrm{MC}}, q)$ and $n_{\mathcal{A}} < N_{\mathrm{MC}}$, $\mathbb{P}(n \le n_{\mathcal{A}}) = 1 - F_{\mathrm{Beta}(n_{\mathcal{A}}+1, N_{\mathrm{MC}}-n_{\mathcal{A}})}(q)$. Hence $\overline{q}(\mathcal{A})$ in (A.79) is the value of q at which $n_{\mathcal{A}}$ or fewer events have probability 0.05. When $n_{\mathcal{A}} = N_{\mathrm{MC}}$, $\overline{q}(\mathcal{A})$ is one. For $n_{\mathcal{A}} = 0$, $F_{\mathrm{Beta}(1,N_{\mathrm{MC}})}(q) = 1 - (1-q)^{N_{\mathrm{MC}}}$, so (A.79) gives

$$\overline{q}(\mathcal{A}) = 1 - 0.05^{1/N_{\mathrm{MC}}},$$

which equals 0.0149 for $N_{\mathrm{MC}} = 200$, the upper confidence bound of 1.5% for zero possible events in the tables.

Assumption checks Fix a decision time t with $c_t < c_{\max}$, and let $\sigma := \sigma_{r_t}$ and $\tau := \tau_{r_t}(c_t)$ bound the monitoring period of the test with the supporting test index r_t defined in (A.91). As in Appendix A.E-C, $d^{\mathrm{V}}_t(c_t; s, s+1)$ is the risk-increase term of the deployed variant at s, capped at 1. For V-C the terms are at most $\bar{P}(c_t) \le 1/2$, so the cap does not apply. Since $\max_s \min\{1, \cdot\} = \min\{1, \max_s \cdot\}$ and a failed required check sets both sides to 1, (A.30) and (A.47) give

$$d^{(r_t)}_t(c_t) = d^{\mathrm{V}}_t(c_t; \sigma, \tau) = \max_{\sigma \le s < \tau} d^{\mathrm{V}}_t(c_t; s, s+1). \tag{1.14}$$

Let s^* minimize $\widehat{p}_s(c_t)$ over $\sigma \le s < \tau$. Then (A.83) and (1.14) give

$$\begin{aligned} e_{\mathrm{A2},t} &= \widehat{p}_t(c_t) - \widehat{p}_{s^*}(c_t) - \max_{\sigma \le s < \tau} d_t^{\mathrm{V}}(c_t; s, s+1) \\ &\le \widehat{p}_t(c_t) - \widehat{p}_{s^*}(c_t) - d_t^{\mathrm{V}}(c_t; s^*, s^*+1) \le e_{\mathrm{pw},t}. \end{aligned}$$

This proves $e_{\mathrm{A2},t} \le e_{\mathrm{pw},t}$ for the bound excesses of Appendix A.E-C. Moreover, $e_{\mathrm{A2},t} = \max_{\sigma \le s < \tau}[\widehat{p}_t(c_t) - \widehat{p}_s(c_t)] - d_t^{(r_t)}(c_t)$. Hence $e_{\mathrm{A2},t} \le 0$ is (A.23) of Assumption A.2 at t for every s of the monitoring period, with $\widehat{p}$ in place of p, and $e_{\mathrm{pw},t} \le 0$ requires the same of each risk-increase term at its own s. For a bound factor $\phi < 1$, every term in (1.14) is scaled by ϕ, and these relations still hold. A check fails if an excess exceeds the tolerance $\varepsilon_{\mathrm{chk}} = 10^{-12}$. The checks compare estimates, so a passed check supports Assumption A.2 at the checked decisions but does not prove it. The frequencies $\widehat{q}_{\mathrm{pw}}$ and $\widehat{q}_{\mathrm{A2}}$ count the trajectories with a failed check. The support condition of Assumption A.1, the nonemptiness of $\mathcal{U}(x_t, c_{\max})$ and the residual of (A.3) are recorded at every decision. The support check fails if $Y_t > c_{\max} + \varepsilon_{\mathrm{chk}}$, and the residual check fails below $-\varepsilon_{\mathrm{num}}$, with the numerical tolerance $\varepsilon_{\mathrm{num}} := 10^{-10}$. The indicator $I_{\mathrm{CBF},k,t}$ in (A.81) uses the same tolerance, $h(x_{k,t+1}) < (1-\gamma)h(x_{k,t}) - \varepsilon_{\mathrm{num}}$. An empty $\mathcal{U}(x_t, c_{\max})$ ends the trajectory, so the support condition and the residual are recorded only at executed transitions. Each trajectory also records its stopping reason.

What the tables report Each row is one method in one scenario and setting. Its entries are the means $\pm$ standard deviations over the trajectories of the metrics of Appendix A.E-B and Section 1.2, and the counts and upper confidence bounds above. Each mean (1.15) is over its own set of trajectories. Means over empty sets and standard deviations over fewer than two trajectories are undefined. Contact is detected only at the decision times, and the control intervention is in unscaled input coordinates. Section 2.1 gives the key to the columns.

1.2 Secondary Metrics

The secondary metrics complement those of Appendix A.E-B, with the same notation. For a quantity X_k defined on a set $\mathcal{K}_X$ of trajectories, the mean is

$$\widehat{m}_X := \frac{1}{|\mathcal{K}_X|} \sum_{k \in \mathcal{K}_X} X_k. \tag{1.15}$$

With the set $\mathcal{S}_{k,t}$ of certified candidates and the deployable set $\mathcal{D}_{k,t}$ in (A.31), the *no-certificate fraction* $\widehat{q}^{+}_{\mathrm{nocert}}$, the *no-deployable-margin fraction* $\widehat{q}^{+}_{\mathrm{nodep}}$ and the *smaller-margin-use fraction* $\widehat{q}^{+}_{\mathrm{use}}$ are the decision fractions of

$$\begin{aligned} I_{\mathrm{nocert},k,t} &:= \mathbf{1}_{\{\mathcal{S}_{k,t} = \emptyset\}}, \\ I_{\mathrm{nodep},k,t} &:= \mathbf{1}_{\{\mathcal{S}_{k,t} \neq \emptyset, \mathcal{D}_{k,t} = \emptyset\}}, \\ I_{\mathrm{use},k,t} &:= \mathbf{1}_{\{c_{k,t} < c_{\max}\}}. \end{aligned} \tag{1.16}$$

The first decision with a margin below $c_{\max}$ and the *estimated peak exceedance risk* are

$$\begin{aligned} \tau_k^{\mathrm{use}} &:= \inf\{0 \le t < N_k : c_{k,t} < c_{\max}\}, \\ \widehat{p}_k^{\max} &:= \max_{0 \le t < N_k,\ c_{k,t} < c_{\max}} \widehat{p}_{k,t}(c_{k,t}). \end{aligned} \tag{1.17}$$

With $\mathcal{K}_{\mathrm{use}} := \{k : \tau_k^{\mathrm{use}} < \infty\}$ and $\mathcal{K}_{p^{\max}} := \{k : \widehat{p}_k^{\max} \text{ is defined}\}$, the mean *first-use time* and the mean estimated peak exceedance risk are

$$\begin{aligned} \widehat{m}_{\tau^{\mathrm{use}}} &:= \frac{T_s}{|\mathcal{K}_{\mathrm{use}}|} \sum_{k \in \mathcal{K}_{\mathrm{use}}} \tau_k^{\mathrm{use}}, \\ \widehat{m}_{p^{\max}} &:= \frac{1}{|\mathcal{K}_{p^{\max}}|} \sum_{k \in \mathcal{K}_{p^{\max}}} \widehat{p}_k^{\max}. \end{aligned} \tag{1.18}$$

The certificate count and the minimum barrier value are

$$\begin{aligned} n_{\mathrm{cert},k} &:= \#\{(c,r) : \tau_{k,r}(c) \le N_k\}, \\ h_k^{\min} &:= \min_{0\le t\le N_k} h(x_{k,t}), \end{aligned} \tag{1.19}$$

with means (1.15) $\widehat{m}_{\mathrm{cert}}$ and $\widehat{m}_{h^{\min}}$ over all trajectories. With the exceedance indicator $Z_{k,t}(c)$ and the nominal input u_t^{nom} in (A.4), define

$$\begin{aligned} I_{Y,k,t} &:= Z_{k,t}(c_{k,t}), \\ m_{Y,k} &:= \frac{1}{N_k}\sum_{t<N_k} Y_{k,t}, \\ m_{\|u-u^{\mathrm{nom}}\|^2,k} &:= \frac{1}{N_k}\sum_{t<N_k} \|u_{k,t}-u_{k,t}^{\mathrm{nom}}\|_2^2. \end{aligned} \tag{1.20}$$

The decision fraction of I_Y is the *barrier-loss exceedance fraction* $\widehat{q}_Y^{+}$, and the means over $\mathcal{K}_+$ of $m_{Y,k}$ and $m_{\|u-u^{\mathrm{nom}}\|^2,k}$ give the *normalized mean barrier loss* $\widehat{m}_Y/c_{\max}$ and the *control intervention* $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}$. With the body center $y_{k,t}$ and the clearance ψ of Appendix A.E-D, the *mean minimum sampled body clearance* $\widehat{m}_{\psi^{\min}}$ is the mean (1.15) of $\psi_k^{\min} := \min_{1\le t\le N_k}\psi(y_{k,t})$ over $\mathcal{K}_+$.

1.3 Parameters, Nominal Input and Lipschitz Constants

Table 1 gives the parameters of both scenarios and the input bounds, and Table 2 gives the obstacle coordinates.

Nominal input and safety filter In the Corridor, the nominal input is $u_t^{\mathrm{nom}} = (v^{\mathrm{nom}}, \omega^{\mathrm{nom}})$ with $v^{\mathrm{nom}} := v_{\mathrm{c}}\max\{\cos\tilde{\theta}, 0\}$ and $\omega^{\mathrm{nom}} := \mathrm{clip}(k_\theta\tilde{\theta})$, where $\tilde{\theta}$ is the heading error to the goal. In the Maze, $\tilde{\theta}$ is the heading error to the point one look-ahead distance beyond the point nearest to y on a fixed reference curve. The nearest-point search uses a nondecreasing progress index and a forward window of at most 400 sampled points. With d_{goal} the distance to the goal, $u_t^{\mathrm{nom}} = (a^{\mathrm{nom}}, \nu^{\mathrm{nom}})$ with $a^{\mathrm{nom}} := \mathrm{clip}(k_v(v^{\mathrm{des}} - v))$, $\nu^{\mathrm{nom}} := \mathrm{clip}(k_\omega(\omega^{\mathrm{des}} - \omega))$, $\omega^{\mathrm{des}} := \mathrm{clip}(k_\theta\tilde{\theta})$ and $v^{\mathrm{des}} := v_{\mathrm{c}}\max\{\cos\tilde{\theta}, 0\}\min\{1, d_{\mathrm{goal}}/(0.6\,\mathrm{m}) + 0.3\}$. The safety filter (A.4) uses the objective $(v - v^{\mathrm{nom}})^2 + w_\omega(\omega - \omega^{\mathrm{nom}})^2$ in the Corridor and $(a - a^{\mathrm{nom}})^2 + w_\nu(\nu - \nu^{\mathrm{nom}})^2$ in the Maze instead of $\|u - u_t^{\mathrm{nom}}\|^2$. In the Corridor, the filter searches 41 equally spaced yaw rates and the nominal yaw rate, selecting the closest feasible speed analytically for each. In the Maze, it minimizes the objective over a 41×25 equally spaced acceleration and yaw-acceleration grid, augmented by the respective nominal components. Each clip saturates at the corresponding bound of Table 1, which also gives the feedback coefficients k_θ, k_ω, k_v, the cruise speed v_{c}, the look-ahead distance and the weights w_ω and w_ν.

Lipschitz constants of the predicted nose point If the absolute speed is at most $\bar{v}$ throughout each sampling period, as in Appendix A.D-B, then (A.63) and (A.64) bound the induced norms of $\partial_x z$ and $\partial_u z$, and (A.62) holds on every convex operating region.

Fix (x, u) and let $\tilde{t} \in [0, T_s]$ be the time elapsed within one sampling period. With u held constant, (A.84) and (A.85) give the speed and the heading

$$v(\tilde{t}) = v + a\tilde{t}, \qquad \theta(\tilde{t}) = \theta + \omega\tilde{t} + \nu\tilde{t}^2/2, \tag{1.21}$$

with $a = \nu = 0$ in the Corridor, where v and ω are inputs. Hence $\hat{f}(x, u)$ has the heading $\theta^+ = \theta(T_s)$ and the position $y^+ = y + \int_0^{T_s} v(\tilde{t})\, e(\theta(\tilde{t}))\, d\tilde{t}$, and the predicted nose point of Appendix A.D-B is

$$z = y + \int_0^{T_s} v(\tilde{t})\, e(\theta(\tilde{t}))\, d\tilde{t} + l\, e(\theta^+). \tag{1.22}$$

With $e_\perp(\theta) := (-\sin\theta, \cos\theta)$, the derivative of $e(\theta)$ is $e_\perp(\theta)$. By (1.21), $\theta(\tilde{t})$ has the derivatives 1, $\tilde{t}$ and $\tilde{t}^2/2$ in θ,

Table 1: Parameters of both scenarios.

Quantity	Corridor	Maze
Start; goal; goal tolerance	$[0.3, 1.1] \times [0.5, 1.5]$ m, $\lvert\theta\rvert \le 15^\circ$; $(14.4, 1.0)$ m; 0.25 m	$(1.0, 0.75 \pm 0.08)$ m, $\theta = 90^\circ$; $(8.06, 8.25)$ m; 0.15 m
Input bounds	$\lvert v\rvert \le 0.6$ m/s, $\lvert\omega\rvert \le 2$ rad/s	$\lvert a\rvert \le 1.5$ m/s^2, $\lvert\nu\rvert \le 8$ rad/s^2
State bounds; braking a_{br}	–	$v \in [-0.3, 0.8]$ m/s, $\lvert\omega\rvert \le 1.5$ rad/s; 1.2 m/s^2
Cone radius ρ_j; body radius ρ_{body}	0.305 m; 0.22 m	same
Nose distance l	0.15 m	same
Nominal input	$k_\theta = 2$, $v_{\mathrm{c}} = 0.5$ m/s	look-ahead 0.5 m, $k_\theta = 2.5$, $k_\omega = 4$, $k_v = 2$, $v_{\mathrm{c}} = 0.6$ m/s
Filter weight	$w_\omega = 0.0225$	$w_\nu = 0.0225$
Gains Λ	$\{0.2, 0.4, 0.6, 0.8, 1.05\}$	same
K_x^z; K_u^z in (A.63), (A.64)	1.022; 0.102	1.067; 0.0204
$L_{\mathrm{geom}}(c)$ in (A.57) over Γ	$[3.09, 6.69] \times 10^{-3}$ m^{-1}	$[5.13, 11.14] \times 10^{-3}$ m^{-1}
L_x^{geom}; L_u^{geom} in (A.67)	6.84×10^{-3}; 6.80×10^{-4}	1.19×10^{-2}; 2.27×10^{-4}
ACI k_{ACI}; N^{win}; initial margin	0.02; 500; $c_{\max}$	0.005; 500; $c_{\max}$
QT k_{QT}; initial margin	$0.3\, c_{\max}$; $c_{\max}$	$0.02\, c_{\max}$; $c_{\max}$
Tolerances $\varepsilon_{\mathrm{num}}$; $\varepsilon_{\mathrm{chk}}$	10^{-10}; 10^{-12}	same

Table 2: Obstacle coordinates in meters: cone centers and walls $[\underline{z}_1, \bar{z}_1] \times [\underline{z}_2, \bar{z}_2]$.

Object	Coordinates
Corridor cones	$(2.40, 0.45)$, $(4.49, 1.51)$, $(6.20, 0.55)$, $(7.55, 1.55)$, $(8.90, 0.55)$, $(10.40, 1.35)$, $(11.85, 0.46)$, $(13.25, 1.45)$
Maze walls	$[2.650, 3.350] \times [0, 6.750]$, $[5.650, 6.350] \times [2.250, 9]$, $[8.650, 9] \times [0, 6.750]$, $[3.350, 4.400] \times [3.975, 5.025]$, $[7.600, 8.650] \times [4.931, 5.269]$, $[7.600, 8.650] \times [6.281, 6.619]$
Maze cones	$(1.62, 1.90)$, $(1.58, 4.87)$, $(1.80, 6.10)$, $(1.15, 7.80)$, $(4.20, 8.40)$, $(4.90, 7.25)$, $(5.08, 2.72)$, $(5.03, 2.03)$, $(7.77, 0.93)$, $(6.98, 2.47)$, $(7.83, 4.05)$, $(8.05, 6.95)$, $(6.70, 8.30)$

ω and ν, and $v(\tilde{t})$ has the derivatives 1 and $\tilde{t}$ in v and a. Differentiating (1.22) under the integral gives $\partial_y z = I_2$ and

$$\begin{aligned} \partial_\theta z &= \textstyle\int_0^{T_s} v(\tilde{t})\, e_\perp(\theta(\tilde{t}))\, d\tilde{t} + l\, e_\perp(\theta^+), & \partial_v z &= \textstyle\int_0^{T_s} e(\theta(\tilde{t}))\, d\tilde{t}, \\ \partial_\omega z &= \textstyle\int_0^{T_s} \tilde{t}\, v(\tilde{t})\, e_\perp(\theta(\tilde{t}))\, d\tilde{t} + l T_s\, e_\perp(\theta^+), & \partial_a z &= \textstyle\int_0^{T_s} \tilde{t}\, e(\theta(\tilde{t}))\, d\tilde{t}, \\ \partial_\nu z &= \textstyle\int_0^{T_s} \frac{\tilde{t}^2}{2}\, v(\tilde{t})\, e_\perp(\theta(\tilde{t}))\, d\tilde{t} + \frac{l T_s^2}{2}\, e_\perp(\theta^+). \end{aligned} \tag{1.23}$$

Since $\lvert v(\tilde{t})\rvert \le \bar{v}$, $\lVert e\rVert = \lVert e_\perp\rVert = 1$ and $\int_0^{T_s} \tilde{t}^k\, d\tilde{t} = T_s^{k+1}/(k+1)$, bounding the norm of each integral in (1.23) by the integral of the norm gives

$$\begin{aligned} \lVert\partial_\theta z\rVert &\le l + \bar{v} T_s, & \lVert\partial_v z\rVert &\le T_s, \\ \lVert\partial_\omega z\rVert &\le l T_s + \bar{v} T_s^2/2, & \lVert\partial_a z\rVert &\le T_s^2/2, \\ \lVert\partial_\nu z\rVert &\le l T_s^2/2 + \bar{v} T_s^3/6. \end{aligned} \tag{1.24}$$

Let J be the matrix of the columns of $\partial_x z$ other than $\partial_y z$, that is, $J = \partial_\theta z$ in the Corridor and $J = [\partial_\theta z, \partial_v z, \partial_\omega z]$ in the Maze, and let $\partial_u z = [\partial_v z, \partial_\omega z]$ in the Corridor and $\partial_u z = [\partial_a z, \partial_\nu z]$ in the Maze. Since $[I_2, J][I_2, J]^\top =$

$I_2 + JJ^\top$ and the spectral norm is at most the Frobenius norm $\|\cdot\|_F$,

$$\|\partial_x z\|_2^2 = 1 + \|J\|_2^2 \le 1 + \|J\|_F^2, \qquad \|\partial_u z\|_2^2 \le \|\partial_u z\|_F^2, \tag{1.25}$$

where each squared Frobenius norm is the sum of the squared column norms. Substituting (1.24) into (1.25) yields $\|\partial_x z\|_2 \le K_x^z$ and $\|\partial_u z\|_2 \le K_u^z$, with the constants (A.63) in the Corridor and (A.64) in the Maze. For two state-input pairs in a convex operating region, the segment between them lies in the region, and integrating the Jacobian of z along it gives (A.62). This proves the claim.

In the Maze, $v(\tilde{t})$ is affine in $\tilde{t}$ by (1.21), so $|v(\tilde{t})| \le \bar{v}$ holds when the current speed v and the predicted speed $v + aT_s$ satisfy the limits of Table 1. The nominal one-step predictor $\hat{f}$ of the Maze evaluates the integral in (1.22) with the 12-point Gauss–Legendre rule on $[0, T_s]$, so the derivatives of the computed z are the same rule applied to the integrands in (1.23). The rule has positive weights and integrates 1, $\tilde{t}$ and $\tilde{t}^2$ exactly. Hence (1.24), and with it the constants (A.64), also hold for the computed nose point.

1.4 Temporal Variation

The variable $\upsilon_{k,t}$ is the normalized robustness margin $c_{k,t}/c_{\max}$ or the estimated exceedance risk $\widehat{p}_{k,t}(c_{k,t})$, and $N_{\mathrm{var}} := 200$. The set $\mathcal{K}_\upsilon$ contains the trajectories with at least N_{var} decisions and finite $\upsilon_{k,t}$ for all $t < N_{\mathrm{var}}$. For $k \in \mathcal{K}_\upsilon$, the trajectory statistics and their means (1.15) are

$$\begin{aligned} \bar{\upsilon}_k &:= \frac{1}{N_{\mathrm{var}}} \sum_{t=0}^{N_{\mathrm{var}}-1} \upsilon_{k,t}, \qquad V_{\upsilon,k} := \frac{1}{N_{\mathrm{var}}-1} \sum_{t=0}^{N_{\mathrm{var}}-1} (\upsilon_{k,t} - \bar{\upsilon}_k)^2, \\ J_{\upsilon,k} &:= \Big[\frac{1}{N_{\mathrm{var}}-1} \sum_{t=1}^{N_{\mathrm{var}}-1} (\upsilon_{k,t} - \upsilon_{k,t-1})^2\Big]^{1/2}, \qquad \widehat{V}_\upsilon := \frac{1}{|\mathcal{K}_\upsilon|} \sum_{k \in \mathcal{K}_\upsilon} V_{\upsilon,k}, \quad \widehat{J}_\upsilon := \frac{1}{|\mathcal{K}_\upsilon|} \sum_{k \in \mathcal{K}_\upsilon} J_{\upsilon,k}. \end{aligned} \tag{1.26}$$

They give the *temporal variance* $\widehat{V}_\upsilon$ and the *root-mean-square change* $\widehat{J}_\upsilon$.

2 Simulation Figures and Tables

This section collects the tables and figures of the simulations of Appendix A.F at the *default setting* $(\alpha_0, \alpha, \delta, K) = (0.15, 0.10, 0.05, 50)$ and at S1–S5 (Table 3), together with heatmaps over the main grid. The data and the scripts are in the repository[1].

2.1 Compared Parameters and Table Key

Table 3 gives the parameter values. The columns of the result tables are the quantities of Appendix A.E. The tables of the δ, K, b, and ϕ comparisons omit the event counts, which the sweeps (Section 2.2.3) show.

Grid cells, cases and series A *grid cell* is one of the 375 settings of the main grid (Table 3) with one variant in one scenario. A *case* is the default setting or one of S1–S5 with one variant in one scenario at one value of each compared parameter below, and a *series* is a set of grid cells or cases that share the variant, the scenario, and all parameters except those varied. Every grid cell and case uses $N_{\mathrm{MC}} = 200$ trajectories, and all methods share the initial states and, for each L_μ, the disturbance realizations. Each comparison at the default or a selected setting has *matched baselines*, the baselines at the same α_0 and L_μ. The *detailed checks* (Section 2.8) compare the event counts with the bounds of Theorems A.1–A.3 and evaluate the checks of Appendix A.E-C in the 192 cases formed by the default and the selected settings, the four variants, both scenarios, both values of L_μ, and two test periods (K of the setting and $K = \infty$).

[1] https://github.com/omidmirza/ecbf-ablation

Compared parameters

- *Margin insufficiency α and CBF-violation probability α_0:* A larger α shrinks the null hypothesis (A.13), which raises each factor in (A.14) and speeds up certification, but reduces $\alpha_0 - \alpha$ in the deployment conditions (A.25) and (A.31). A larger α_0 relaxes $p_t(c_t) \leq \alpha_0$.
- *Error confidence δ:* A larger δ lowers the certification thresholds $1/\delta_r$ of Appendix A.F-B, while the bound (A.22) on the expected certification time, with δ replaced by δ_r, depends on δ_r only through $\log(|\Lambda|/\delta_r)$.
- *Test period K:* A smaller K starts tests more often, but a test starting at a given time then has a larger index r and a smaller δ_r. By (2.2) in Section 2.5, certificates of V-U, V-Δ, and V-L from test r are deployable for at most 100 decisions after σ_r at $L_\mu = 0.0005$ and $\alpha_0 - \alpha = 0.05$, while the V-C bound (A.72) relaxes this limit.
- *Budget normalizer b:* A larger b lowers only the error levels δ_r and thus raises every threshold $1/\delta_r$.
- *Bound factor ϕ* (Appendix A.F-B): Values $\phi < 1$ give bounds that need not satisfy Assumption A.2, so their results carry no guarantee.
- *Distribution-drift bound L_μ:* A larger L_μ makes the magnitude distribution (A.88) change faster and raises the drift term Δ_t (A.46c) in every risk-increase term.

Display conventions Event counts are those of the possible events (Appendix A.E-A), except the count range of QT at $\alpha_0 = 0.15$ in the Maze (Tables 5 and 21), which runs from the confirmed to the possible events, and the confirmed false-certification counts for S* (Table 8). All upper confidence bounds use possible events. The upper confidence bounds $\overline{q}$ (A.79) are rounded up. A count is consistent with its guarantee, and is marked ✓, if its upper confidence bound is at most the bound of the guarantee it checks, δ for finite K (Theorem A.3) and δ_1 for $K = \infty$ (Theorems A.1 and A.2). Zero possible events give the upper confidence bound 1.5%, so no count is consistent with a guarantee bound below 1.5%. For $\phi < 1$, no guarantee applies, and δ and δ_1 serve only for comparison. The upper confidence bounds are not adjusted for the number of settings or for the selection of S1–S5. Counts of trajectories with a CBF or safe-set violation carry no bound. Box plots show the median and the quartiles, and the whiskers extend to the most extreme values within 1.5 interquartile ranges of the quartiles (distribution figures) or to the 5th and 95th percentiles (sweep figures).

Key to all tables. $\pm$: SD; $^+$: decision fraction; ↓/↑: lower/higher is better; counts: trajectories of N_{MC} with a possible event, or the range from confirmed to possible events (Appendix A.E-A), except $\widehat{q}_{\mathrm{cert}}$ of S*, which gives confirmed events (Table 8), 95% upper confidence bound (%) from possible events in parentheses; ✓: bound at most δ, or δ_1 for $K = \infty$; N-A: not applicable; bold: best mean of the four variants at one setting (ties all bold; never SDs, $\widehat{m}_{\mathrm{cert}}$, $\widehat{m}_{\tau^{\mathrm{use}}}$, $\widehat{m}_{p^{\max}}$); a/m: a trajectories with an excess above $\varepsilon_{\mathrm{chk}}$ among the m with a checked decision (a alone if $m = N_{\mathrm{MC}}$), bound over all N_{MC}.

Table 3: Compared parameter values.

Comparison	Varied values	Fixed values
Main grid	$\alpha_0 = 0.10$: $\alpha = 0.05$ $\alpha_0 = 0.15$: $\alpha \in \{0.05, 0.10\}$ $\alpha_0 = 0.20$: $\alpha \in \{0.05, 0.10, 0.15\}$ $\alpha_0 = 0.25$: $\alpha \in \{0.05, 0.10, 0.15, 0.20\}$ $\alpha_0 = 0.30$: $\alpha \in \{0.05, 0.10, 0.15, 0.20, 0.25\}$ $\delta \in \{0.025, 0.05, 0.10, 0.15, 0.20\}$ $K \in \{10, 50, 100, 200, \infty\}$ methods: V-U, V-Δ, V-L, V-C	$\phi = 1$ $L_\mu = 0.0005$
Baselines	methods: Unfiltered, $c = 0$, $c_{\max}$, ACI, QT $\alpha_0 \in \{0.15, 0.30\}$ ACI also at $\alpha_0 \in \{0.10, 0.20, 0.25\}$ with $L_\mu = 0.0005$ (in the repository) $L_\mu \in \{0.0005, 0.005\}$	no certification parameters
Default setting	$(\alpha_0, \alpha, \delta, K) = (0.15, 0.10, 0.05, 50)$ methods: V-U, V-Δ, V-L, V-C	$\phi = 1$ $L_\mu = 0.0005$ (part of the main grid)
Selected settings	settings $(\alpha_0, \alpha, \delta, K)$: S1: $(0.30, 0.25, 0.20, 50)$, V-C S2: $(0.30, 0.25, 0.15, 50)$, V-C S3: $(0.30, 0.20, 0.20, 50)$, V-L S4: $(0.30, 0.25, 0.025, 100)$, V-Δ S5: $(0.30, 0.25, 0.10, 10)$, V-C methods: V-U, V-Δ, V-L, V-C	$\phi = 1$ $L_\mu = 0.0005$ (part of the main grid)
Budget normalizer b	$b \in \{\kappa, 2.5, 3, 4\}$ settings: default setting, S1–S5 methods: V-U, V-Δ, V-L, V-C	K of the setting $\phi = 1$ $L_\mu = 0.0005$
Bound factor ϕ and distribution-drift bound L_μ	$\phi \in \{0, 0.25, 0.5, 0.75, 1\}$ $L_\mu \in \{0.0005, 0.005\}$ (ϕ and L_μ crossed) settings: default setting, S1–S5 methods: V-U, V-Δ, V-L, V-C	K of the setting
Test period K (detailed checks)	K of the setting and $K = \infty$ $L_\mu \in \{0.0005, 0.005\}$ (K and L_μ crossed) settings: default setting, S1–S5 methods: V-U, V-Δ, V-L, V-C	$\phi = 1$

2.2 Default and Selected Settings

Tables 4 and 5 compare all nine methods at the default setting, and Tables 6 and 7 give S1–S5 with all four variants.

Table 4: Corridor, all metrics of Appendices A.E-A and A.E-B for the baselines and for the default setting with all four variants. $N_{\mathrm{MC}} = 200$ trajectories per method.

Corridor **Baselines** $\alpha_0 = 0.15,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$	$\widehat{q}_{\mathrm{dep}}\downarrow$
Unfiltered	N-A 0.17±0.04 0.00±0.00	30.69±13.51 N-A 22.32±10.04	N-A N-A N-A	−0.27±0.05 −0.02±0.01 70.00±0.00	N-A
$c = 0$	0.00±0.00 0.17±0.02 0.66±0.07	28.00±2.99 49.65±2.89 15.27±4.17	100.0±0.0 0.00±0.00 78.8±8.6	−0.03±0.01 0.10±0.01 32.45±0.77	200 (100.0)
$c_{\max}$	1.00±0.00 0.17±0.01 1.00±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.05±0.01 0.17±0.01 35.91±0.92	0 (1.5)
ACI	0.49±0.04 0.17±0.02 0.82±0.09	9.02±1.17 15.26±0.53 0.05±0.17	98.2±0.0 0.60±0.00 33.4±8.6	0.01±0.01 0.14±0.01 34.00±0.87	200 (100.0)
QT	0.52±0.04 0.17±0.02 0.86±0.09	8.34±0.97 14.38±0.32 0.02±0.10	99.0±0.8 0.10±0.00 49.7±8.0	0.01±0.01 0.14±0.01 34.17±0.89	200 (100.0)

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$						
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$	$\widehat{q}_{\mathrm{dep}}\downarrow$ $\widehat{q}_{\mathrm{cert}}\downarrow$
default setting: $(\alpha_0, \alpha, \delta, K) = (0.15, 0.10, 0.05, 50)$						
V-U	1.00±0.00 **0.17**±0.01 1.00±0.08	**0.01**±0.07 **0.03**±0.10 **0.00**±0.00	1.9±1.5 5.83±2.25 2.1±1.7	**17.9**±6.5 80.2±5.4 9.65±2.33	**0.05**±0.01 **0.17**±0.01 35.91±0.91	0 (1.5) ✓ 0 (1.5) ✓
V-Δ	1.00±0.00 **0.17**±0.01 1.00±0.08	**0.01**±0.07 **0.03**±0.10 **0.00**±0.00	1.9±1.5 5.83±2.25 2.1±1.7	**17.9**±6.5 80.2±5.4 9.65±2.33	**0.05**±0.01 **0.17**±0.01 35.91±0.91	0 (1.5) ✓ 0 (1.5) ✓
V-L	0.96±0.02 **0.17**±0.01 0.98±0.08	0.30±0.32 0.49±0.40 **0.00**±0.00	25.3±9.6 6.64±2.97 3.4±2.0	18.0±6.5 56.7±8.6 9.53±2.18	0.04±0.01 **0.17**±0.01 35.75±0.92	0 (1.5) ✓ 0 (1.5) ✓
V-C	**0.83**±0.05 **0.17**±0.01 **0.93**±0.09	2.12±1.18 3.37±1.63 **0.00**±0.00	**79.2**±6.2 6.45±2.37 5.7±2.2	18.3±6.6 **2.5**±0.7 9.39±2.38	0.03±0.01 0.16±0.01 **35.14**±0.91	0 (1.5) ✓ 0 (1.5) ✓

Table 5: Maze, as in Table 4.

Maze **Baselines** $\alpha_0 = 0.15,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$					
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$	$\widehat{q}^+_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^+_Y\downarrow(\%)$ $\widehat{q}^+_{\text{unsafe}}\downarrow(\%)$	$\widehat{q}^+_{\text{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$ $\widehat{T}_{\text{mis}}\downarrow(\text{s})$	$\widehat{q}_{\text{dep}}\downarrow$
Unfiltered	N-A 0.17±0.02 0.00±0.00	55.68±1.58 N-A 53.27±0.95	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.16±0.14	N-A
$c = 0$	0.00±0.00 0.17±0.02 0.32±0.02	24.93±2.57 49.79±2.97 3.86±2.16	100.0±0.0 0.00±0.00 66.8±12.7	−0.01±0.00 0.21±0.01 48.67±0.27	200 (100.0)
$c_{\max}$	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.22±0.01 54.44±0.32	0 (1.5)
ACI	0.49±0.05 0.17±0.02 0.50±0.03	7.45±1.47 14.94±1.34 0.01±0.06	97.7±0.0 1.20±0.00 27.6±7.3	0.01±0.00 0.21±0.01 51.16±0.40	200 (100.0)
QT	0.72±0.02 0.17±0.02 0.61±0.02	2.78±0.88 6.40±0.81 0.00±0.00	99.6±0.1 0.20±0.00 15.3±3.7	0.02±0.00 0.21±0.01 52.52±0.33	85–87 (49.6)

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$						
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$	$\widehat{q}^+_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^+_Y\downarrow(\%)$ $\widehat{q}^+_{\text{unsafe}}\downarrow(\%)$	$\widehat{q}^+_{\text{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^+_{\text{nocert}}\downarrow(\%)$ $\widehat{q}^+_{\text{nodep}}\downarrow(\%)$ $\widehat{m}_{\text{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$ $\widehat{T}_{\text{mis}}\downarrow(\text{s})$	$\widehat{q}_{\text{dep}}\downarrow$ $\widehat{q}_{\text{cert}}\downarrow$
default setting: $(\alpha_0, \alpha, \delta, K)$ = (0.15, 0.10, 0.05, 50)						
V-U	1.00±0.00 **0.17**±0.02 0.81±0.02	**0.00**±0.00 **0.00**±0.00 **0.00**±0.00	0.0±0.0 N-A N-A	**24.4**±8.9 75.6±8.9 6.59±1.93	**0.03**±0.00 **0.22**±0.01 54.44±0.32	0 (1.5) ✓ 0 (1.5) ✓
V-Δ	1.00±0.00 **0.17**±0.02 0.81±0.02	**0.00**±0.00 **0.00**±0.00 **0.00**±0.00	0.0±0.0 N-A N-A	**24.4**±8.9 75.6±8.9 6.59±1.93	**0.03**±0.00 **0.22**±0.01 54.44±0.32	0 (1.5) ✓ 0 (1.5) ✓
V-L	0.96±0.02 **0.17**±0.02 0.77±0.03	0.17±0.28 0.55±0.51 **0.00**±0.00	26.2±12.2 13.80±6.99 2.9±1.8	24.5±8.9 49.3±9.2 6.57±1.96	**0.03**±0.00 0.21±0.01 54.15±0.38	0 (1.5) ✓ 0 (1.5) ✓
V-C	**0.86**±0.05 **0.17**±0.02 **0.70**±0.04	0.77±0.73 2.36±1.25 **0.00**±0.00	**72.6**±8.8 13.28±4.84 4.7±1.9	24.8±9.0 **2.6**±0.7 6.49±1.88	0.02±0.00 0.21±0.01 **53.40**±0.44	0 (1.5) ✓ 0 (1.5) ✓

Table 6: Corridor, all metrics of Appendices A.E-A and A.E-B for the baselines and for S1–S5 with all four variants. $N_{\mathrm{MC}} = 200$ trajectories per method.

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$	$\widehat{q}_{\mathrm{dep}}\downarrow$
Unfiltered	N-A 0.17±0.04 0.00±0.00	30.69±13.51 N-A 22.32±10.04	N-A N-A N-A	−0.27±0.05 −0.02±0.01 70.00±0.00	N-A
$c = 0$	0.00±0.00 0.17±0.02 0.66±0.07	28.00±2.99 49.65±2.89 15.27±4.17	100.0±0.0 0.00±0.00 78.8±8.6	−0.03±0.01 0.10±0.01 32.45±0.77	200 (100.0)
$c_{\max}$	1.00±0.00 0.17±0.01 1.00±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.05±0.01 0.17±0.01 35.91±0.92	0 (1.5)
ACI	0.22±0.04 0.17±0.02 0.73±0.08	17.68±1.59 30.21±0.72 2.74±1.43	99.1±0.0 0.30±0.00 57.4±10.2	−0.01±0.01 0.12±0.01 33.13±0.81	200 (100.0)
QT	0.30±0.03 0.17±0.02 0.78±0.09	15.19±1.31 26.54±1.03 0.61±0.52	99.7±0.1 0.10±0.00 60.1±11.5	−0.00±0.00 0.13±0.01 33.44±0.83	200 (100.0)

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$						
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$	$\widehat{q}_{\mathrm{dep}}\downarrow$ $\widehat{q}_{\mathrm{cert}}\downarrow$
S1: $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.20, 50)$						
V-U	0.79±0.03 **0.17**±0.01 0.93±0.08	**2.57**±0.88 4.71±1.12 **0.01**±0.15	67.7±3.3 1.53±0.31 16.2±4.0	**4.3**±0.9 28.0±3.2 31.50±3.27	**0.02**±0.01 **0.16**±0.01 35.06±0.83	0 (1.5) ✓ 1 (2.4) ✓
V-Δ	0.79±0.03 **0.17**±0.01 0.93±0.08	**2.57**±0.88 **4.70**±1.12 **0.01**±0.15	67.7±3.3 1.53±0.31 16.2±4.0	**4.3**±0.9 28.0±3.2 31.48±3.21	**0.02**±0.01 **0.16**±0.01 35.06±0.83	0 (1.5) ✓ 1 (2.4) ✓
V-L	0.63±0.05 **0.17**±0.02 0.85±0.08	5.86±1.20 9.48±1.53 0.04±0.21	92.2±1.5 1.53±0.31 17.8±3.7	4.4±0.9 3.4±1.2 30.97±3.33	0.01±0.01 0.14±0.01 34.38±0.87	0 (1.5) ✓ 0 (1.5) ✓
V-C	**0.50**±0.06 **0.17**±0.02 **0.80**±0.09	9.74±2.12 15.10±2.94 0.18±0.41	**92.3**±1.1 1.53±0.31 21.1±3.3	4.5±0.9 **3.2**±0.8 30.46±3.36	0.01±0.01 0.13±0.01 **33.86**±0.89	0 (1.5) ✓ 3 (3.9) ✓
S2: $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.15, 50)$						
V-U	0.81±0.03 **0.17**±0.01 0.94±0.08	**2.32**±0.81 **4.29**±1.08 **0.01**±0.15	65.0±3.4 1.74±0.32 15.3±3.8	**4.9**±0.9 30.1±3.3 30.95±3.27	**0.03**±0.01 **0.16**±0.01 35.12±0.84	0 (1.5) ✓ 1 (2.4) ✓
V-Δ	0.81±0.03 **0.17**±0.01 0.94±0.08	**2.32**±0.81 **4.29**±1.08 **0.01**±0.15	65.0±3.4 1.74±0.32 15.3±3.8	**4.9**±0.9 30.1±3.3 30.95±3.26	**0.03**±0.01 **0.16**±0.01 35.12±0.84	0 (1.5) ✓ 1 (2.4) ✓
V-L	0.64±0.05 **0.17**±0.02 0.86±0.09	5.54±1.19 8.94±1.47 0.03±0.21	**91.7**±1.2 1.74±0.32 17.0±3.6	5.0±0.9 3.3±0.9 30.41±3.49	0.02±0.01 0.14±0.01 34.43±0.88	0 (1.5) ✓ 1 (2.4) ✓
V-C	**0.52**±0.06 **0.17**±0.02 **0.81**±0.09	9.35±1.96 14.52±2.74 0.14±0.36	**91.7**±1.2 1.74±0.32 20.3±3.3	5.1±0.9 **3.2**±0.8 29.80±3.27	0.01±0.01 0.13±0.01 **33.92**±0.88	0 (1.5) ✓ 1 (2.4) ✓

Table 6 (continued).

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\text{m},\ T_s = 0.1\,\text{s},\ N_{\text{MC}} = 200$						
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\text{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\text{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\text{nodep}}\downarrow(\%)$ $\widehat{m}_{\text{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$ $\widehat{T}_{\text{mis}}\downarrow(\text{s})$	$\widehat{q}_{\text{dep}}\downarrow$ $\widehat{q}_{\text{cert}}\downarrow$
S3: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.20, 0.20, 50)						
V-U	0.67±0.05 **0.17**±0.02 0.87±0.09	**5.03**±1.15 **7.89**±1.54 **0.02**±0.20	**90.9**±1.6 2.04±0.54 14.0±3.6	**5.9**±1.6 3.3±0.8 24.57±3.19	**0.02**±0.01 **0.15**±0.01 34.53±0.90	0 (1.5) ✓ 7 (6.5) ✓
V-Δ	0.67±0.05 **0.17**±0.02 0.87±0.09	5.04±1.19 7.91±1.57 **0.02**±0.20	**90.9**±1.7 2.04±0.54 14.0±3.6	**5.9**±1.6 3.3±0.8 24.56±3.17	**0.02**±0.01 **0.15**±0.01 34.53±0.90	0 (1.5) ✓ 7 (6.5) ✓
V-L	0.62±0.06 **0.17**±0.02 0.84±0.09	6.38±1.57 10.03±2.14 0.04±0.21	**90.9**±1.7 2.04±0.54 15.3±3.6	**5.9**±1.6 **3.2**±0.8 24.33±3.09	0.01±0.01 0.14±0.01 34.28±0.90	0 (1.5) ✓ 7 (6.5) ✓
V-C	**0.58**±0.07 **0.17**±0.02 **0.83**±0.09	7.44±2.16 11.73±3.19 0.07±0.27	**90.9**±1.7 2.04±0.54 16.6±3.8	**5.9**±1.6 **3.2**±0.8 24.28±3.09	0.01±0.01 0.14±0.01 **34.18**±0.91	0 (1.5) ✓ 7 (6.5) ✓
S4: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.025, 100)						
V-U	0.92±0.02 **0.17**±0.01 0.97±0.08	**0.75**±0.52 **1.44**±0.71 **0.00**±0.00	29.8±3.3 2.74±0.47 11.2±3.4	**7.7**±1.3 62.5±3.0 16.38±1.82	**0.03**±0.01 **0.17**±0.01 35.62±0.88	0 (1.5) ✓ 0 (1.5) ✓
V-Δ	0.92±0.02 **0.17**±0.01 0.97±0.08	**0.75**±0.52 **1.44**±0.71 **0.00**±0.00	29.8±3.3 2.74±0.47 11.2±3.4	**7.7**±1.3 62.5±3.0 16.38±1.82	**0.03**±0.01 **0.17**±0.01 35.62±0.88	0 (1.5) ✓ 0 (1.5) ✓
V-L	0.82±0.03 **0.17**±0.01 0.92±0.08	2.48±0.91 4.12±1.18 **0.00**±0.02	58.7±2.7 2.74±0.47 13.8±3.2	7.8±1.3 33.6±2.5 15.88±1.88	0.02±0.01 0.15±0.01 35.07±0.90	0 (1.5) ✓ 0 (1.5) ✓
V-C	**0.58**±0.05 **0.17**±0.02 **0.83**±0.09	7.74±1.71 11.99±2.17 0.05±0.19	**88.9**±1.4 2.74±0.47 18.3±2.9	8.0±1.3 **3.1**±0.7 15.15±1.83	0.01±0.01 0.14±0.01 **34.14**±0.88	0 (1.5) ✓ 0 (1.5) ✓
S5: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.10, 10)						
V-U	0.77±0.04 **0.17**±0.01 0.91±0.08	**2.87**±0.82 **4.77**±0.91 **0.01**±0.15	89.1±3.4 1.99±0.40 13.6±4.0	**5.7**±1.1 5.3±3.2 122.99±13.66	**0.03**±0.01 **0.15**±0.01 34.92±0.89	0 (1.5) ✓ 1 (2.4) ✓
V-Δ	0.77±0.04 **0.17**±0.01 0.91±0.08	**2.87**±0.82 **4.77**±0.91 **0.01**±0.15	89.1±3.4 1.99±0.40 13.6±4.0	**5.7**±1.1 5.3±3.2 122.99±13.67	**0.03**±0.01 **0.15**±0.01 34.92±0.89	0 (1.5) ✓ 1 (2.4) ✓
V-L	0.66±0.05 **0.17**±0.02 0.87±0.09	5.14±1.09 8.21±1.32 0.02±0.20	**91.0**±1.4 1.99±0.40 15.7±3.6	**5.7**±1.1 3.3±0.8 121.62±14.51	0.02±0.01 **0.15**±0.01 34.51±0.88	0 (1.5) ✓ 0 (1.5) ✓
V-C	**0.53**±0.06 **0.17**±0.02 **0.81**±0.09	8.94±1.88 13.84±2.67 0.12±0.36	**91.0**±1.4 1.99±0.40 19.6±3.2	5.8±1.2 **3.2**±0.8 119.71±13.69	0.01±0.01 0.13±0.01 **33.97**±0.88	0 (1.5) ✓ 0 (1.5) ✓

Table 7: Maze, as in Table 6.

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$					
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$	$\widehat{q}^+_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^+_{Y}\downarrow(\%)$ $\widehat{q}^+_{\text{unsafe}}\downarrow(\%)$	$\widehat{q}^+_{\text{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$ $\widehat{T}_{\text{mis}}\downarrow(\text{s})$	$\widehat{q}_{\text{dep}}\downarrow$
Unfiltered	N-A 0.17±0.02 0.00±0.00	55.68±1.58 N-A 53.27±0.95	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.16±0.14	N-A
$c = 0$	0.00±0.00 0.17±0.02 0.32±0.02	24.93±2.57 49.79±2.97 3.86±2.16	100.0±0.0 0.00±0.00 66.8±12.7	−0.01±0.00 0.21±0.01 48.67±0.27	200 (100.0)
$c_{\max}$	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.22±0.01 54.44±0.32	0 (1.5)
ACI	0.22±0.04 0.17±0.02 0.39±0.02	15.84±1.93 29.86±1.85 0.48±0.65	98.8±0.0 0.60±0.00 47.1±8.6	−0.00±0.00 0.21±0.01 49.71±0.32	199 (100.0)
QT	0.50±0.03 0.17±0.02 0.50±0.02	8.04±1.31 15.68±1.10 0.05±0.17	99.6±0.0 0.20±0.00 33.8±7.6	0.01±0.00 0.21±0.01 51.18±0.32	131 (71.1)

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$						
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$	$\widehat{q}^+_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^+_{Y}\downarrow(\%)$ $\widehat{q}^+_{\text{unsafe}}\downarrow(\%)$	$\widehat{q}^+_{\text{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^+_{\text{nocert}}\downarrow(\%)$ $\widehat{q}^+_{\text{nodep}}\downarrow(\%)$ $\widehat{m}_{\text{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$ $\widehat{T}_{\text{mis}}\downarrow(\text{s})$	$\widehat{q}_{\text{dep}}\downarrow$ $\widehat{q}_{\text{cert}}\downarrow$
S1: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.20, 50)						
V-U	0.97±0.01 **0.17**±0.02 0.80±0.02	**0.15**±0.25 **0.68**±0.53 **0.00**±0.00	11.3±2.0 3.04±0.63 11.1±5.3	**5.6**±1.2 83.1±1.6 23.50±2.71	**0.03**±0.00 **0.21**±0.01 54.36±0.32	0 (1.5) ✓ 0 (1.5) ✓
V-Δ	0.97±0.01 **0.17**±0.02 0.80±0.02	**0.15**±0.26 0.71±0.54 **0.00**±0.00	12.2±2.1 3.04±0.63 11.1±5.2	**5.6**±1.2 82.2±1.8 23.44±2.67	**0.03**±0.00 **0.21**±0.01 54.34±0.32	0 (1.5) ✓ 0 (1.5) ✓
V-L	0.62±0.05 **0.17**±0.02 0.56±0.03	4.77±1.42 9.74±1.93 **0.00**±0.04	91.0±1.4 3.04±0.63 17.2±3.8	5.9±1.2 3.2±0.8 22.91±2.42	0.01±0.00 **0.21**±0.01 51.88±0.42	0 (1.5) ✓ 0 (1.5) ✓
V-C	**0.52**±0.07 **0.17**±0.02 **0.51**±0.04	7.40±2.27 14.52±3.57 0.01±0.06	**91.1**±1.4 3.04±0.63 19.7±3.5	5.9±1.2 **2.9**±0.8 22.83±2.54	0.01±0.00 **0.21**±0.01 **51.25**±0.52	0 (1.5) ✓ 1 (2.4) ✓
S2: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.15, 50)						
V-U	0.98±0.01 **0.17**±0.02 0.80±0.02	**0.13**±0.25 **0.54**±0.46 **0.00**±0.00	9.4±1.9 3.72±1.99 9.6±4.9	**6.4**±1.1 84.3±1.5 23.00±2.66	**0.03**±0.00 **0.21**±0.01 54.36±0.34	0 (1.5) ✓ 0 (1.5) ✓
V-Δ	0.97±0.01 **0.17**±0.02 0.80±0.02	**0.13**±0.25 0.56±0.47 **0.00**±0.00	10.2±2.0 3.50±0.98 9.7±4.9	**6.4**±1.1 83.5±1.7 23.05±2.68	**0.03**±0.00 **0.21**±0.01 54.36±0.32	0 (1.5) ✓ 0 (1.5) ✓
V-L	0.64±0.05 **0.17**±0.02 0.56±0.03	4.59±1.37 9.25±1.84 **0.00**±0.04	**90.3**±1.5 3.45±0.62 16.7±3.9	6.6±1.2 3.1±0.8 22.56±2.38	0.01±0.00 **0.21**±0.01 51.93±0.45	0 (1.5) ✓ 0 (1.5) ✓
V-C	**0.53**±0.08 **0.17**±0.02 **0.51**±0.04	7.28±2.37 13.89±3.46 0.01±0.07	**90.3**±1.5 3.45±0.62 19.2±3.7	6.7±1.2 **2.9**±0.8 22.37±2.50	0.01±0.00 **0.21**±0.01 **51.33**±0.53	0 (1.5) ✓ 0 (1.5) ✓

Table 7 (continued).

Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$	$\widehat{q}_{\mathrm{dep}}\downarrow$ $\widehat{q}_{\mathrm{cert}}\downarrow$
Maze $\phi=1,\ L_\mu=0.0005,\ c_{\max}=0.01\,\mathrm{m},\ T_s=0.2\,\mathrm{s},\ N_{\mathrm{MC}}=200$						
S3: $(\alpha_0,\alpha,\delta,K)$ = (0.30, 0.20, 0.20, 50)						
V-U	0.82±0.04 **0.17**±0.02 0.68±0.03	**1.49**±0.82 **3.70**±1.23 **0.00**±0.00	67.8±5.6 4.01±0.97 11.7±3.8	**7.5**±1.8 24.7±5.3 18.91±2.53	**0.02**±0.00 **0.21**±0.01 53.19±0.38	0 (1.5) ✓ 5 (5.2) ✓
V-Δ	0.82±0.04 **0.17**±0.02 0.68±0.03	1.54±0.84 3.76±1.23 **0.00**±0.00	69.4±5.5 4.01±0.97 11.6±3.9	**7.5**±1.8 23.1±5.1 18.83±2.57	**0.02**±0.00 **0.21**±0.01 53.17±0.37	0 (1.5) ✓ 5 (5.2) ✓
V-L	0.62±0.07 **0.17**±0.02 **0.55**±0.04	5.04±1.93 9.98±2.87 **0.00**±0.06	**89.1**±2.0 4.01±0.97 14.5±3.8	7.7±1.9 3.2±0.9 18.19±2.35	0.01±0.00 **0.21**±0.01 51.86±0.54	0 (1.5) ✓ 4 (4.6) ✓
V-C	**0.61**±0.08 **0.17**±0.02 **0.55**±0.04	5.38±2.19 10.63±3.31 **0.00**±0.06	**89.1**±1.9 4.01±0.97 15.3±3.8	7.7±1.8 **3.1**±0.9 18.26±2.33	0.01±0.00 **0.21**±0.01 **51.79**±0.58	0 (1.5) ✓ 4 (4.6) ✓
S4: $(\alpha_0,\alpha,\delta,K)$ = (0.30, 0.25, 0.025, 100)						
V-U	1.00±0.00 **0.17**±0.01 0.81±0.02	**0.00**±0.00 **0.04**±0.12 **0.00**±0.00	1.3±0.7 7.85±7.25 3.9±3.1	**10.0**±1.6 88.7±1.0 12.86±1.58	**0.03**±0.00 **0.22**±0.01 54.43±0.32	0 (1.5) ✓ 0 (1.5) ✓
V-Δ	1.00±0.00 **0.17**±0.01 0.81±0.02	**0.00**±0.00 0.05±0.13 **0.00**±0.00	1.4±0.8 7.85±7.25 3.9±3.1	**10.0**±1.6 88.5±1.0 12.86±1.58	**0.03**±0.00 **0.22**±0.01 54.43±0.32	0 (1.5) ✓ 0 (1.5) ✓
V-L	0.81±0.04 **0.17**±0.02 0.65±0.03	2.05±0.95 4.27±1.47 **0.00**±0.00	58.5±3.1 5.45±0.85 13.2±3.5	10.3±1.6 31.2±2.7 12.74±1.59	0.02±0.00 0.21±0.01 53.01±0.40	0 (1.5) ✓ 0 (1.5) ✓
V-C	**0.60**±0.06 **0.17**±0.02 **0.54**±0.04	5.75±1.88 11.17±2.50 0.01±0.05	**86.6**±1.8 5.45±0.85 17.2±3.2	10.5±1.6 **2.9**±0.8 12.20±1.44	0.01±0.00 0.21±0.01 **51.67**±0.47	0 (1.5) ✓ 0 (1.5) ✓
S5: $(\alpha_0,\alpha,\delta,K)$ = (0.30, 0.25, 0.10, 10)						
V-U	0.99±0.01 **0.17**±0.02 0.80±0.02	**0.03**±0.12 **0.29**±0.33 **0.00**±0.00	5.4±1.7 3.96±1.03 7.4±4.6	**7.2**±1.4 87.4±1.0 90.06±11.54	**0.03**±0.00 **0.21**±0.01 54.43±0.34	0 (1.5) ✓ 0 (1.5) ✓
V-Δ	0.99±0.01 **0.17**±0.02 0.80±0.02	**0.03**±0.12 **0.29**±0.33 **0.00**±0.00	5.6±1.7 3.96±1.03 7.4±4.6	**7.2**±1.4 87.2±1.0 90.06±11.54	**0.03**±0.00 **0.21**±0.01 54.43±0.34	0 (1.5) ✓ 0 (1.5) ✓
V-L	0.66±0.05 **0.17**±0.02 0.57±0.03	4.20±1.20 8.42±1.51 **0.00**±0.05	**89.3**±1.6 3.93±0.77 15.0±3.7	7.5±1.5 3.2±0.8 85.18±9.56	0.01±0.00 **0.21**±0.01 52.11±0.45	0 (1.5) ✓ 0 (1.5) ✓
V-C	**0.55**±0.07 **0.17**±0.02 **0.52**±0.04	6.76±2.17 13.03±3.13 0.01±0.09	**89.3**±1.7 3.93±0.77 18.5±3.3	7.6±1.5 **3.0**±0.8 83.58±9.71	0.01±0.00 **0.21**±0.01 **51.43**±0.52	0 (1.5) ✓ 0 (1.5) ✓

2.2.1 Setting S*

Table 8 gives the additional Maze setting S*, $(\alpha_0,\alpha,\delta,K)=(0.25,0.225,0.15,50)$, with all four variants. The remaining parameters take their default values. All trajectories reach the goal without contact. The displayed false-certification counts are confirmed events, whereas the upper confidence bounds include possible events. For V-C these counts are 3 and 4, respectively, and the upper confidence bound is 4.6%.

Table 8: Maze, S*, all four variants. False-certification counts are confirmed events, with upper confidence bounds based on possible events.

Maze $\phi = 1$, $L_\mu = 0.0005$, $c_{\max} = 0.01\,\mathrm{m}$, $T_s = 0.2\,\mathrm{s}$, $N_{\mathrm{MC}} = 200$ –: not available; a column keeps its recorded digits where rounding to those of Table 7 is ambiguous						
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\lVert u-u^{\mathrm{nom}}\rVert^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$	$\widehat{q}_{\mathrm{dep}}\downarrow$ $\widehat{q}_{\mathrm{cert}}\downarrow$
S*: $(\alpha_0, \alpha, \delta, K) = (0.25, 0.225, 0.15, 50)$						
V-U	1.00±0.00 **0.170**±0.015 0.808±0.021	**0.00**±0.00 **0.00**±0.00 **0.00**±0.00	0.00±0.00 N-A N-A	**6.88**±1.59 93.1±1.6 20.9±2.5	**0.03**±0.00 – 54.44±0.32	0 (1.5) ✓ 1 (3.2) ✓
V-Δ	1.00±0.00 **0.170**±0.015 0.808±0.021	**0.00**±0.00 **0.00**±0.00 **0.00**±0.00	0.00±0.00 N-A N-A	**6.88**±1.59 93.1±1.6 20.9±2.5	**0.03**±0.00 – 54.44±0.32	0 (1.5) ✓ 1 (3.2) ✓
V-L	0.91±0.02 **0.170**±0.016 0.748±0.026	0.74±0.64 1.89±1.00 **0.00**±0.00	34.19±4.12 3.74±0.86 11.7±4.5	6.95±1.60 58.9±3.9 20.8±2.5	0.02±0.00 – 53.88±0.34	0 (1.5) ✓ 1 (3.2) ✓
V-C	**0.61**±0.06 **0.170**±0.016 **0.549**±0.035	5.17±1.66 10.35±2.21 <0.01±0.04	**89.75**±1.70 3.74±0.86 15.8±3.8	7.23±1.66 **3.0**±0.8 20.3±2.3	0.01±0.00 – **51.76**±0.43	1 (2.4) ✓ 3 (4.6) ✓

2.2.2 Trajectories and Distributions

Figs. 1–12 show the nose-point trajectories of the default setting with V-C and of S1–S5 with their variants. In (a), "Contact" marks the contact boundaries (Appendix A.E-D) of the body center and "$h = 0$" (Maze: "$h = 0$ at $v = 0$") the zero level of the barrier h in (A.86). The inset shows an obstacle passage where the example trajectory passes closer than that of the $c_{\max}$ baseline. Panels (b)–(e), with c_t in (b), Y_t and c_t in (c), the estimated exceedance risk $\widehat{p}_t(c_t)$ defined in (1.8) in (d), and the supporting test index r_t defined in (A.91) of Appendix A.F-A in (e), show the example trajectory, and (b) and (e) also the mean over the trajectories not yet at the goal. In (c), the percentage is the decision fraction q_k^+ (A.80) of $I_{Y,k,t}$ (1.20) of the example trajectory. The example trajectory has the same index k in every map, chosen independently of the outcomes. Figs. 13–18 show the distributions of the per-trajectory values $T_{\mathrm{mis},k}$, $m_{c,k}/c_{\max}$, $q^+_{\mathrm{CBF},k}$, $q^+_{\mathrm{unsafe},k}$ (A.80), (A.81), $\widehat{p}_k^{\max}$ (1.17), and $h_k^{\min}$ (1.19) at the default setting and at S1–S5. Points are trajectories, boxes span the quartiles with the median, and whiskers reach the most extreme values within 1.5 interquartile ranges of the quartiles. Unfiltered control is omitted for scale.

2.2.3 One-Parameter Sweeps

Figs. 19–30 vary one parameter (Section 2.1) per column around the default setting and each of S1–S5: α, δ, K, b, and ϕ, the last at both values of L_μ. The rows show $m_{c,k}/c_{\max}$, $q^+_{\mathrm{CBF},k}$, $q^+_{\mathrm{unsafe},k}$, and $T_{\mathrm{mis},k}$, defined in (A.80) and (A.81) of Appendix A.E-B, as boxes (Section 2.1) with the mean as a marker, and the event counts with bars to their upper confidence bounds (A.79). A row that is zero for every variant is omitted. The level line is N_{MC} times the bound, δ or δ_1, of the guarantee the count checks. ACI and QT are drawn at $L_\mu = 0.0005$ in every column, as lines where their values lie within the axis range (Tables 15 and 16 give them at $L_\mu = 0.005$). Right-hand labels give their values, with ▲/▼ for a value above/below the axis range. At $\phi = 1$, every count in these figures is consistent with its guarantee except at S4 with $K = \infty$, as in the detailed checks (Section 2.8). For $\phi < 1$, no guarantee applies (Section A.VI-C). In the Maze around S3, from $\delta = 0.20$ to 0.025 the $\widehat{m}_c/c_{\max}$ of V-C rises from 0.607 to 0.688 and its $\widehat{q}^+_{\mathrm{CBF}}$ falls from 5.38 to 3.60%, and from $\alpha_0 - \alpha = 0.25$ to 0.05 its $\widehat{m}_c/c_{\max}$ falls from 0.928 to 0.517 and its $\widehat{q}^+_{\mathrm{CBF}}$ rises from 0.27 to 7.40%.

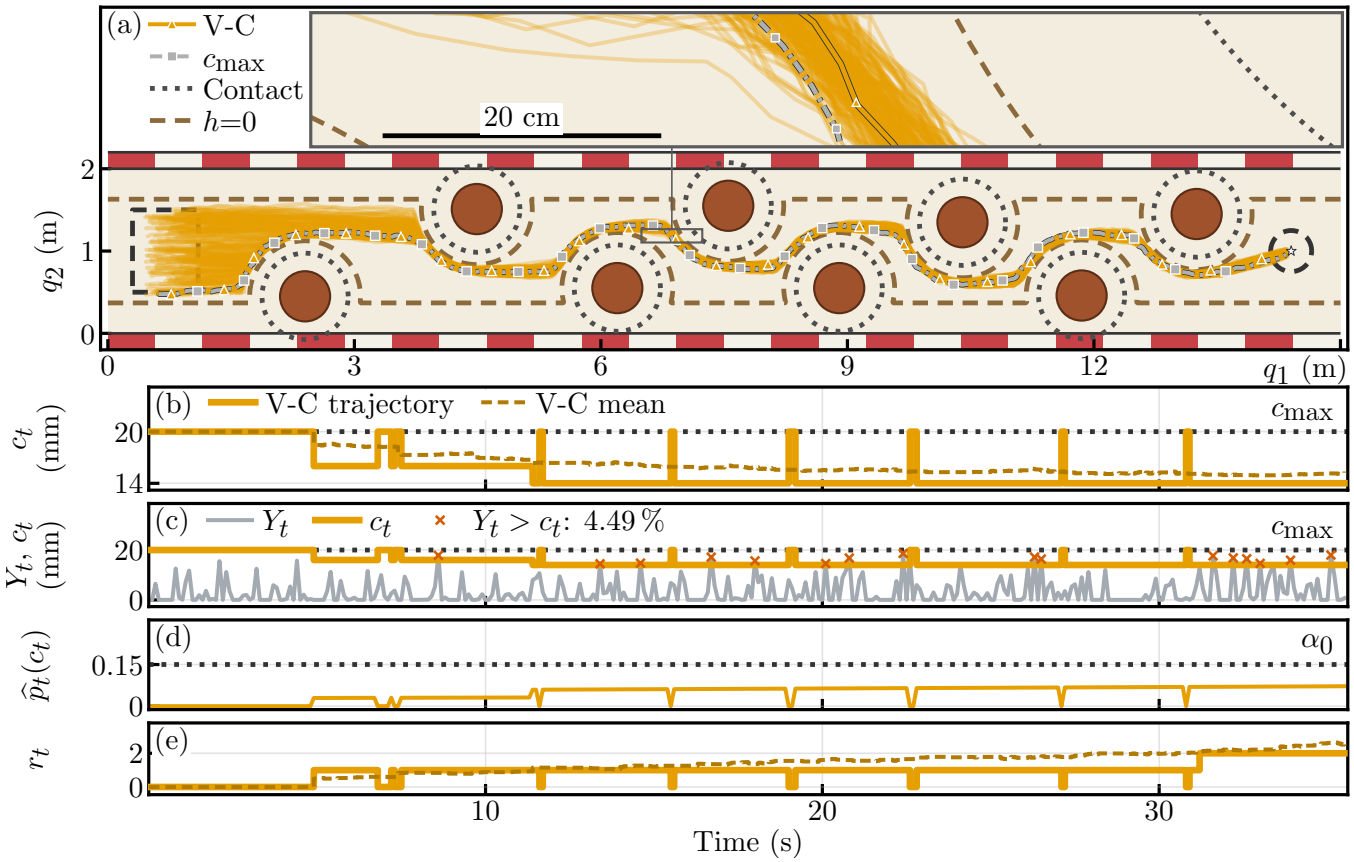


Fig. 1: Corridor, default setting with V-C. Panels in Section 2.2.2.

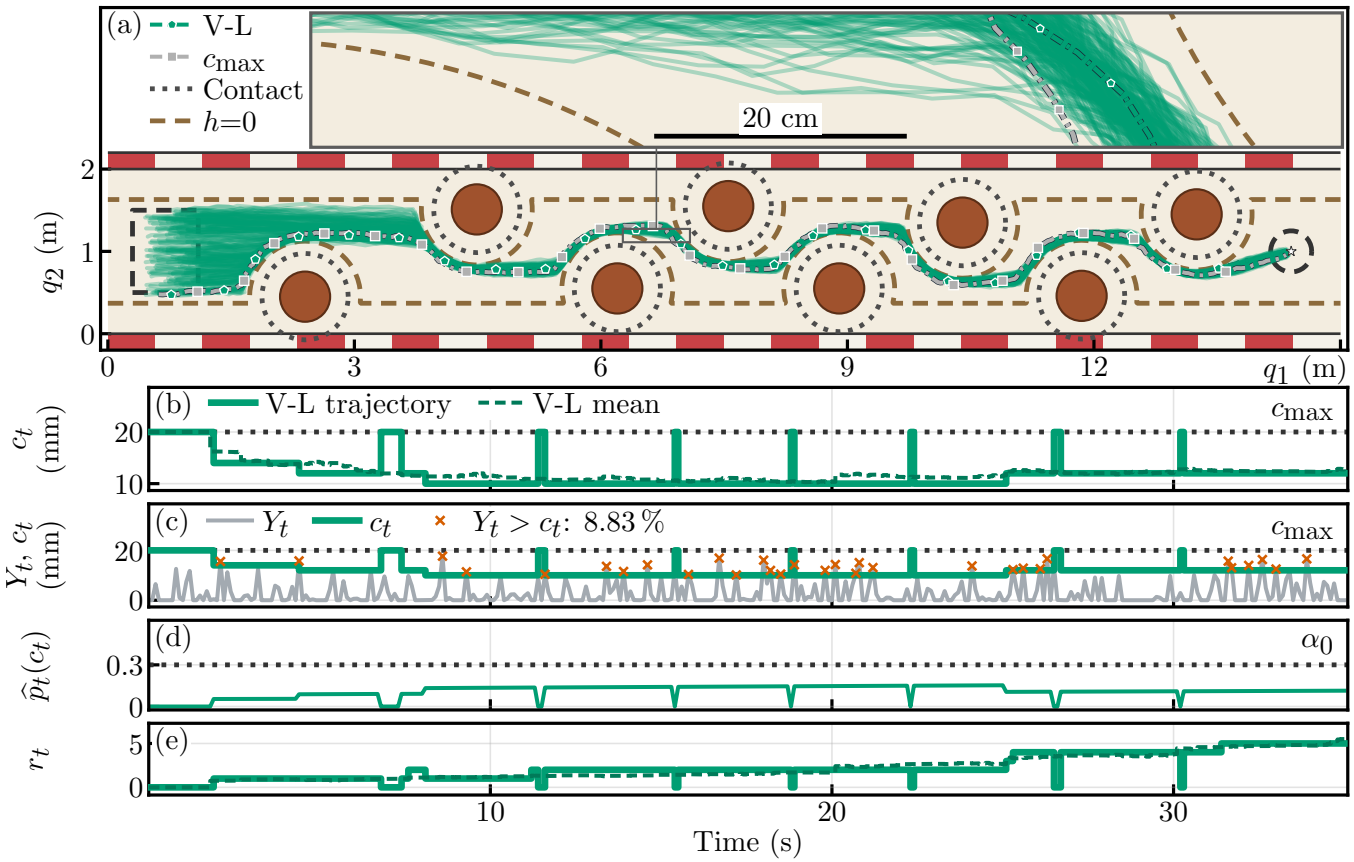


Fig. 2: Corridor, S3 with V-L. Panels as in Fig. 1.

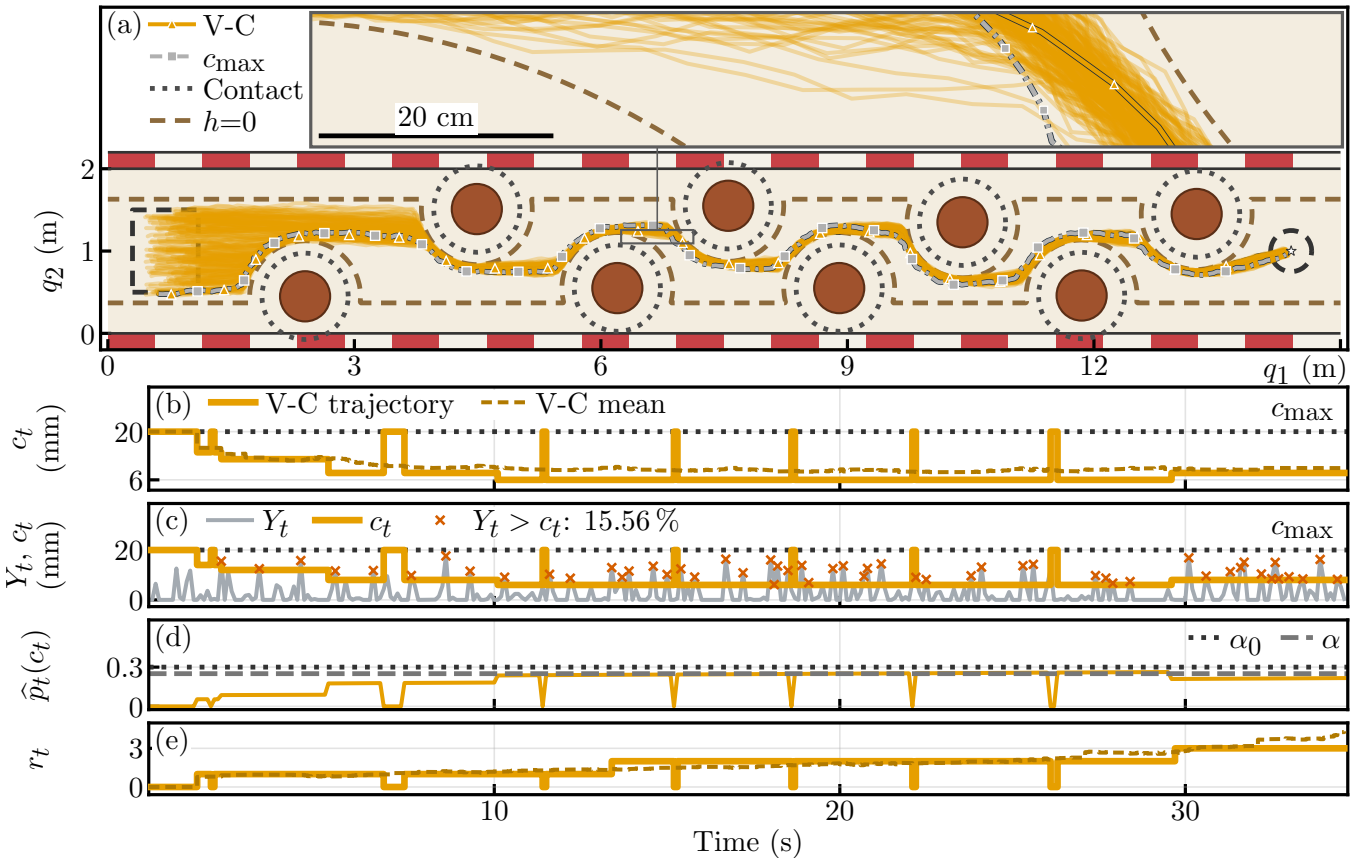


Fig. 3: Corridor, S1 with V-C. Panels as in Fig. 1.

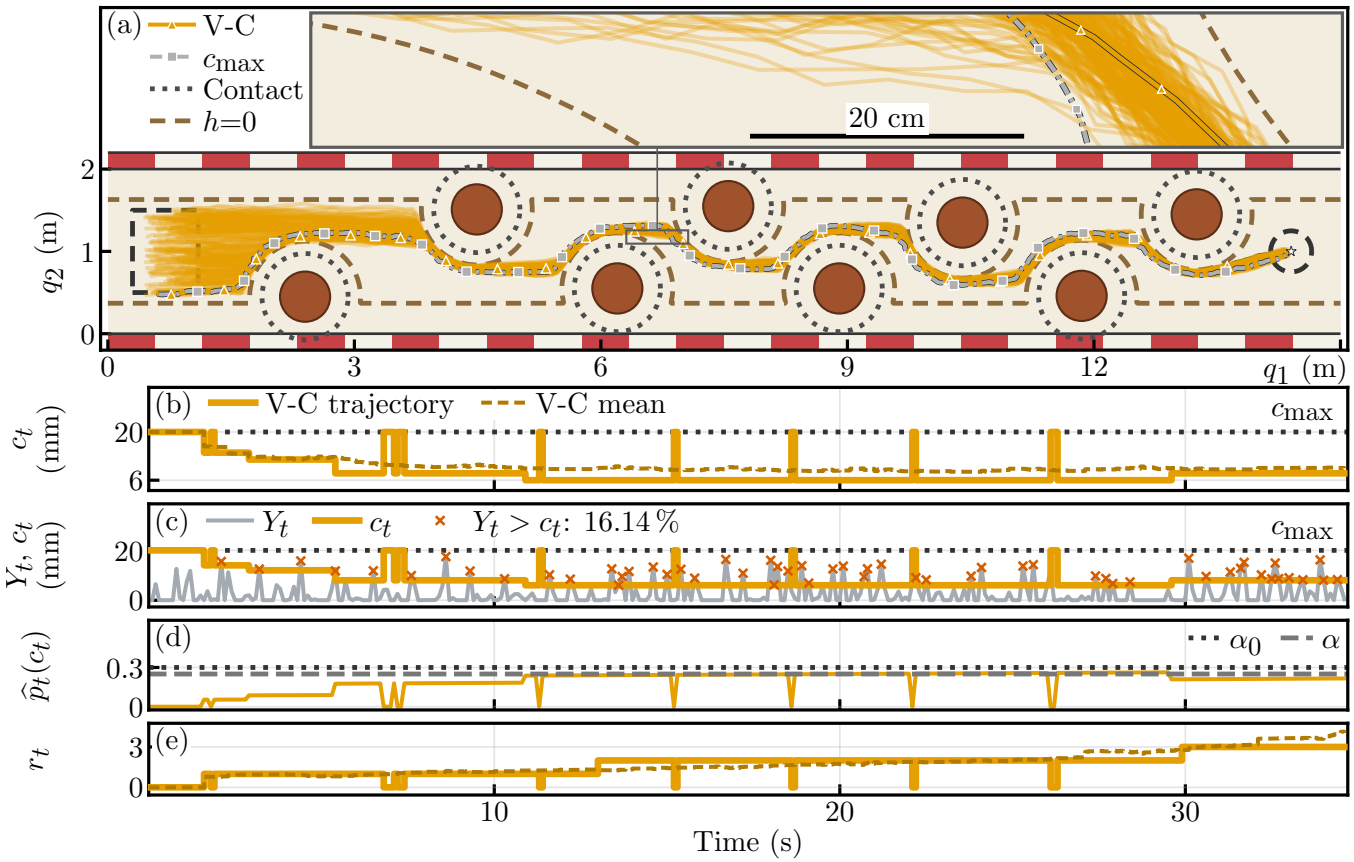


Fig. 4: Corridor, S2 with V-C. Panels as in Fig. 1.

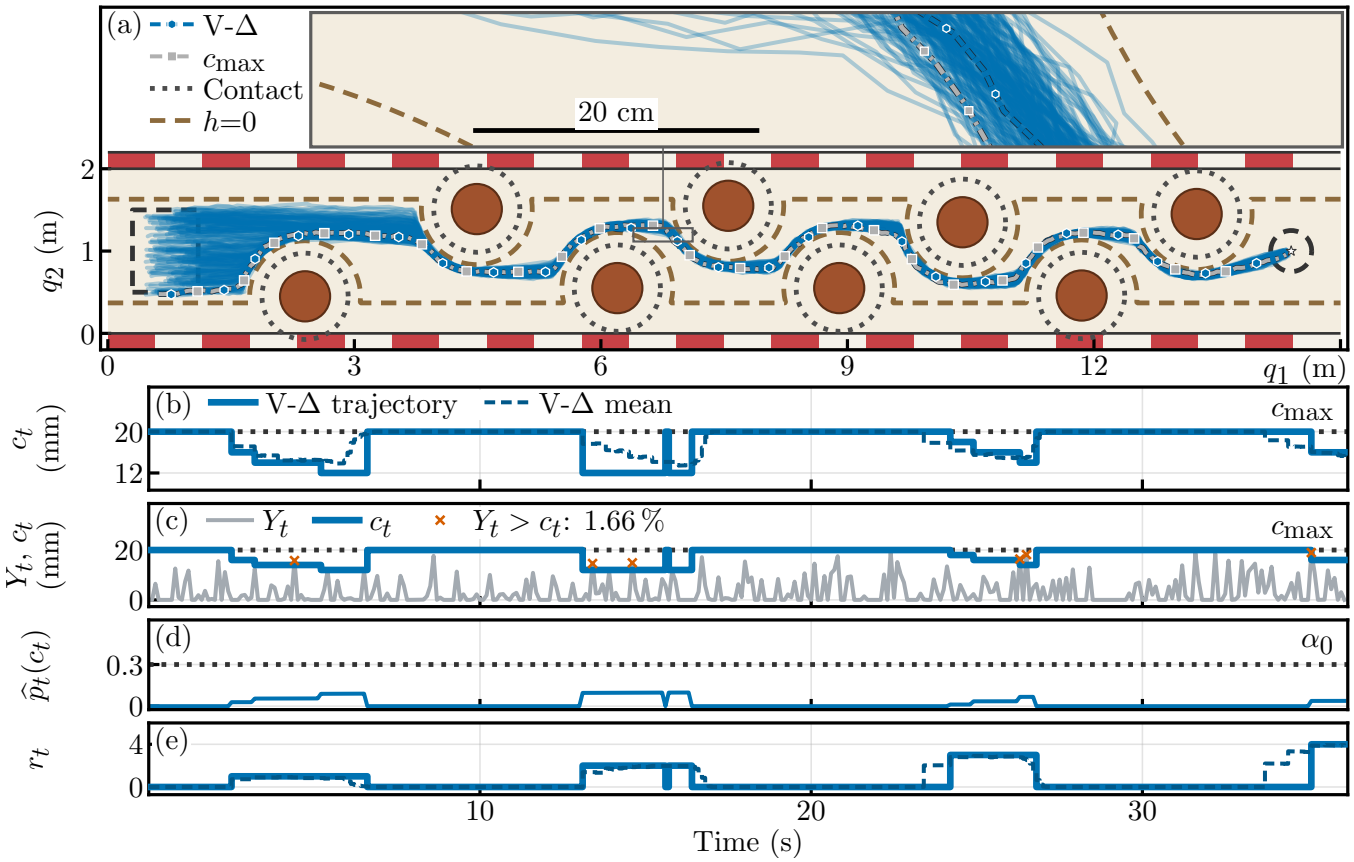


Fig. 5: Corridor, S4 with V-Δ. Panels as in Fig. 1.

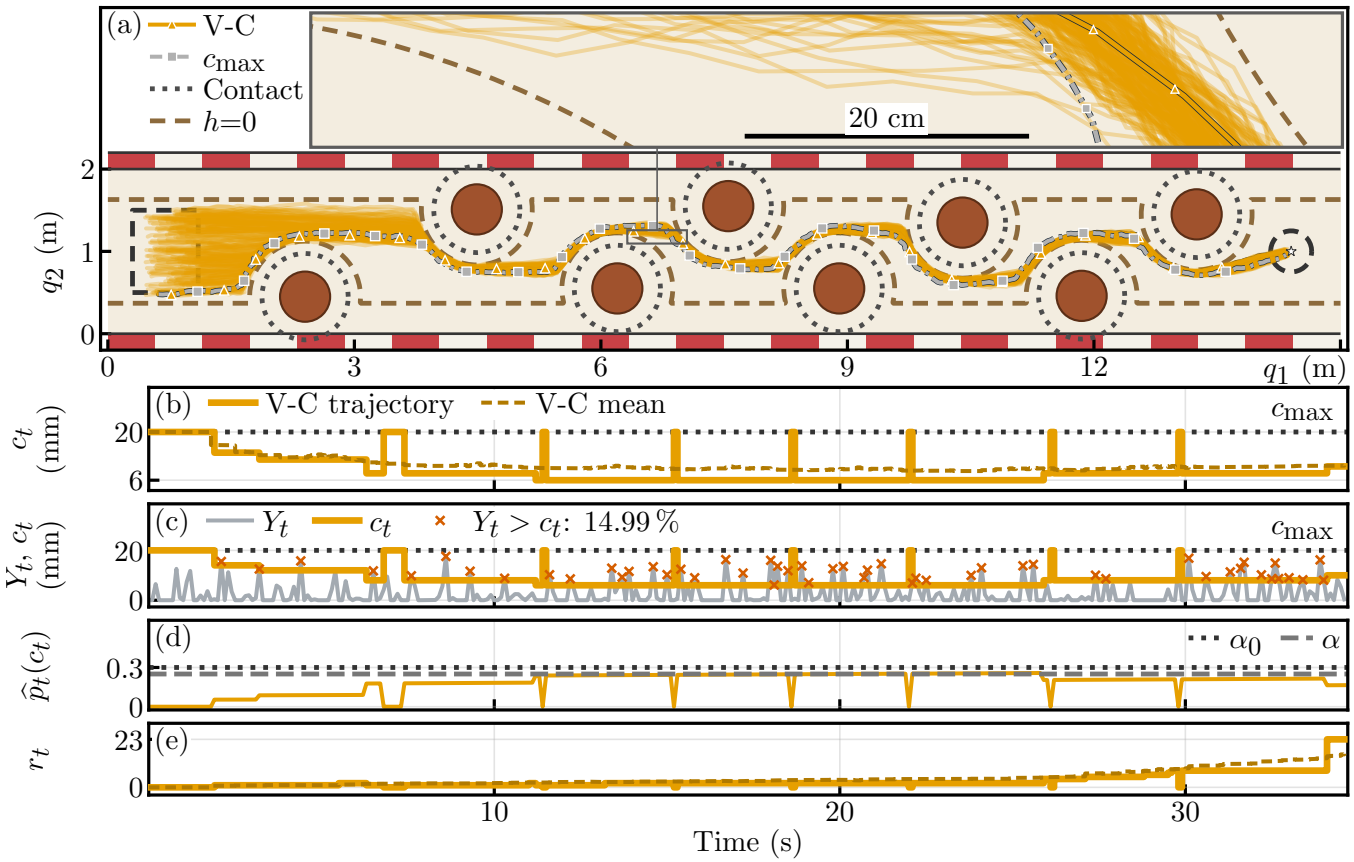


Fig. 6: Corridor, S5 with V-C. Panels as in Fig. 1.

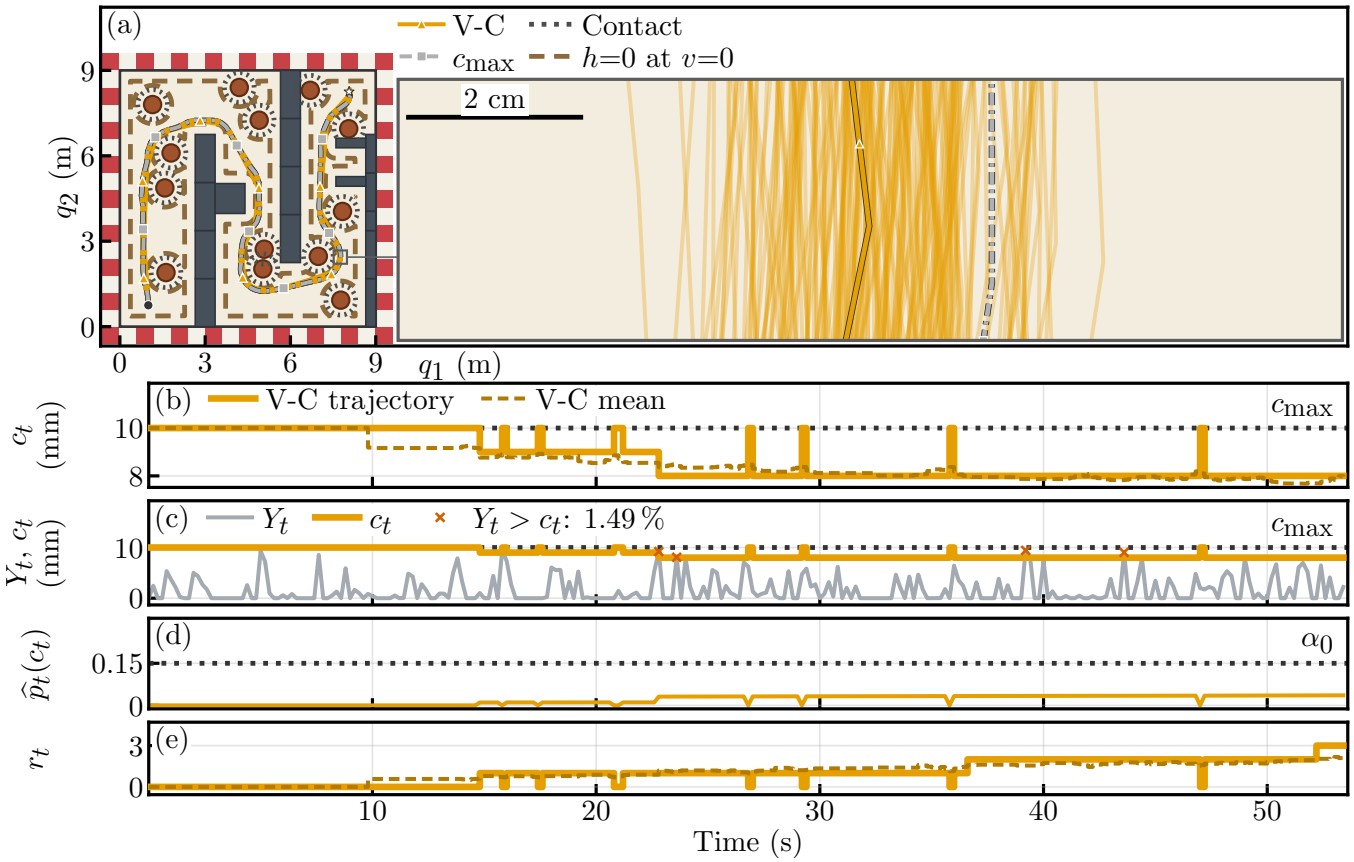


Fig. 7: Maze, default setting with V-C. Panels as in Fig. 1.

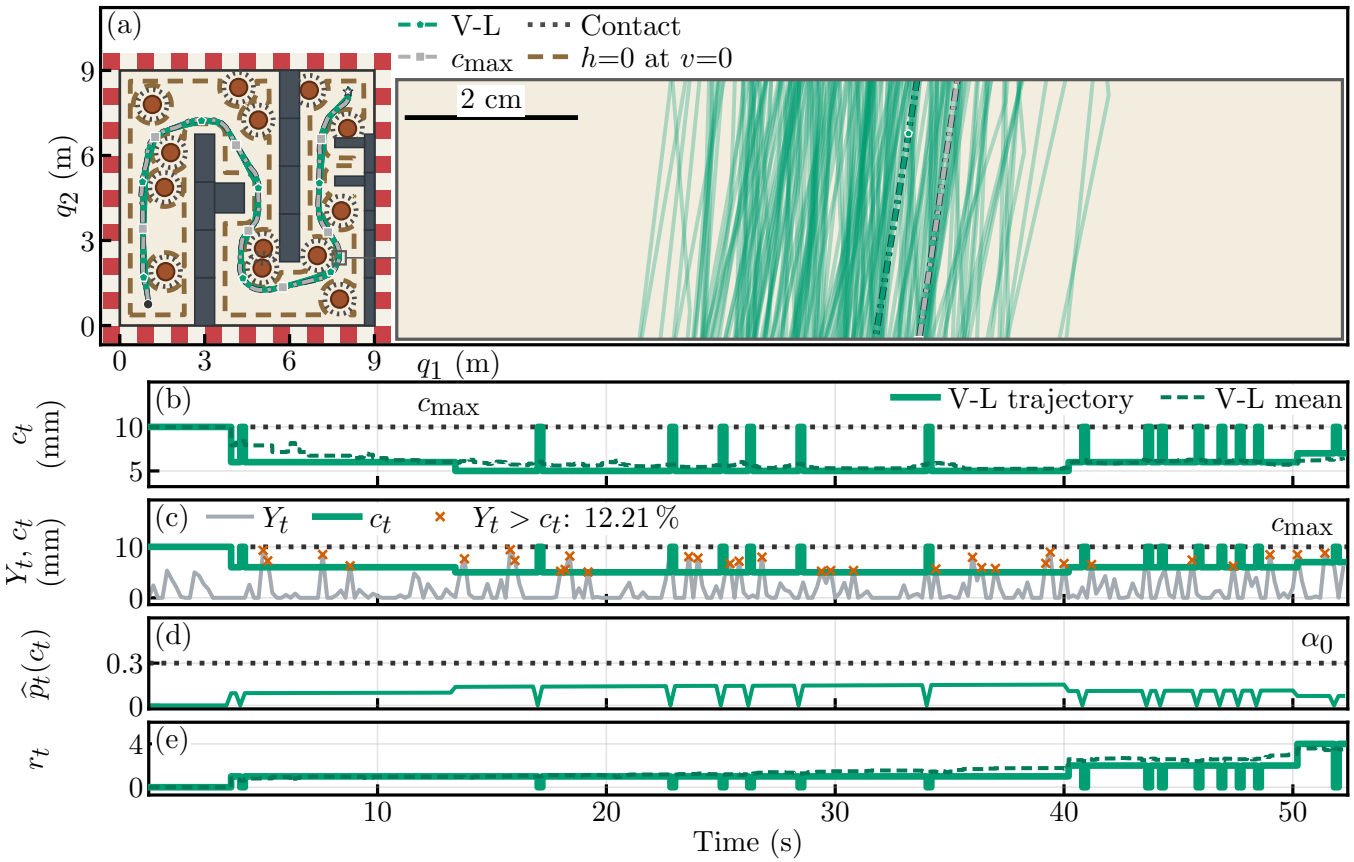


Fig. 8: Maze, S3 with V-L. Panels as in Fig. 1.

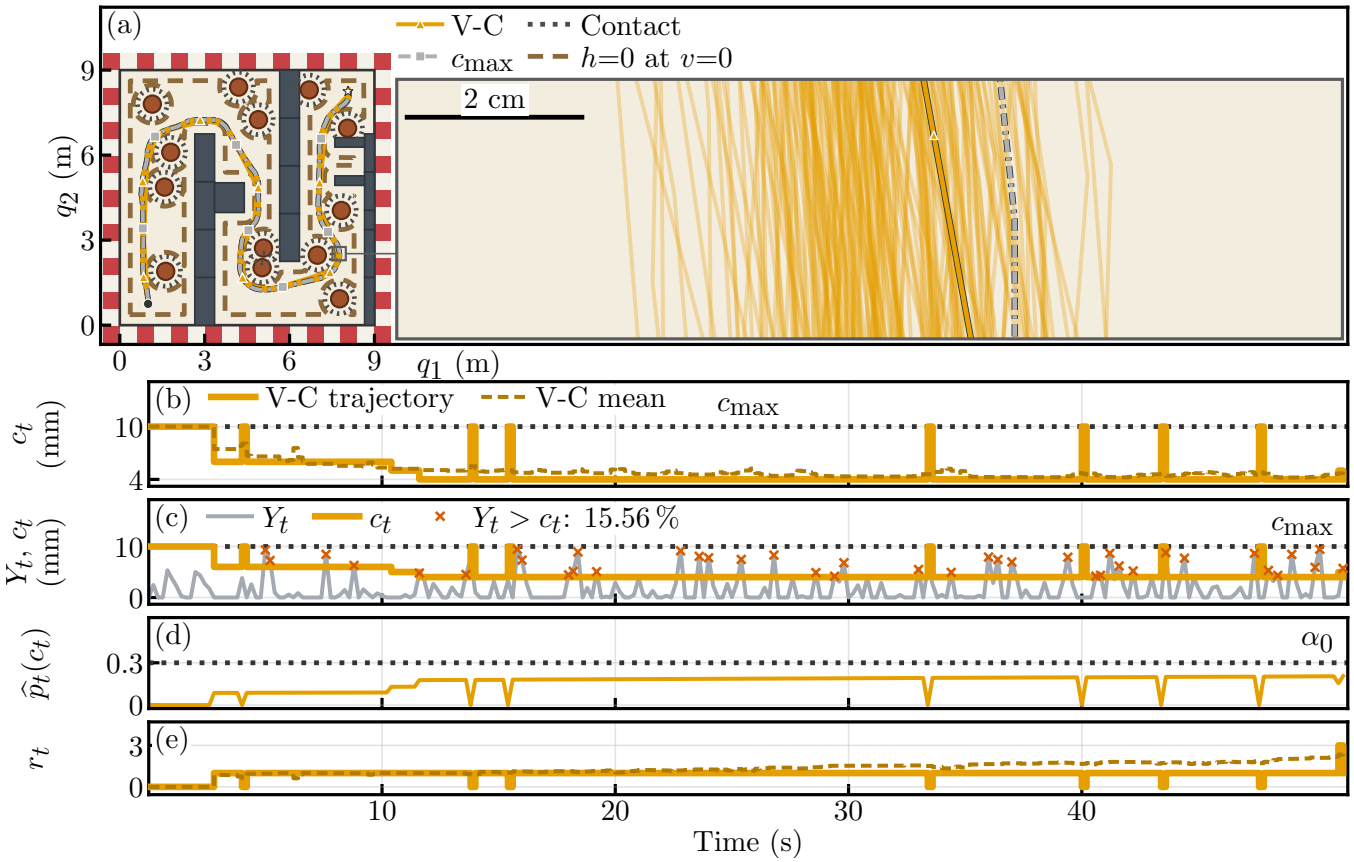


Fig. 9: Maze, S1 with V-C. Panels as in Fig. 1.

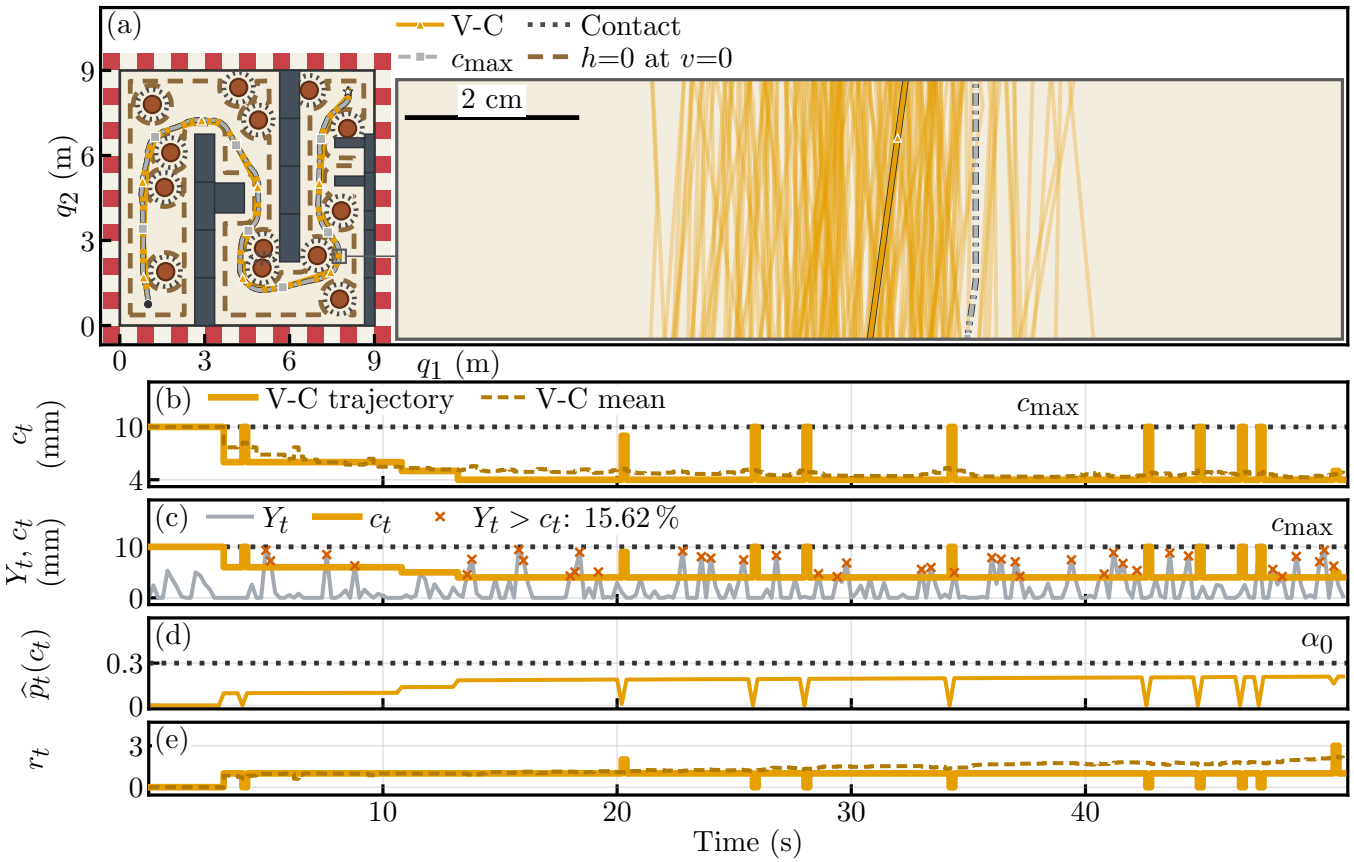


Fig. 10: Maze, S2 with V-C. Panels as in Fig. 1.

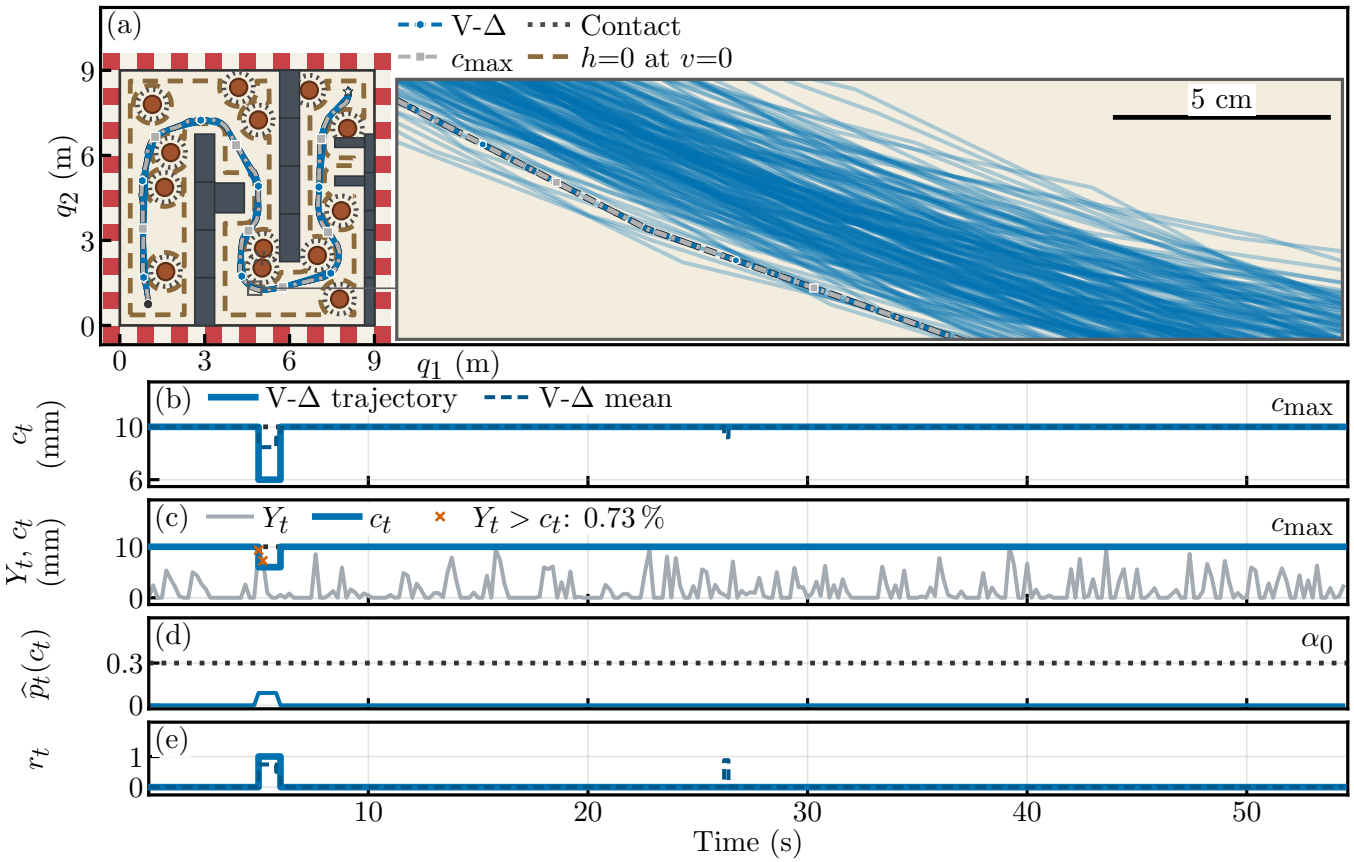


Fig. 11: Maze, S4 with V-Δ. Panels as in Fig. 1. In (a), the example trajectory coincides with that of the $c_{\max}$ baseline.

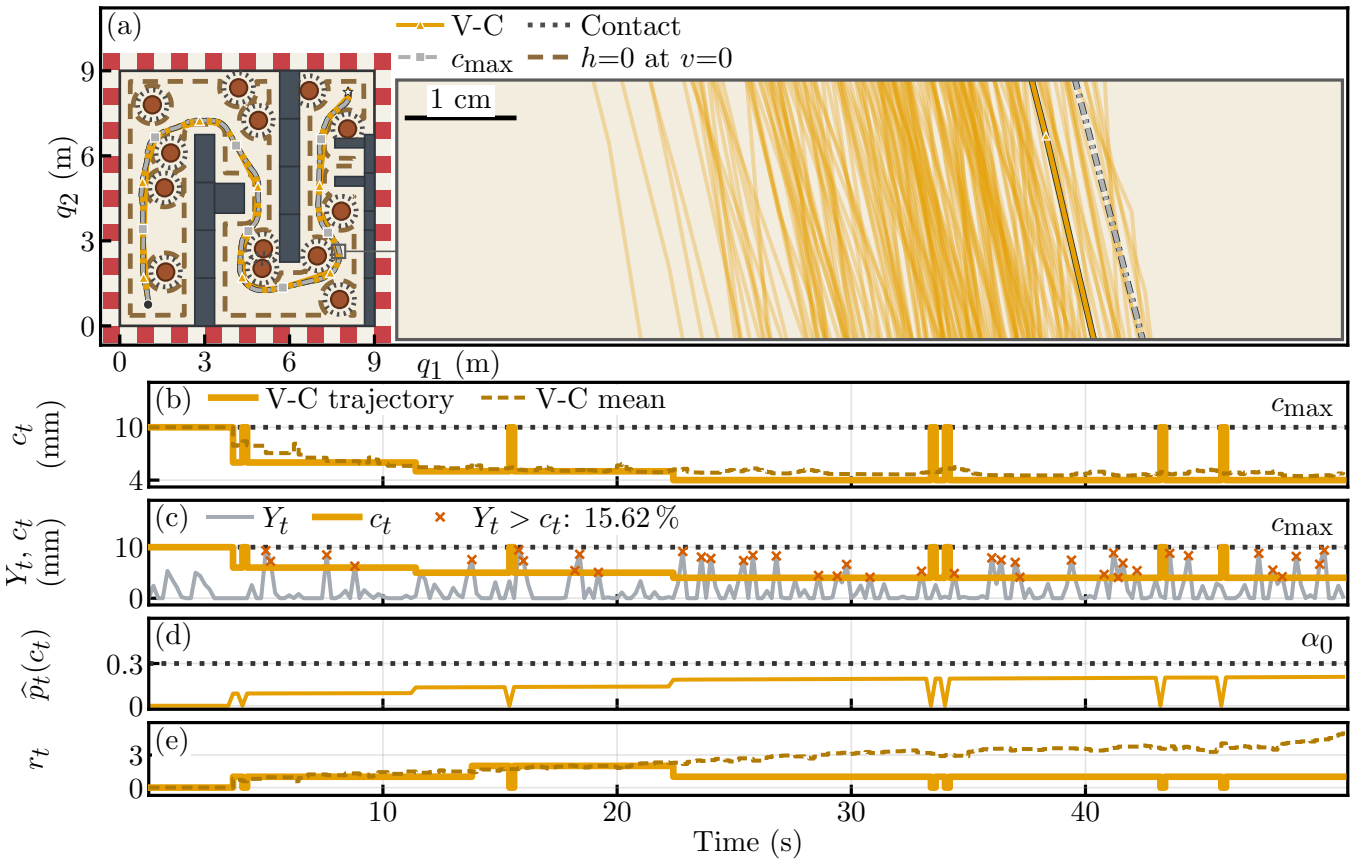


Fig. 12: Maze, S5 with V-C. Panels as in Fig. 1.

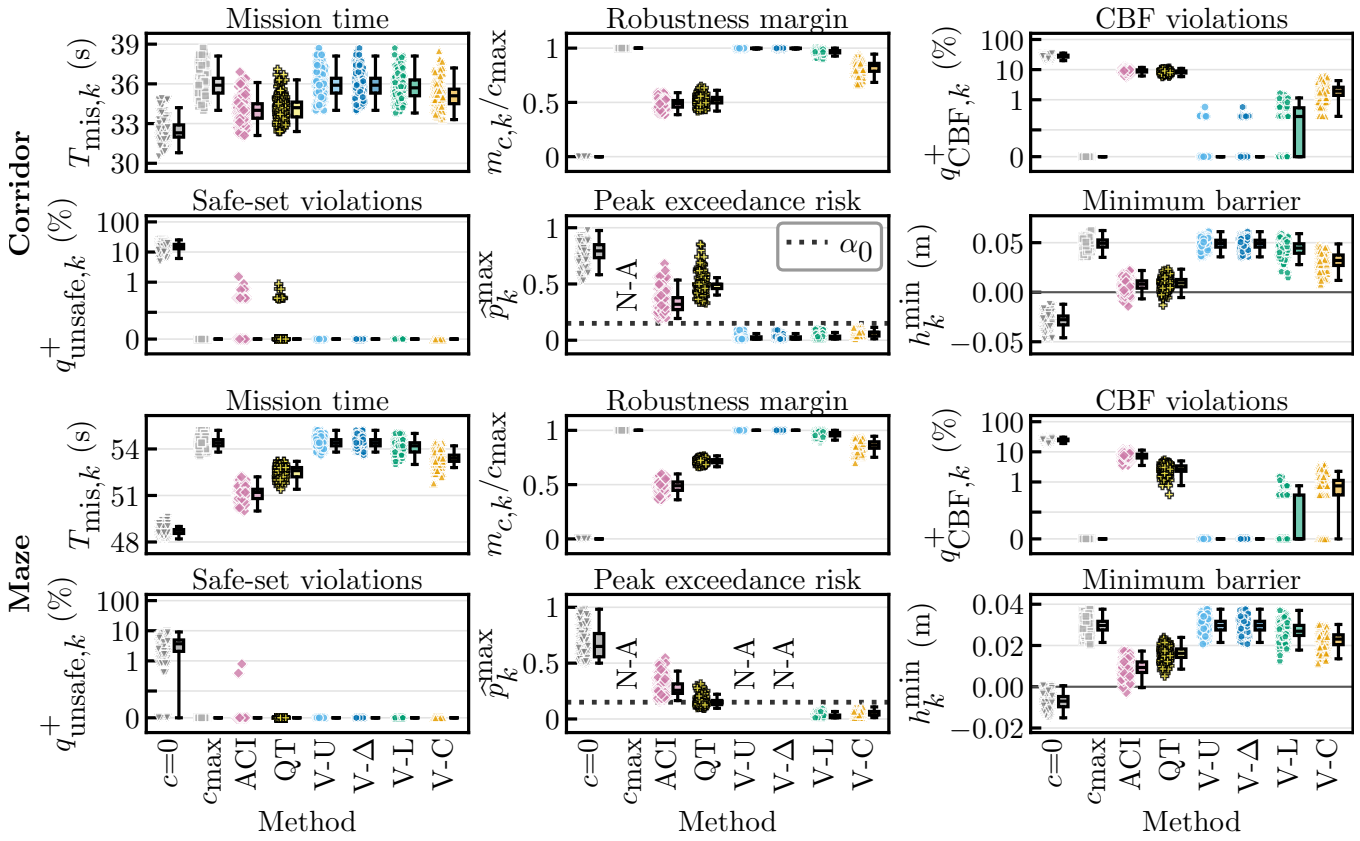


Fig. 13: Corridor (top) and Maze (bottom), default setting: per-trajectory distributions.

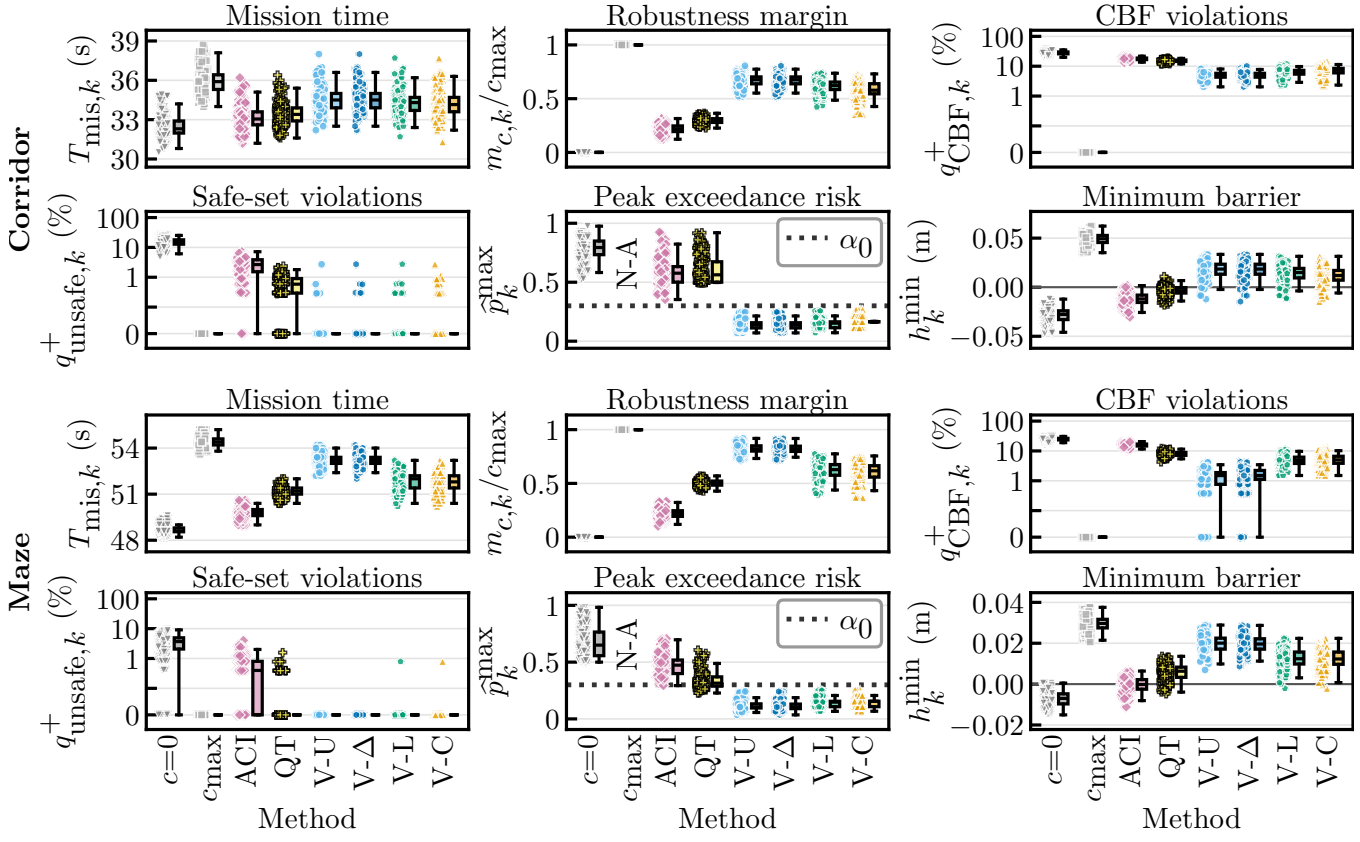


Fig. 14: Corridor and Maze, S3, with panels as in Fig. 13.

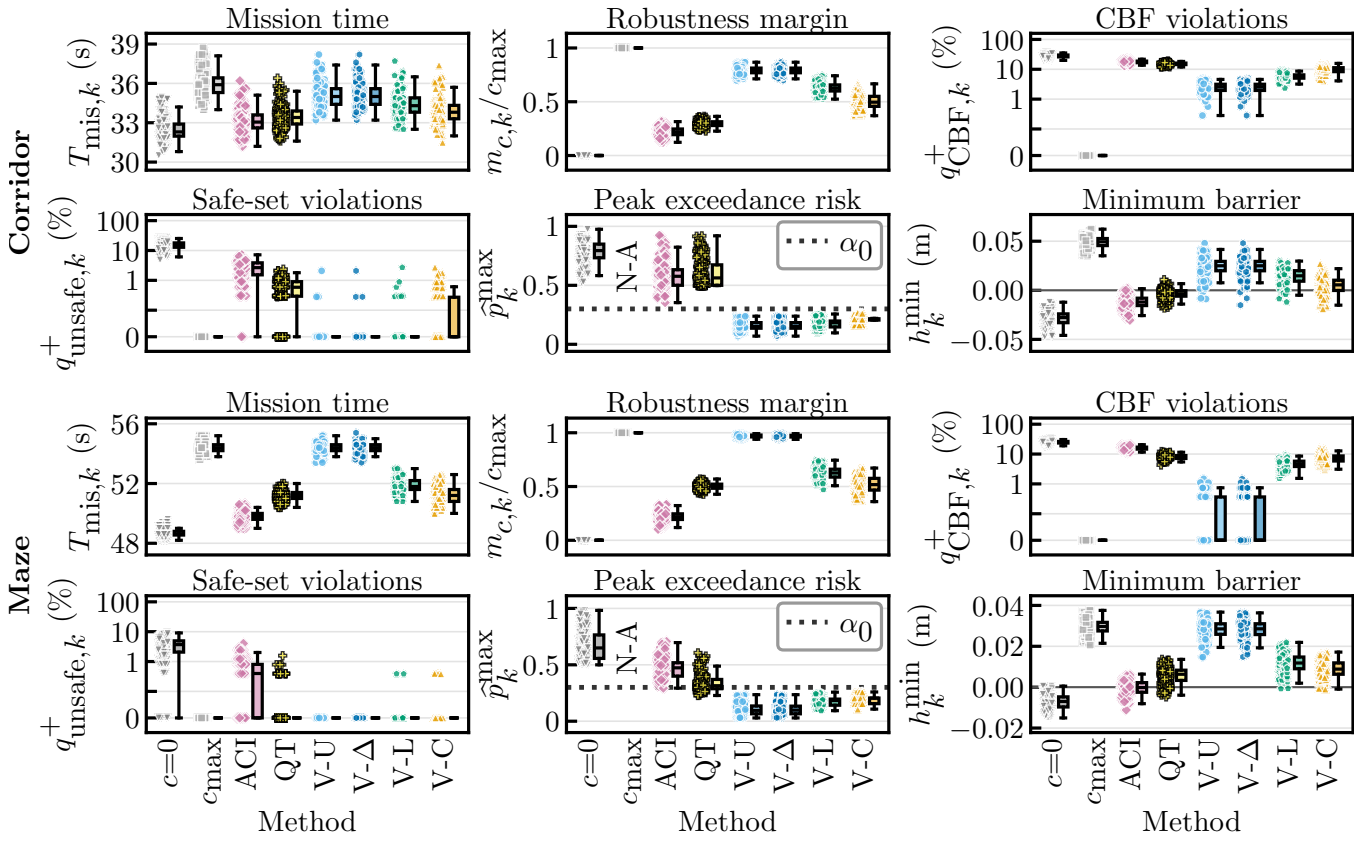


Fig. 15: Corridor and Maze, S1, with panels as in Fig. 13.

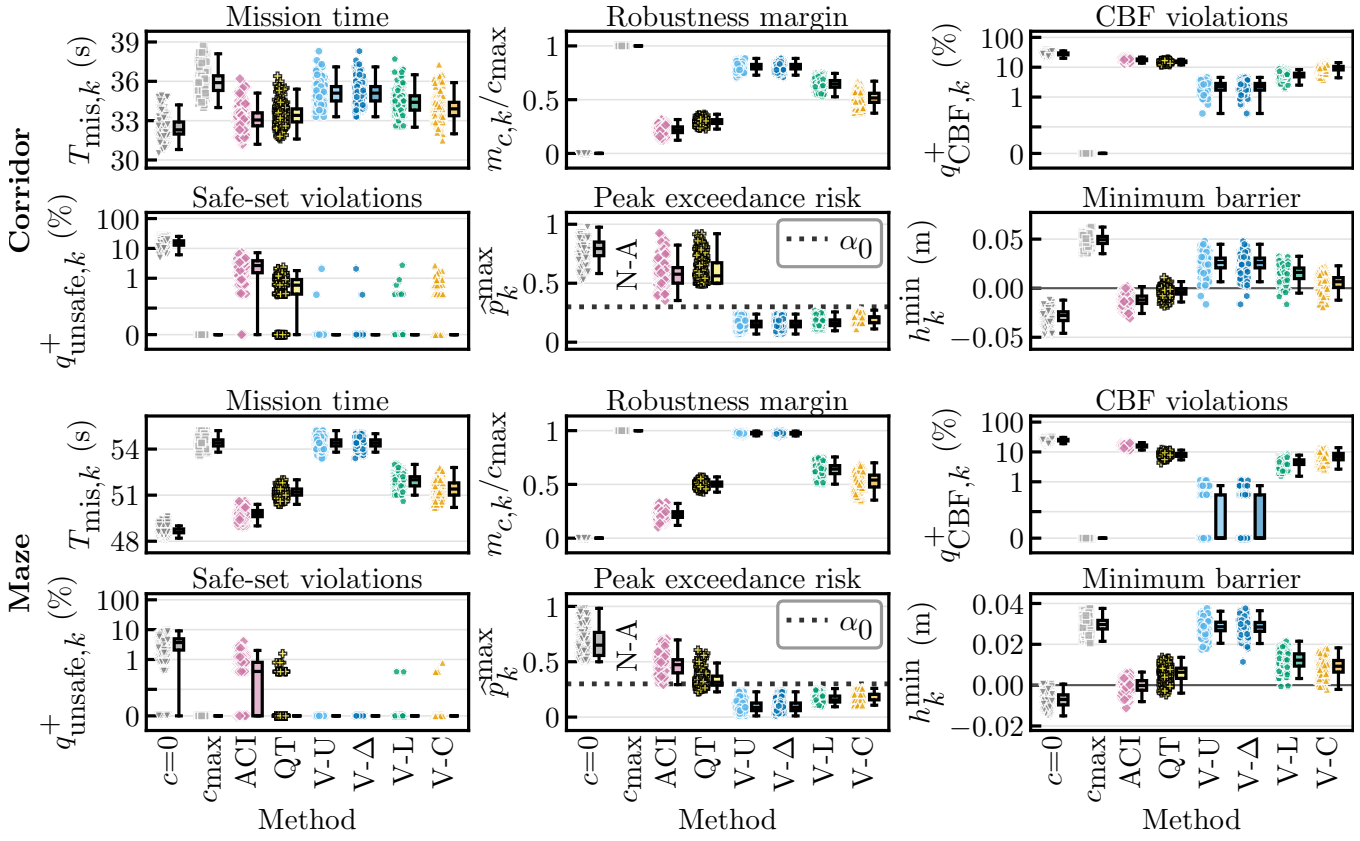


Fig. 16: Corridor and Maze, S2, with panels as in Fig. 13.

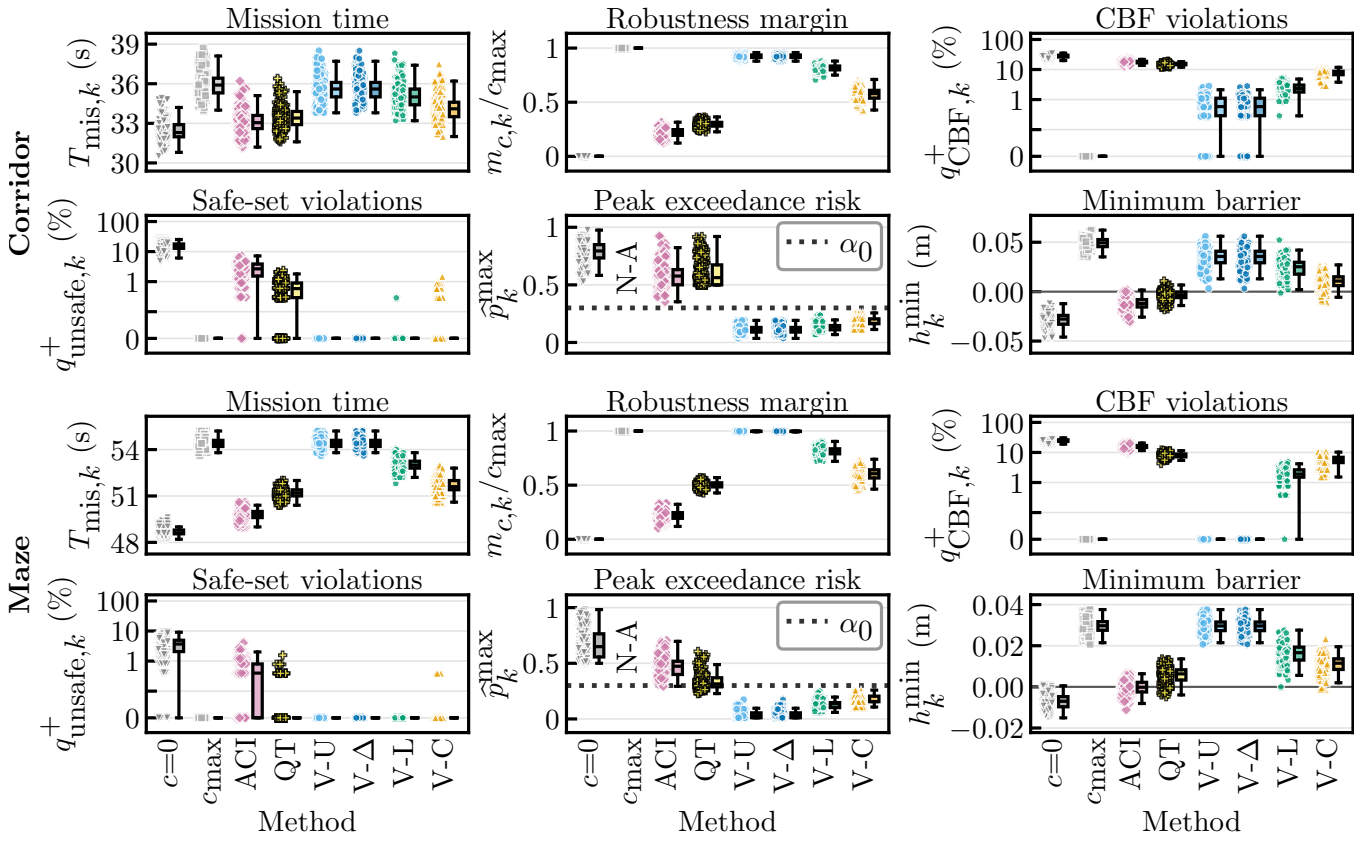


Fig. 17: Corridor and Maze, S4, with panels as in Fig. 13.

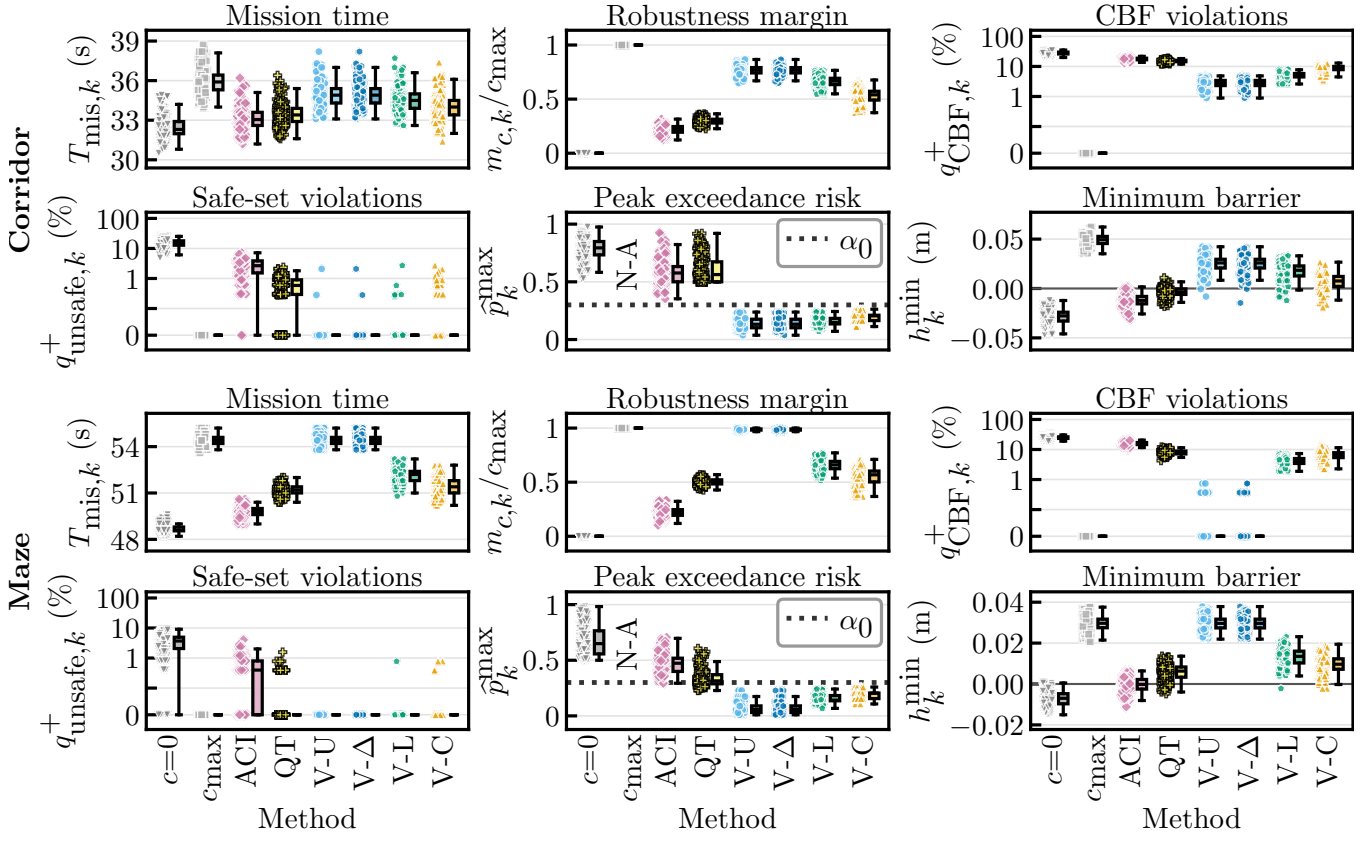


Fig. 18: Corridor and Maze, S5, with panels as in Fig. 13.

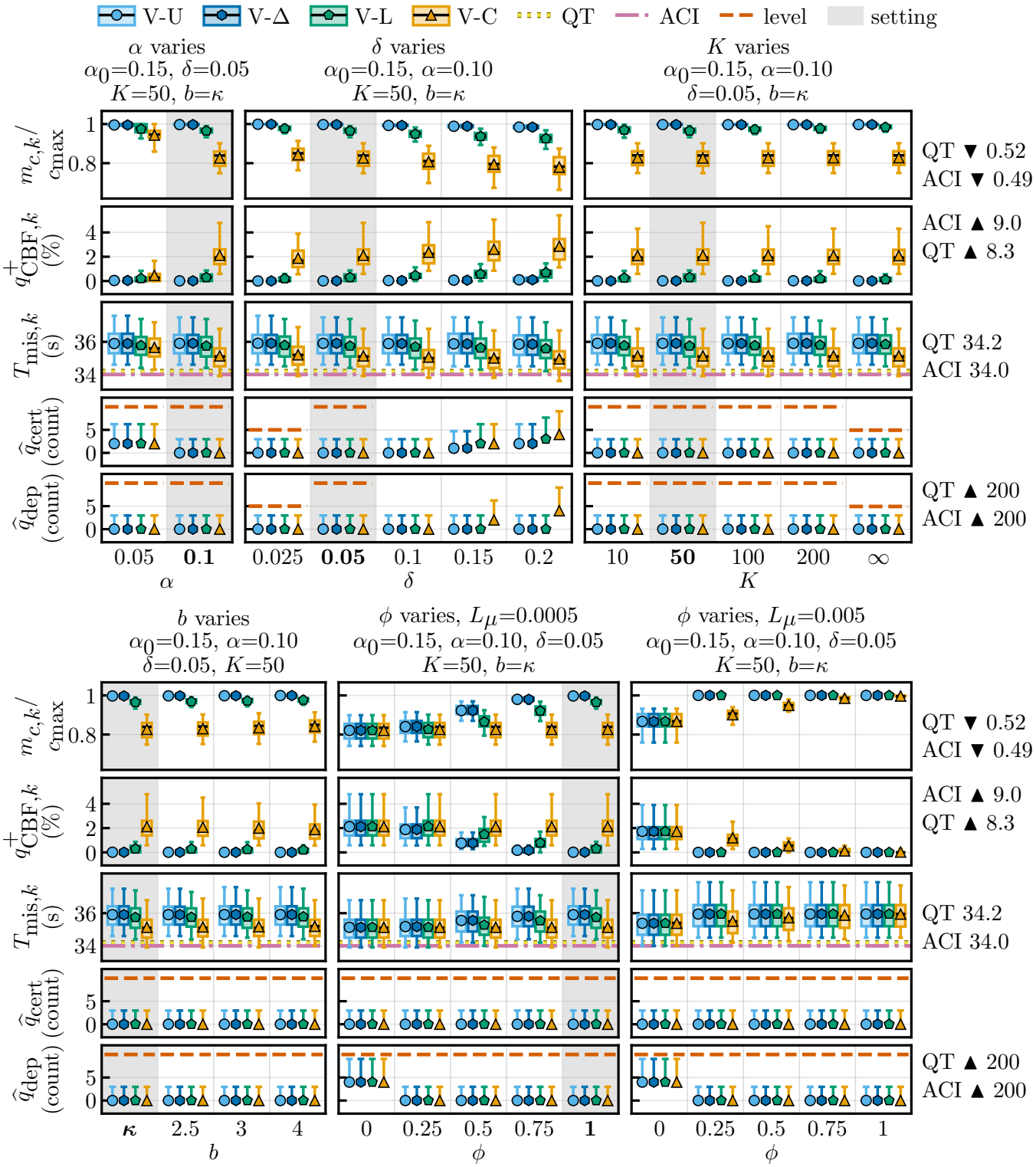


Fig. 19: Corridor, default setting: sweeps, with layout in Section 2.2.3.

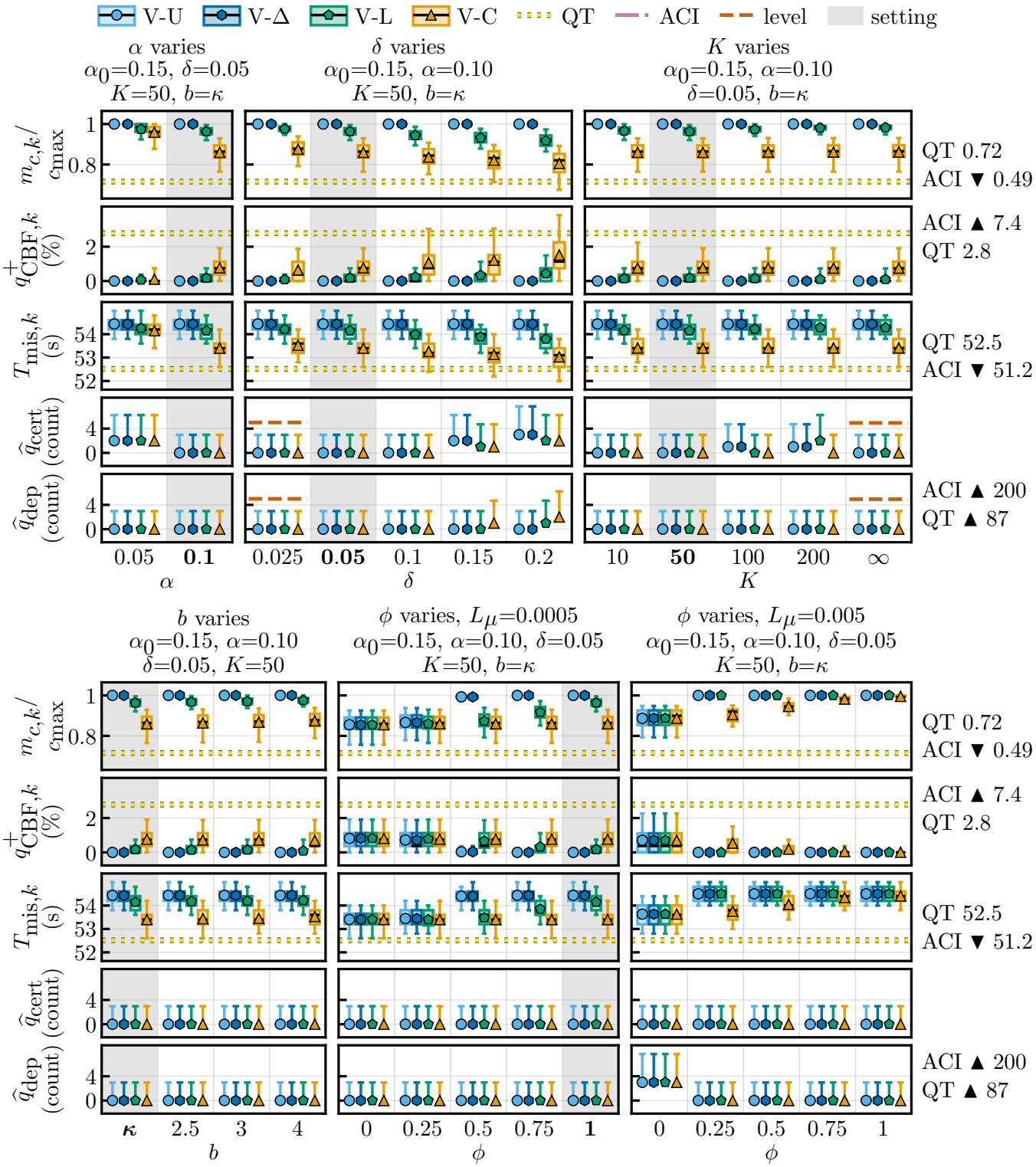


Fig. 20: Maze, default setting: sweeps, with layout in Section 2.2.3.

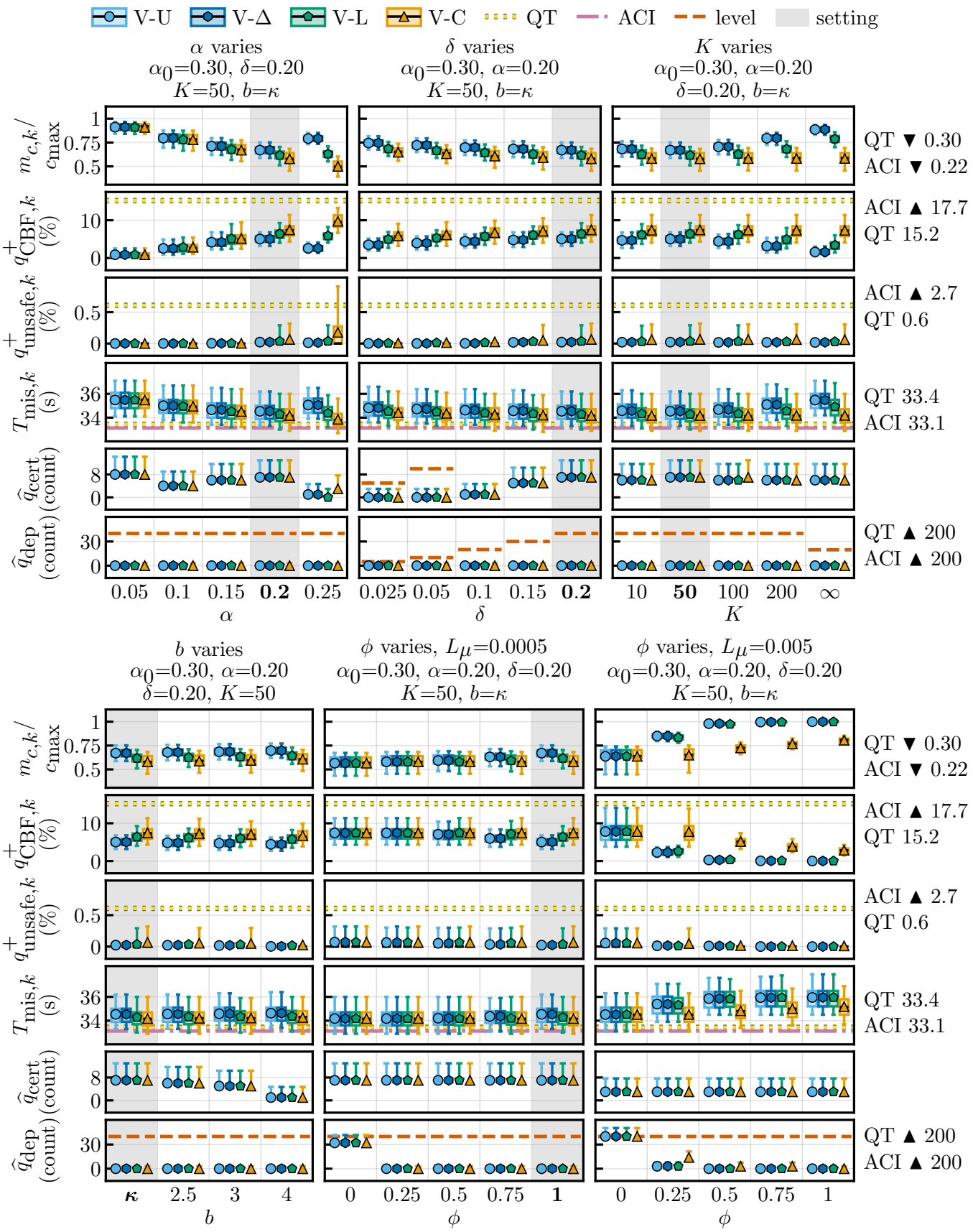


Fig. 21: Corridor, S3: sweeps, with layout in Section 2.2.3.

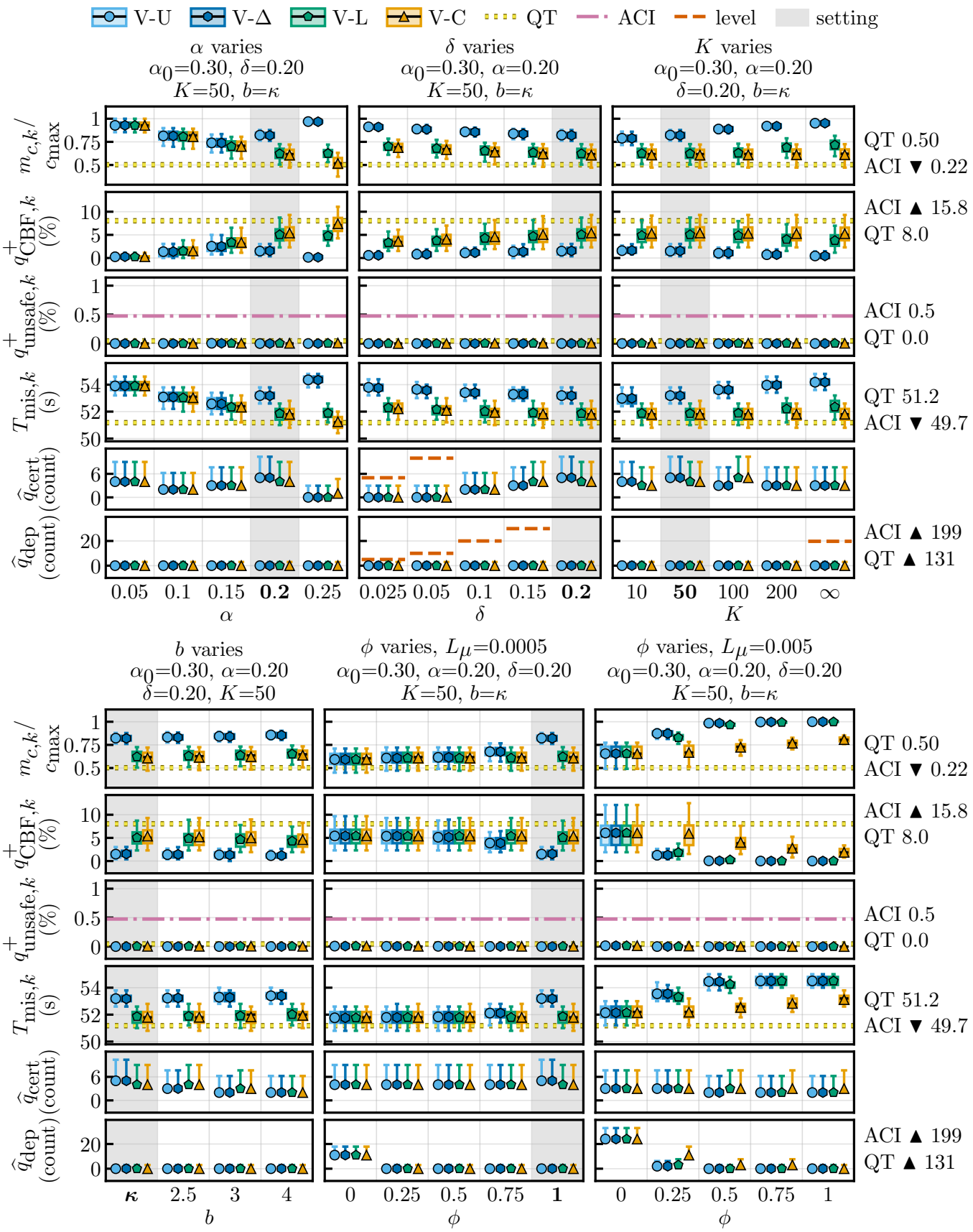


Fig. 22: Maze, S3: sweeps, with layout in Section 2.2.3.

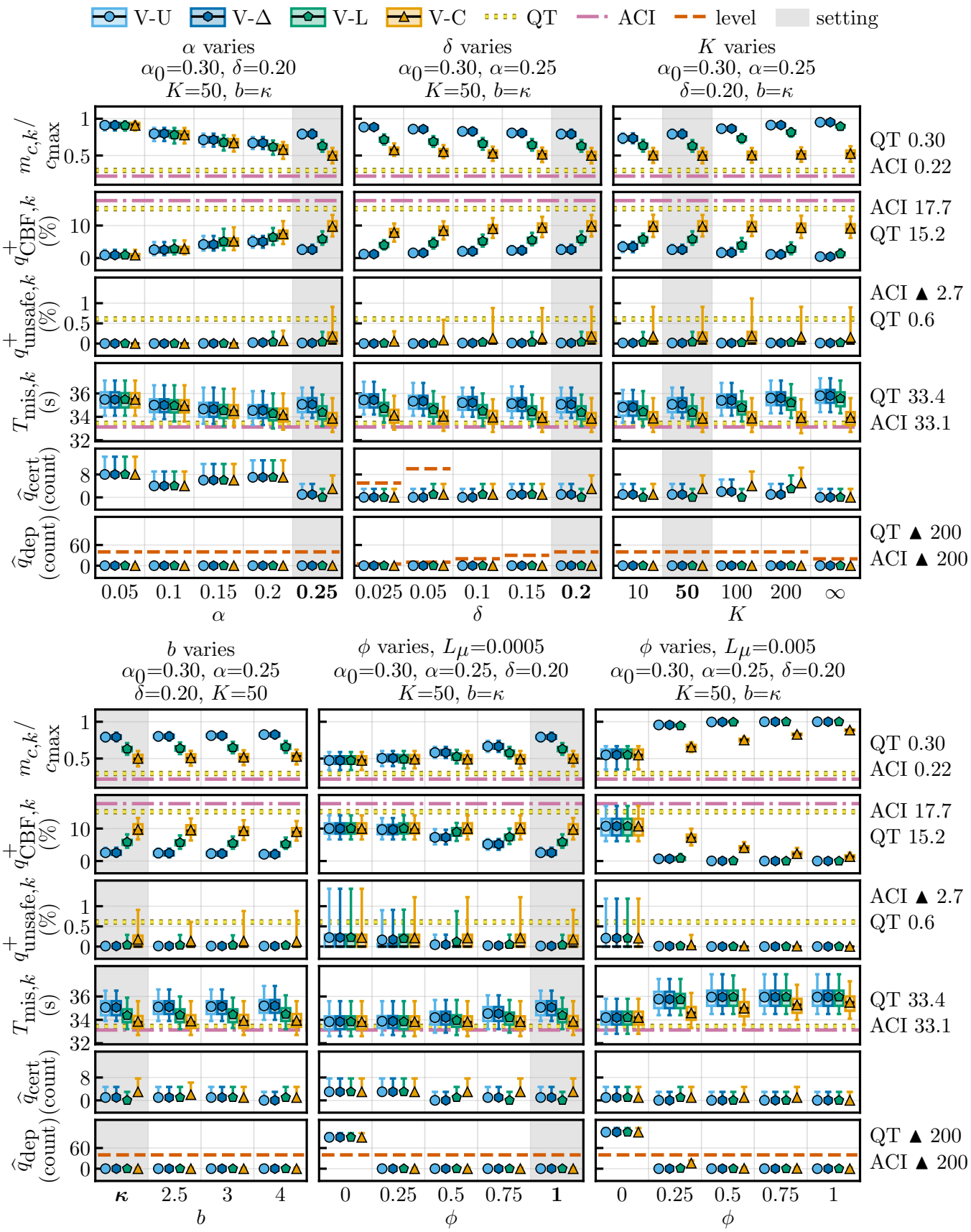


Fig. 23: Corridor, S1: sweeps, with layout in Section 2.2.3.

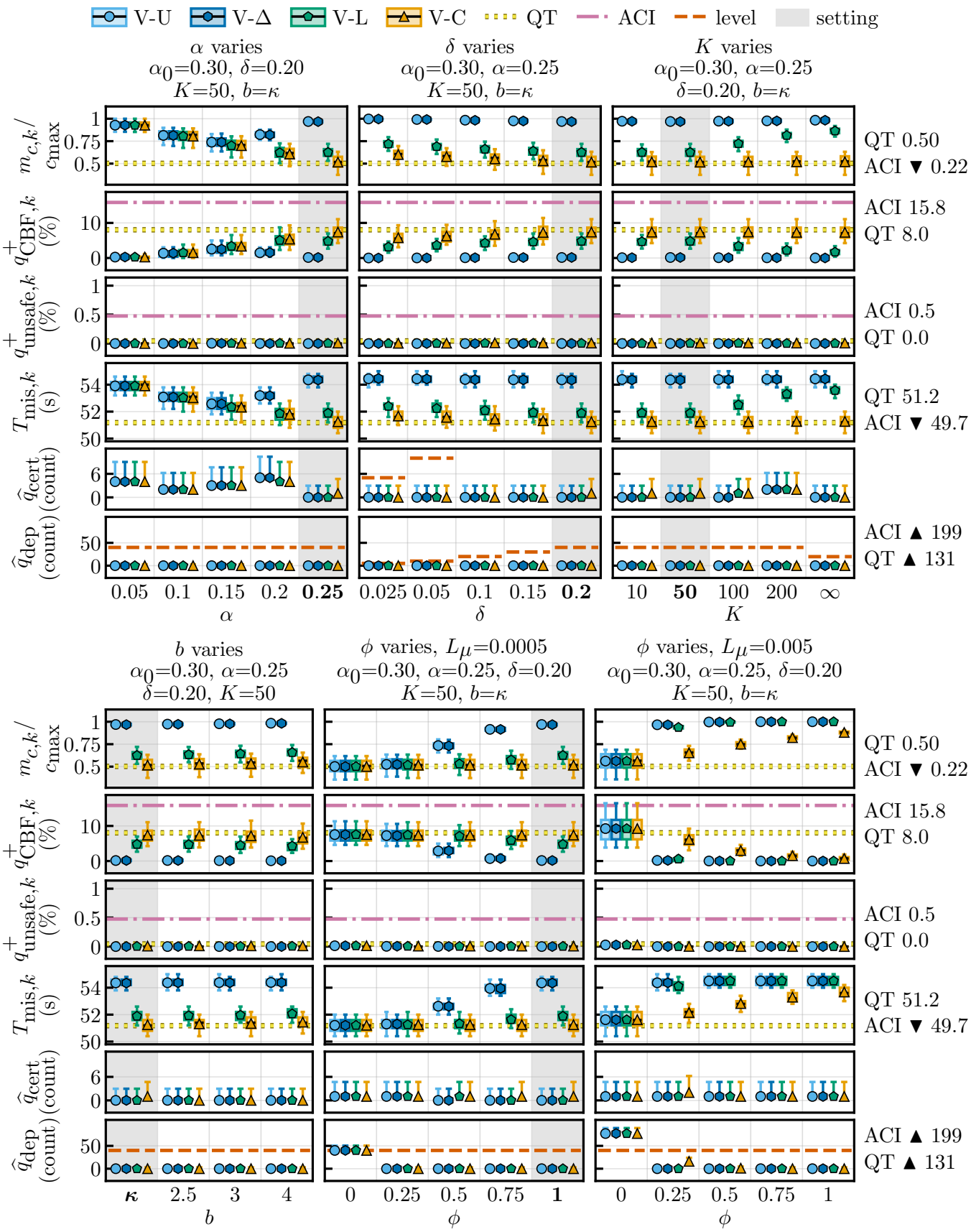


Fig. 24: Maze, S1: sweeps, with layout in Section 2.2.3.

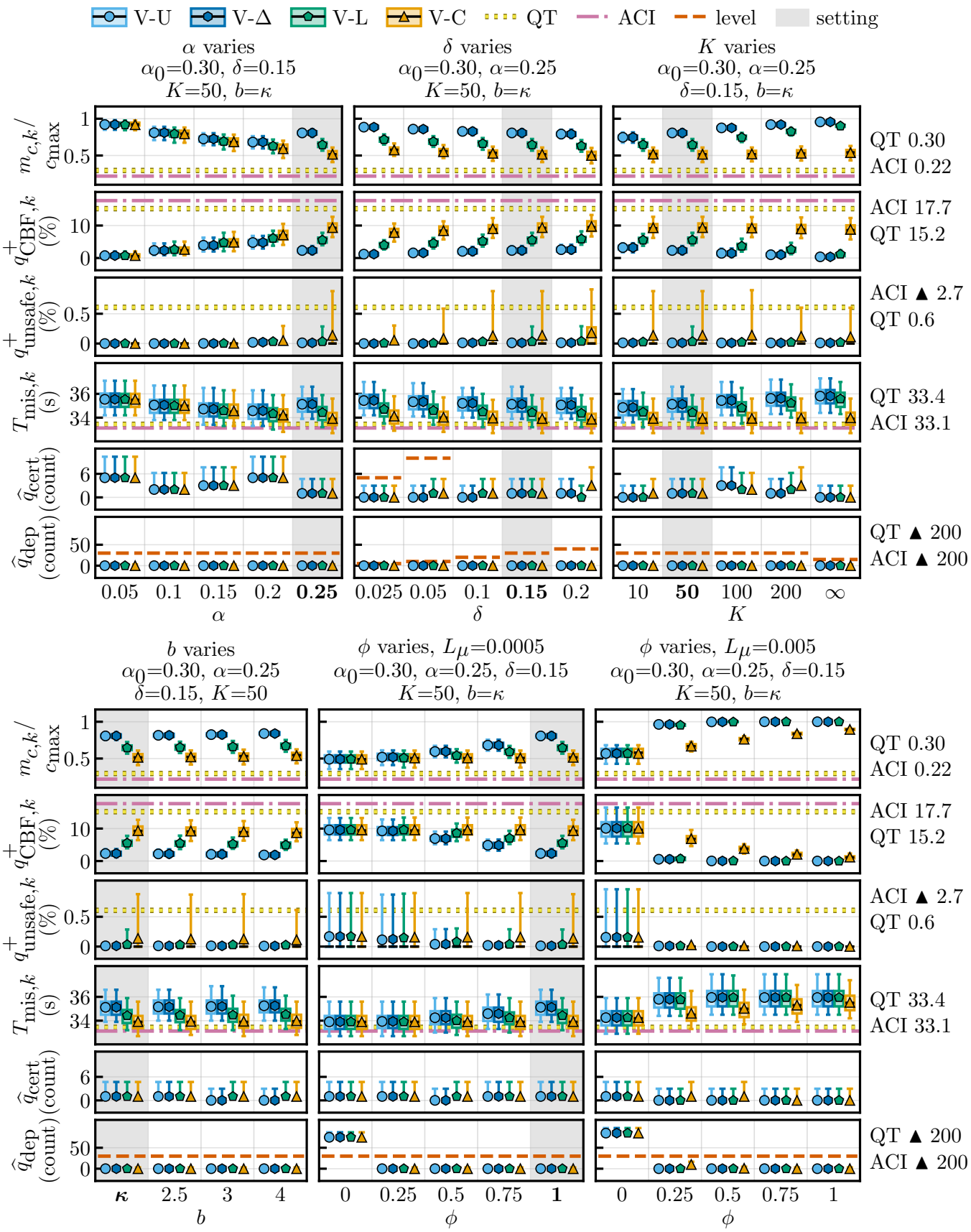


Fig. 25: Corridor, S2: sweeps, with layout in Section 2.2.3.

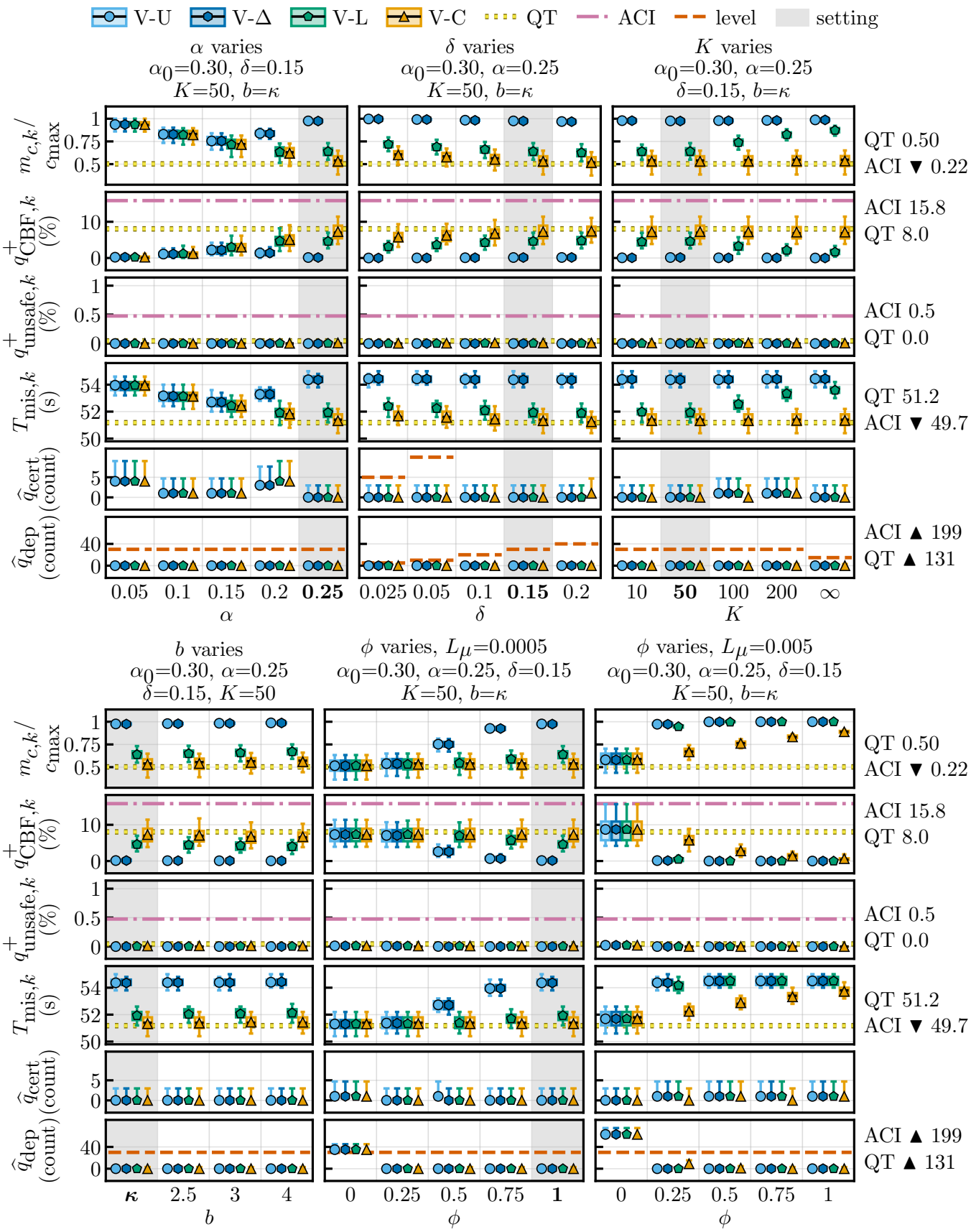


Fig. 26: Maze, S2: sweeps, with layout in Section 2.2.3.

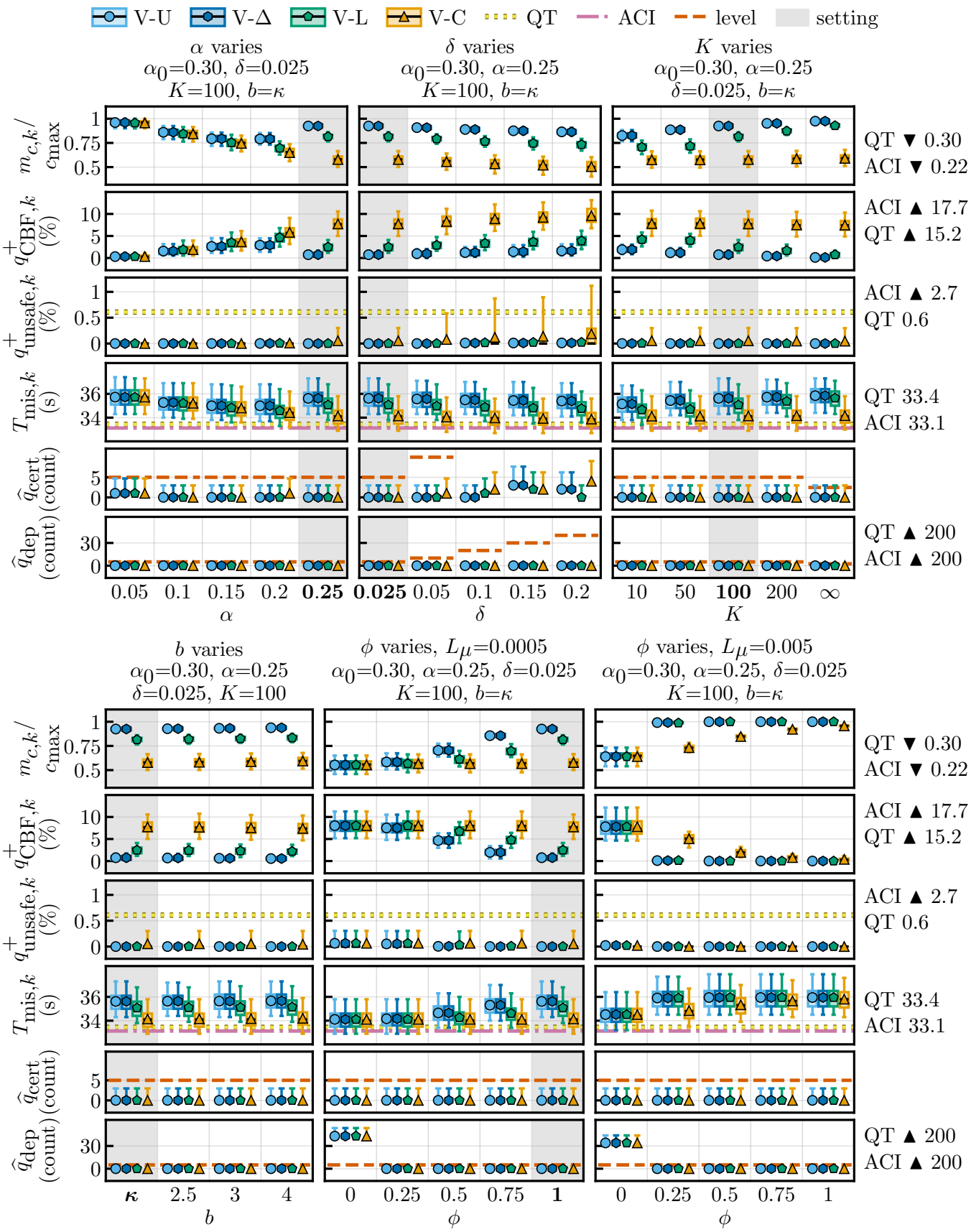


Fig. 27: Corridor, S4: sweeps, with layout in Section 2.2.3.

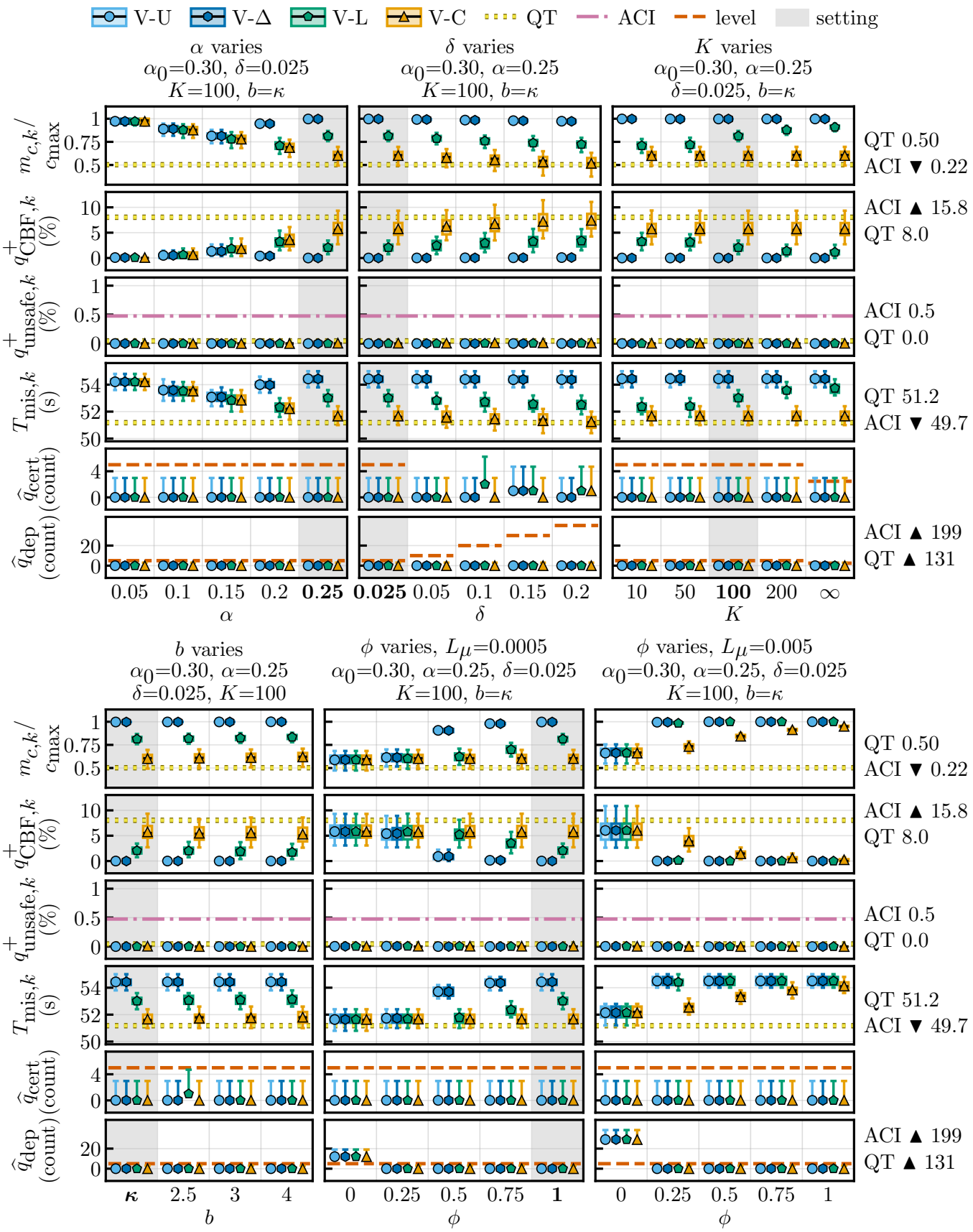


Fig. 28: Maze, S4: sweeps, with layout in Section 2.2.3.

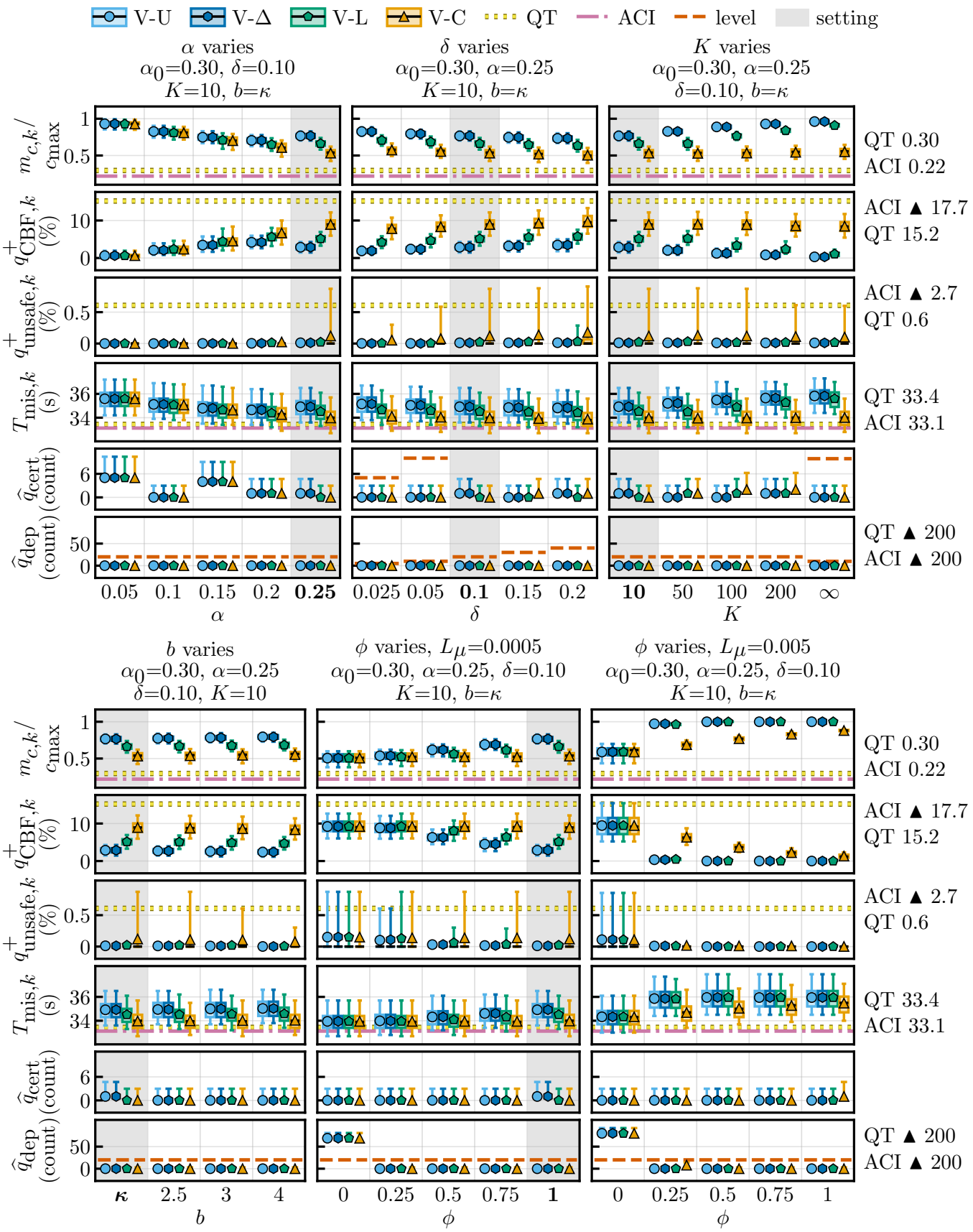


Fig. 29: Corridor, S5: sweeps, with layout in Section 2.2.3.

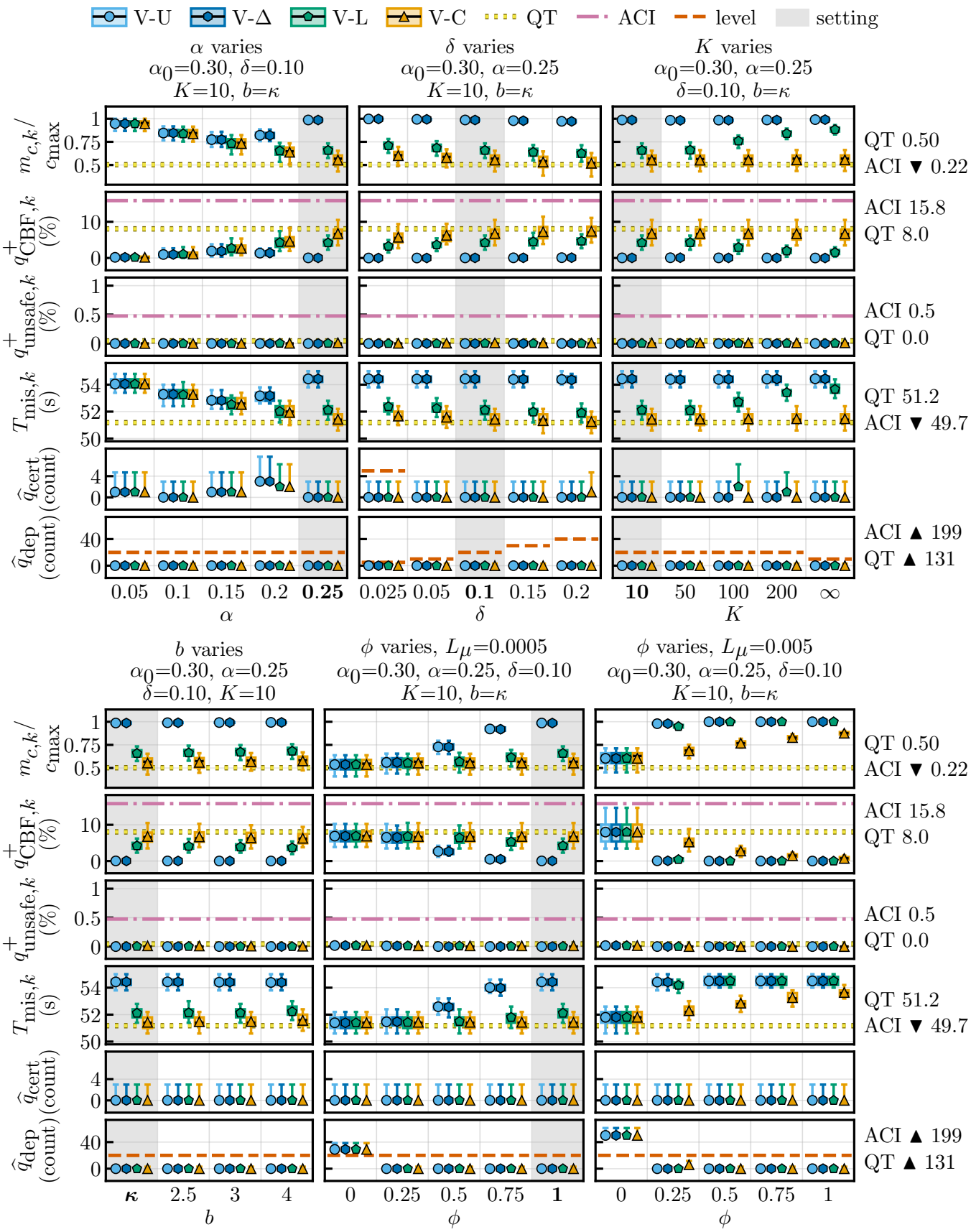


Fig. 30: Maze, S5: sweeps, with layout in Section 2.2.3.

2.3 Margin Insufficiency and CBF-Violation Probability

Figs. 31–41 show heatmaps over the main grid (Table 3) for the four variants: $\widehat{q}^{+}_{\mathrm{CBF}}$ and $\widehat{m}_c/c_{\max}$ (A.80) at $\alpha_0 = 0.15$, and $\widehat{m}_c/c_{\max}$, $\widehat{q}^{+}_{\mathrm{CBF}}$, and $\widehat{q}^{+}_{\mathrm{unsafe}}$ (A.81) at $\alpha_0 = 0.30$, in both scenarios, and $\widehat{q}^{+}_{\mathrm{unsafe}}$ at $\alpha_0 = 0.25$ in the Corridor. Each cell gives the mean with one sample standard deviation below it, with bold as in the table key (Section 2.1). Each color scale covers the range of its metric and scenario over the main grid.

Over the 1500 grid cells of each scenario, the Spearman rank correlation of $\widehat{m}_c/c_{\max}$ is at least 0.99 with $\widehat{T}_{\mathrm{mis}}$ and at most -0.98 with $\widehat{q}^{+}_{\mathrm{CBF}}$. The values of $\widehat{m}_c/c_{\max}$ are ordered like the certificate bounds in (A.73), V-C lowest and V-U highest, for 731 of the 750 pairs of a main-grid setting and a scenario. Those of V-Δ and V-U differ by at most 0.003. Across α_0, $\widehat{m}_c/c_{\max}$ of V-C changes by at most 0.016 at fixed (α, δ, K), and its $\widehat{q}^{+}_{\mathrm{nodep}}$ in (1.16) stays between 1.0% and 3.3% over the main grid. For V-U and V-Δ, $\widehat{q}^{+}_{\mathrm{nodep}}$ reaches 84% in the Corridor and 89% in the Maze. At $(\alpha, \delta, K) = (0.10, 0.05, 50)$, their $\widehat{m}_c/c_{\max}$ in the Corridor falls from 1 to 0.84 as α_0 grows from 0.15 to 0.30, as explained in Section 2.1. For $\alpha_0 \le 0.20$ in the Corridor and $\alpha_0 \le 0.25$ in the Maze, $\widehat{q}^{+}_{\mathrm{unsafe}}$ of every variant is below 0.004%. For every $\alpha_0 \ge 0.15$, δ, K, and scenario, V-C has its smallest $\widehat{m}_c/c_{\max}$ at the largest $\alpha = \alpha_0 - 0.05$, whereas, at $\alpha_0 = 0.30$, V-U, V-Δ, and V-L have it at $\alpha \in \{0.15, 0.20\}$. With $\alpha_0 - \alpha = 0.05$ fixed, $\widehat{m}_c/c_{\max}$ is nonincreasing in α_0 in all 200 series, and the number of trajectories with a safe-set violation at $\alpha_0 = 0.30$ is never below that at $\alpha_0 = 0.10$.

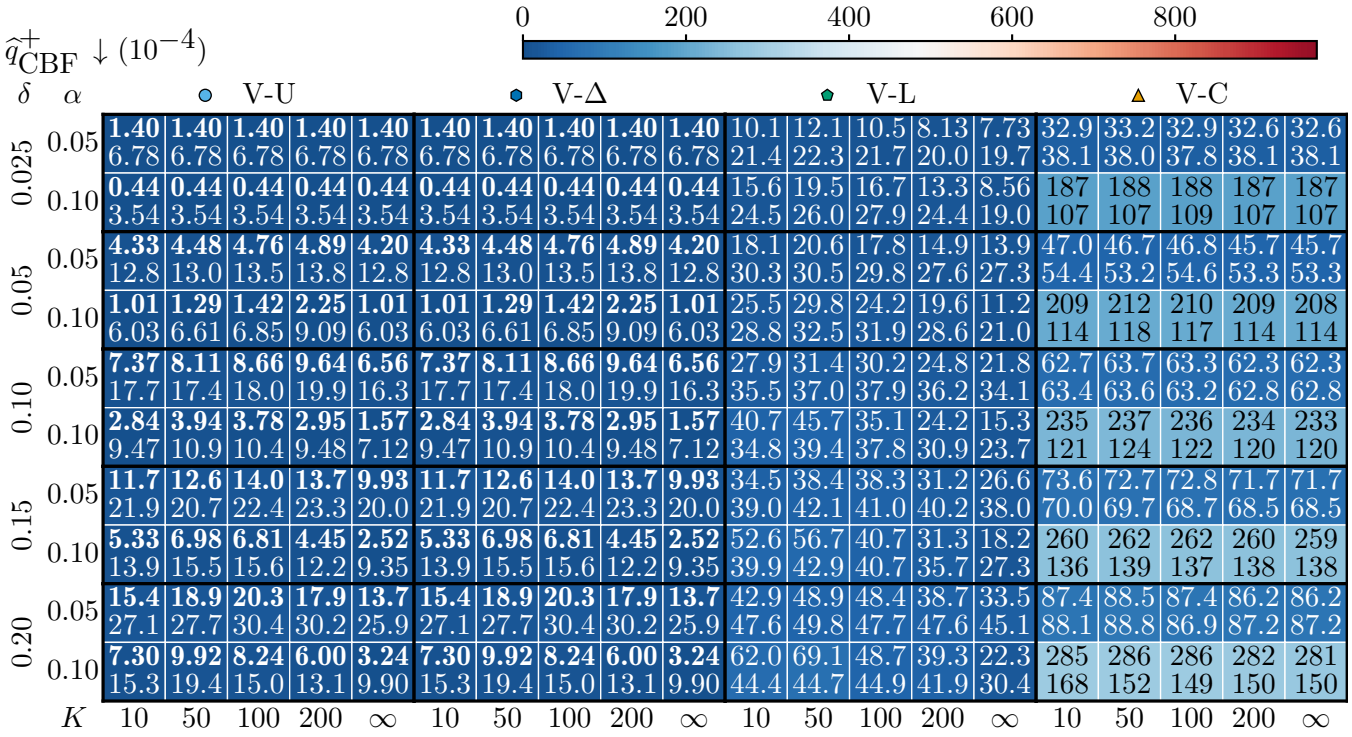


Fig. 31: Corridor, $\widehat{q}^{+}_{\mathrm{CBF}}$ at $\alpha_0 = 0.15$.

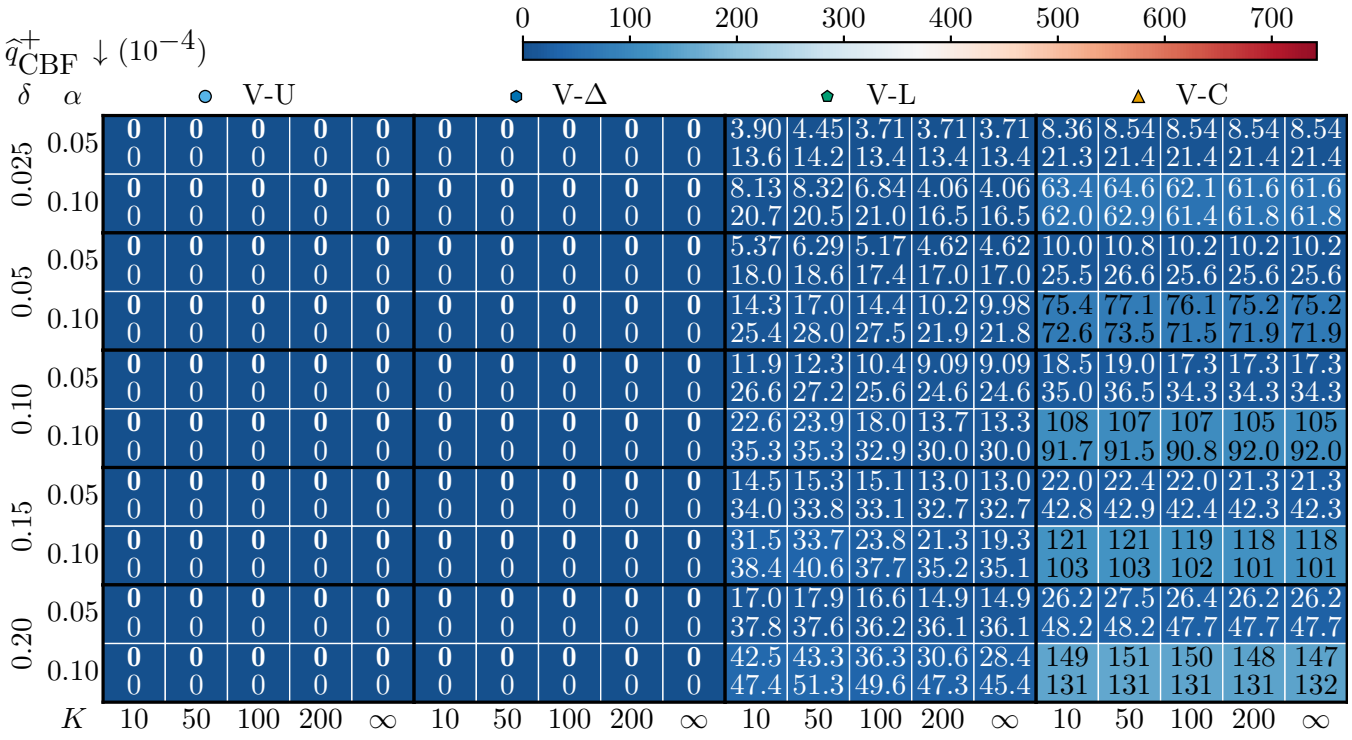


Fig. 32: Maze, $\widehat{q}^{+}_{\mathrm{CBF}}$ at $\alpha_0 = 0.15$.

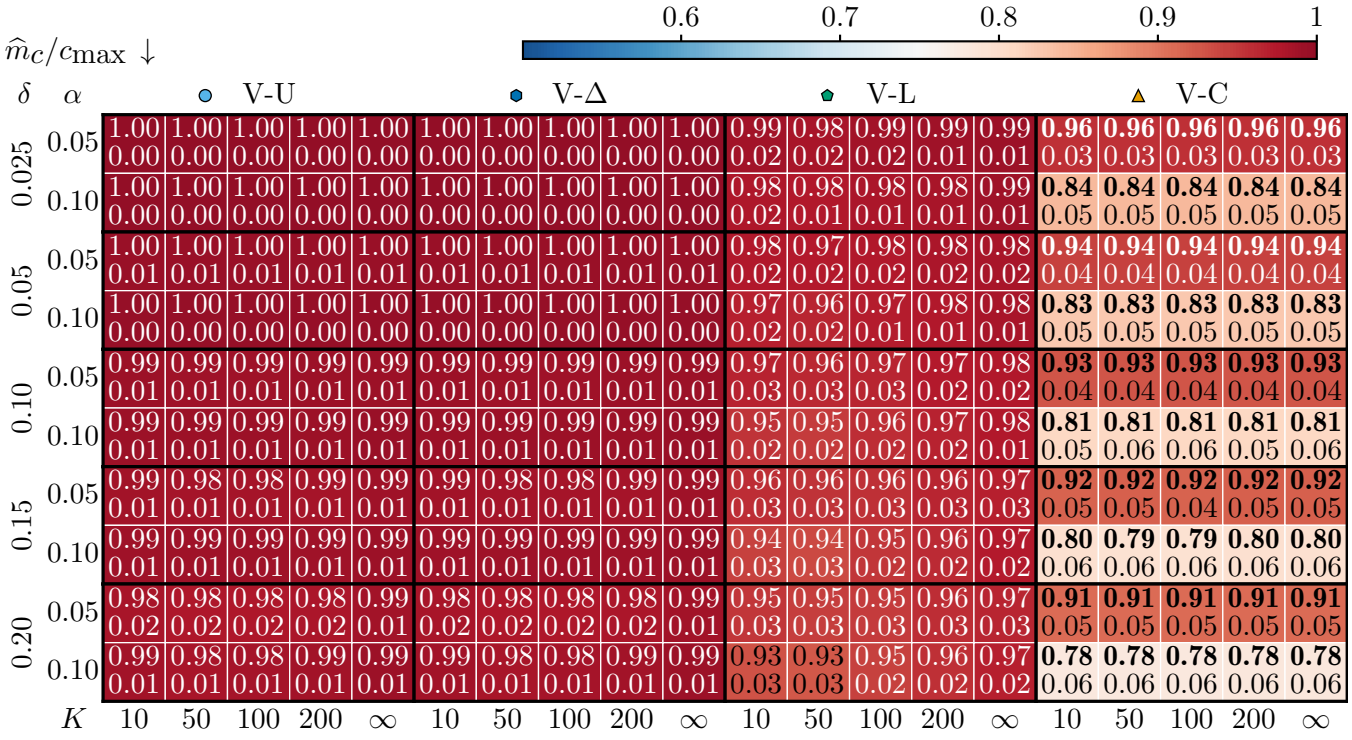


Fig. 33: Corridor, $\widehat{m}_c/c_{\max}$ at $\alpha_0 = 0.15$.

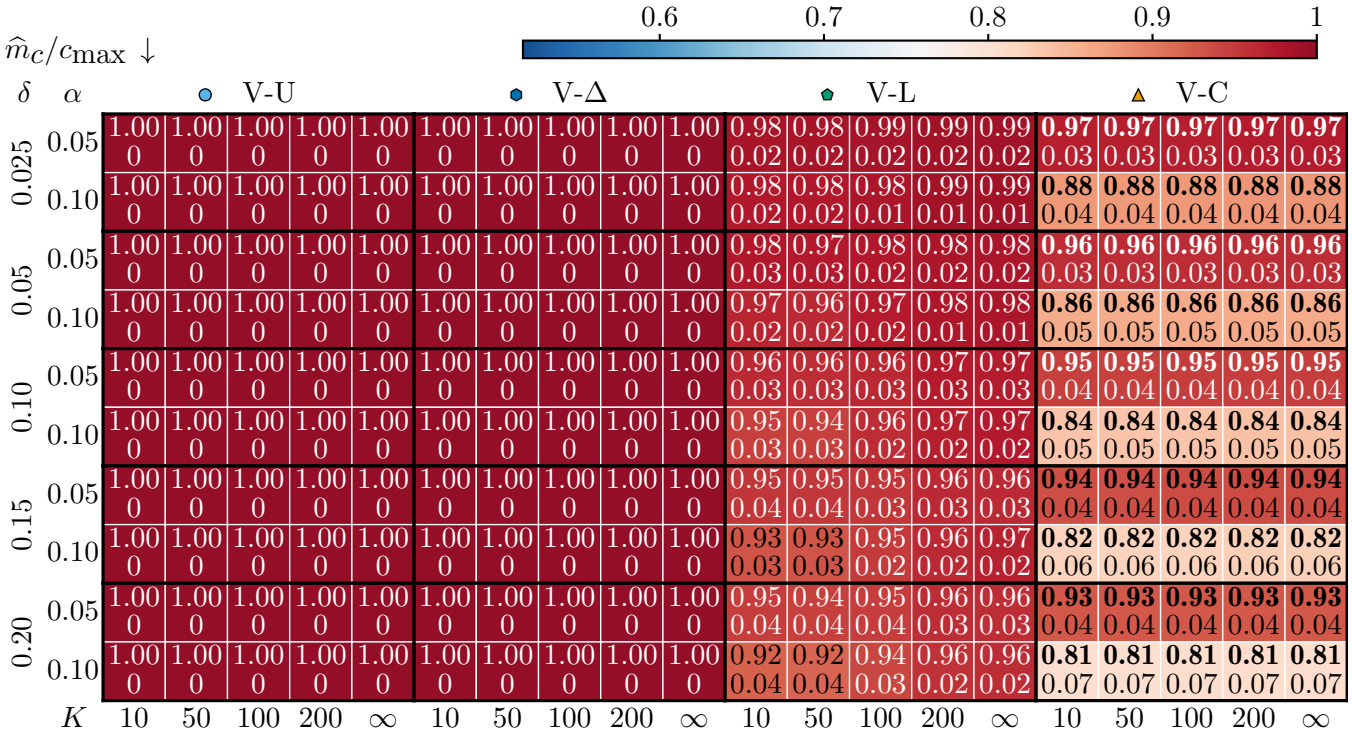


Fig. 34: Maze, $\widehat{m}_c/c_{\max}$ at $\alpha_0 = 0.15$.

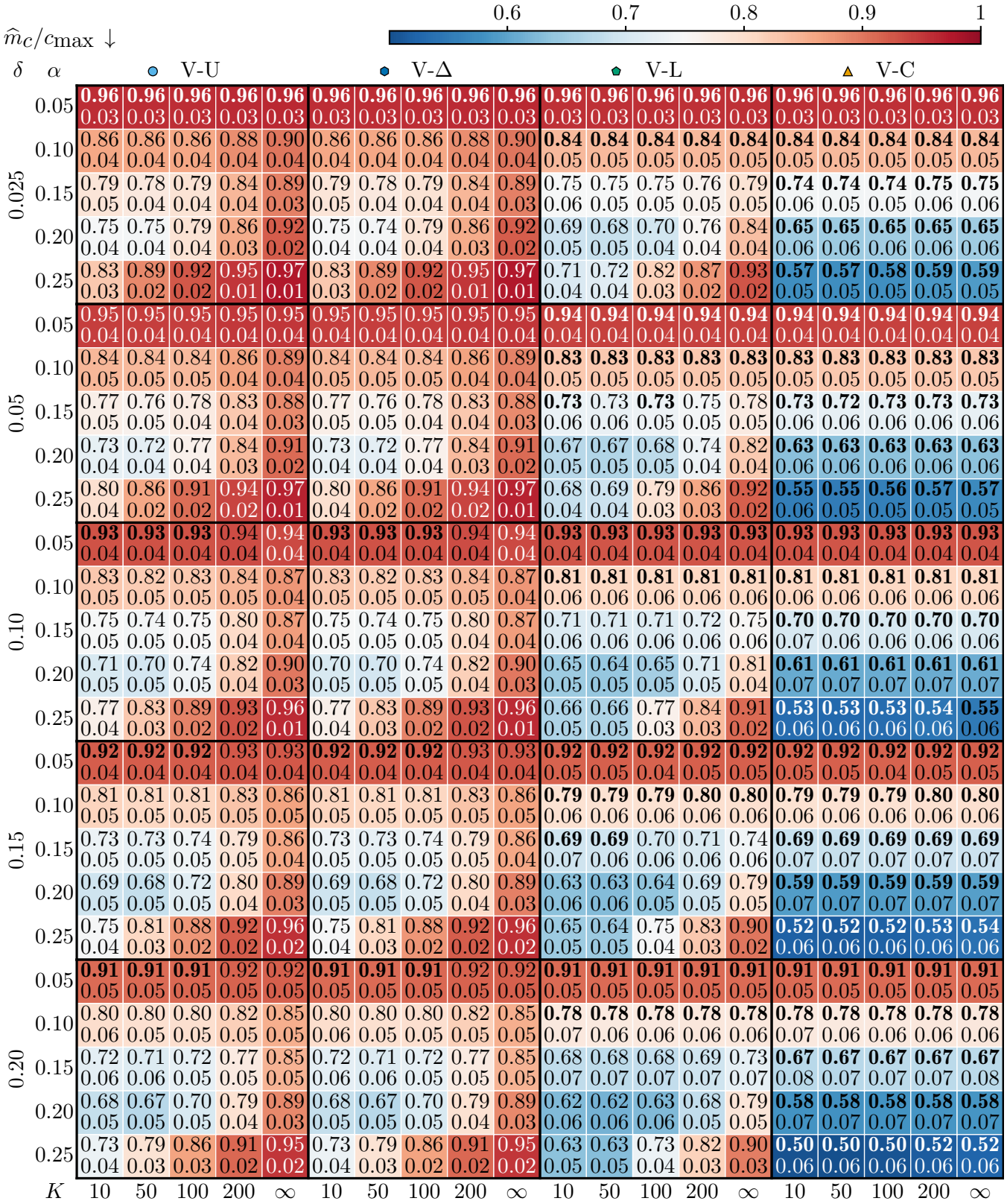


Fig. 35: Corridor, $\widehat{m}_c/c_{\max}$ at $\alpha_0 = 0.30$.

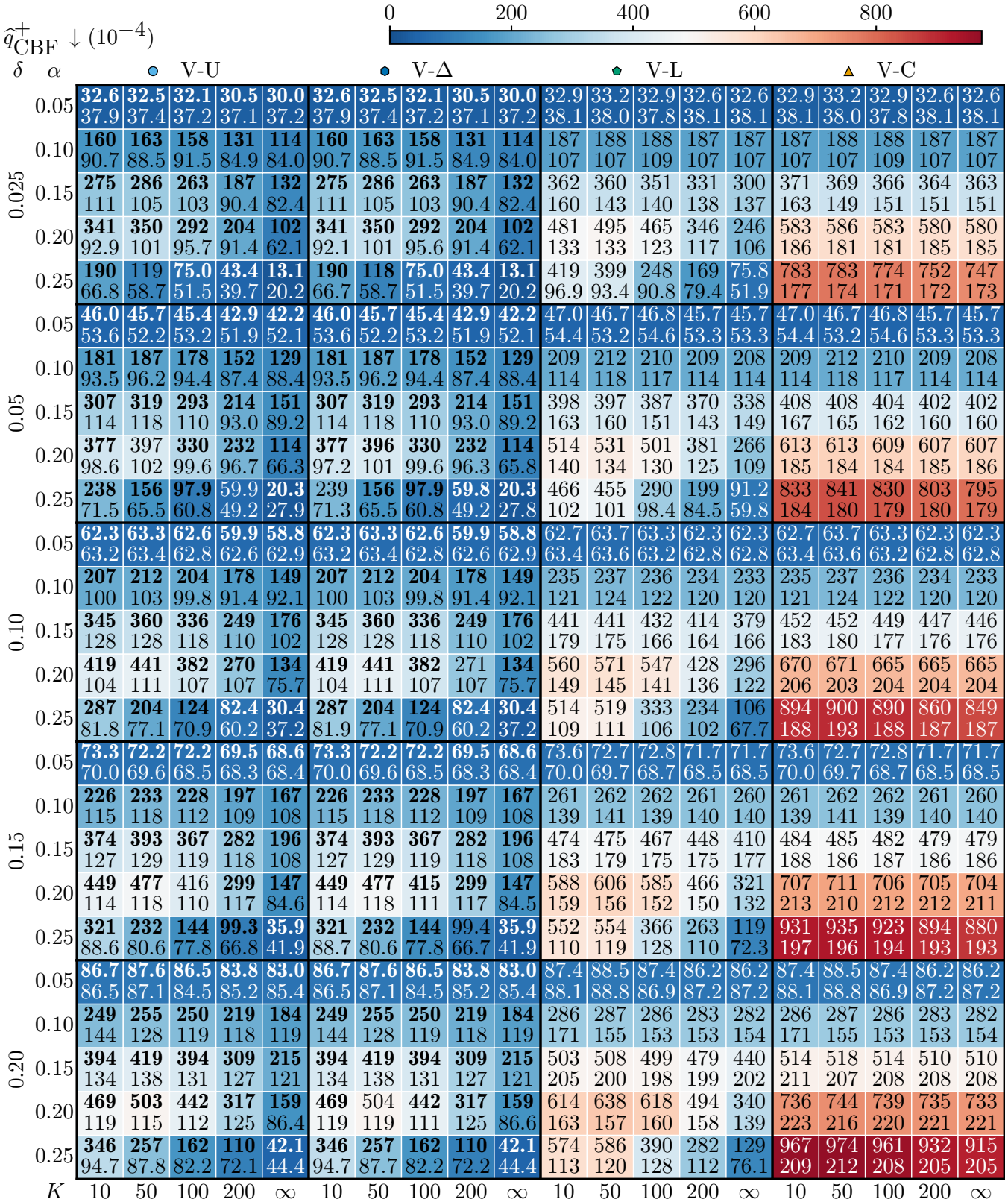


Fig. 36: Corridor, $\widehat{q}^{+}_{\mathrm{CBF}}$ at $\alpha_0 = 0.30$.

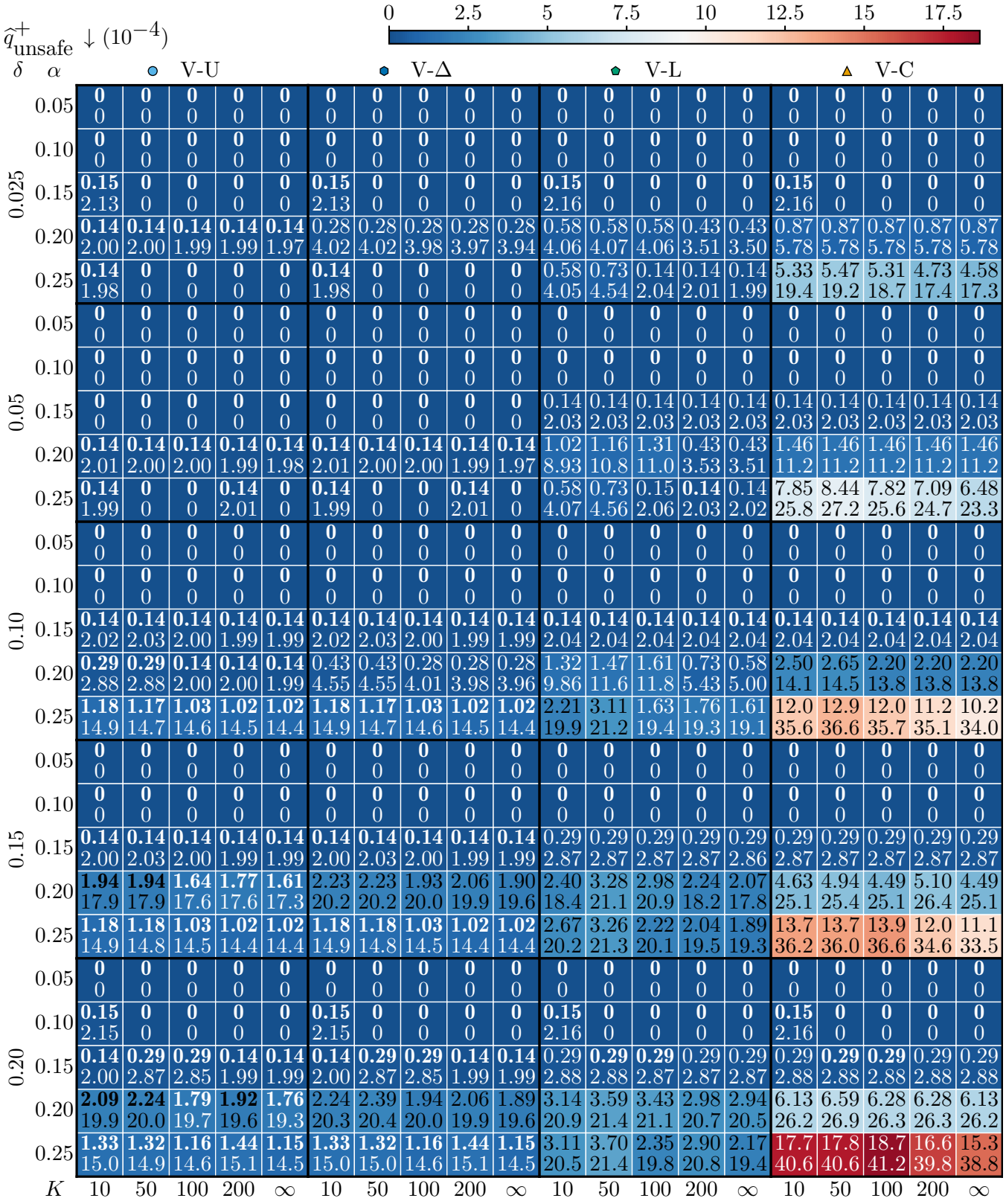


Fig. 37: Corridor, $\widehat{q}^{+}_{\text{unsafe}}$ at $\alpha_0 = 0.30$.

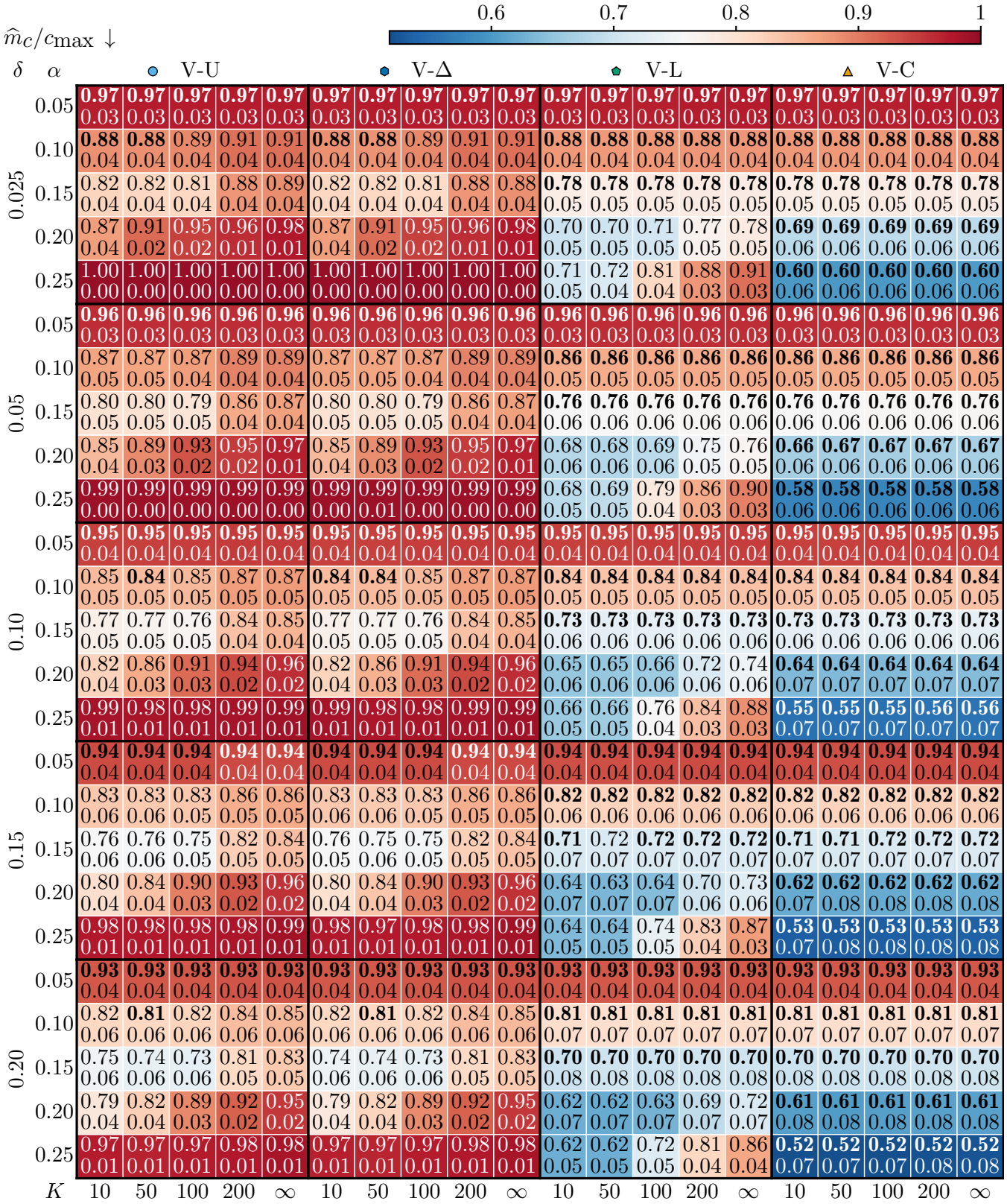


Fig. 38: Maze, $\widehat{m}_c/c_{\max}$ at $\alpha_0 = 0.30$.

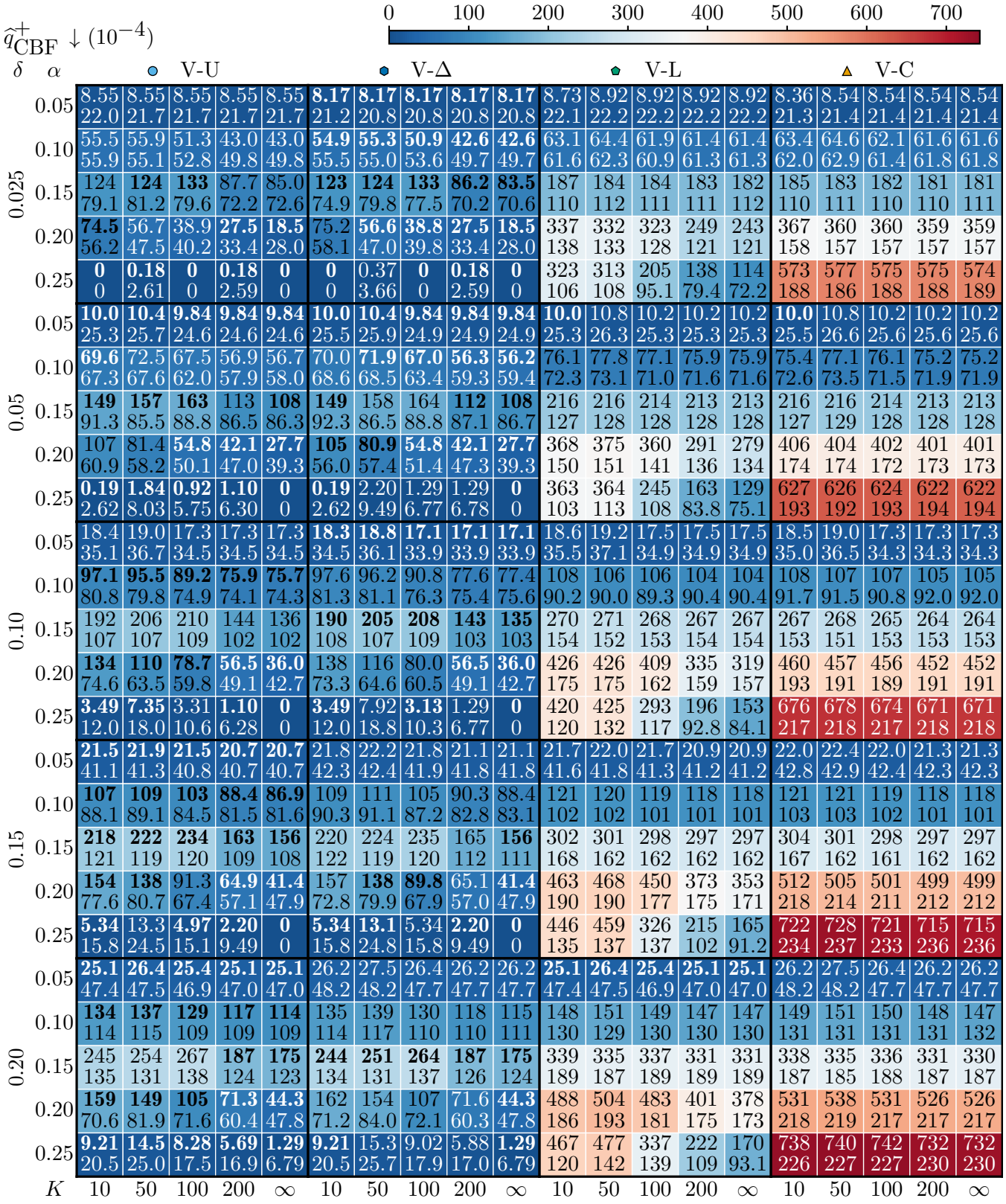


Fig. 39: Maze, $\widehat{q}^{+}_{\mathrm{CBF}}$ at $\alpha_0 = 0.30$.

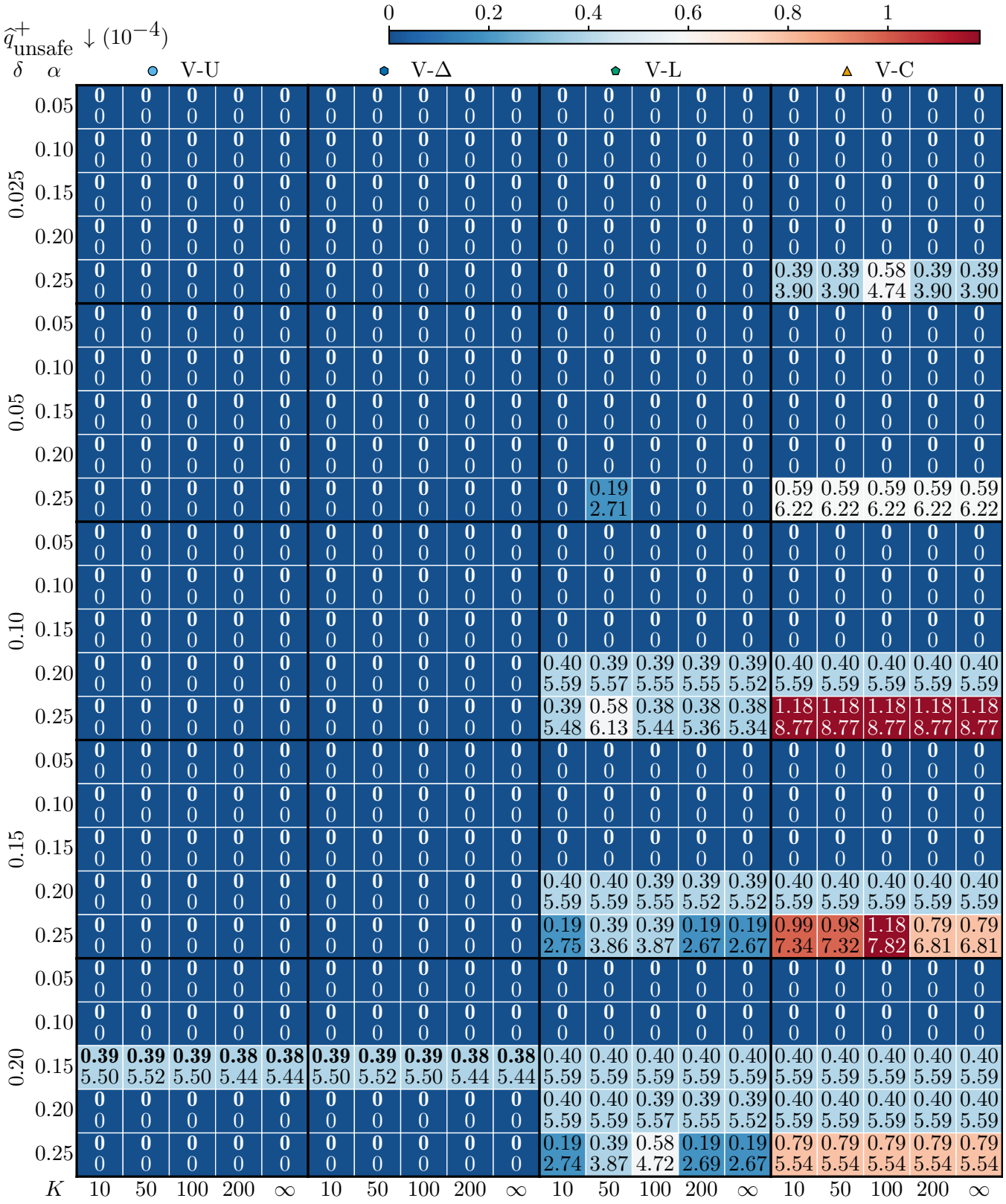


Fig. 40: Maze, $\widehat{q}^{+}_{\text{unsafe}}$ at $\alpha_0 = 0.30$.

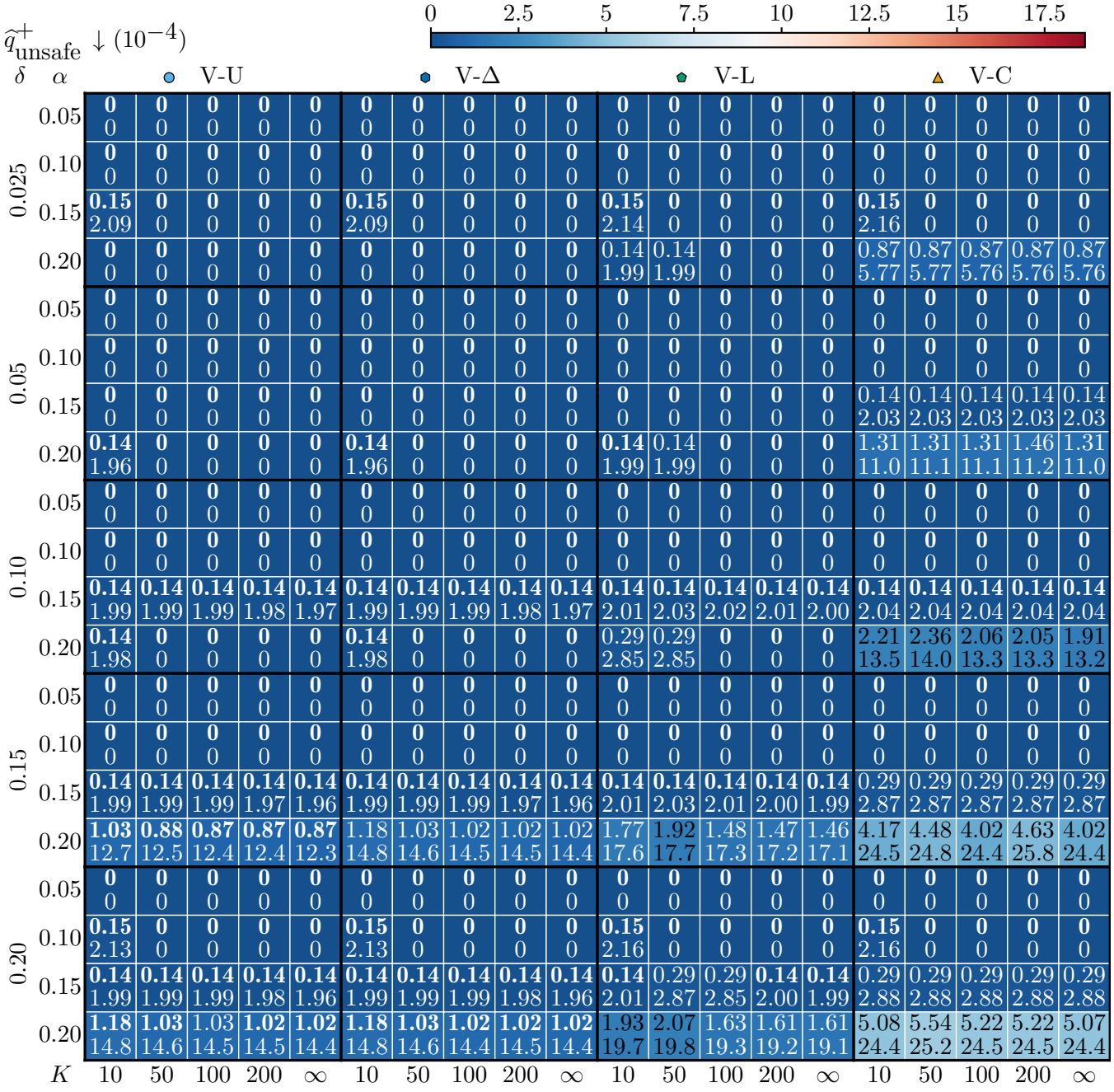


Fig. 41: Corridor, $\widehat{q}^{+}_{\text{unsafe}}$ at $\alpha_0 = 0.25$.

2.4 Error Confidence

Tables 9 and 10 vary the error confidence δ over the values of the main grid at the fixed (α_0, α, K) of the default setting and of S1–S5. S1 and S2 share one series, since they differ only in δ. From $\delta = 0.025$ to 0.20, $\widehat{m}_c/c_{\max}$ decreases in 38 of the 40 series, by at most 0.095, and the other two remain equal to one (V-U and V-Δ in the Maze at the default (α_0, α, K)). At the default (α_0, α, K) with $\delta = 0.20$, V-C has deployment events in both scenarios.

Table 9: Corridor, error confidence δ, default setting and S1–S5.

Corridor **Baselines** $\alpha_0 = 0.15,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04 0.00±0.00	30.69±13.51 N-A 22.32±10.04	N-A N-A N-A	−0.27±0.05 −0.02±0.01 70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02 0.66±0.07	28.00±2.99 49.65±2.89 15.27±4.17	100.0±0.0 0.00±0.00 78.8±8.6	−0.03±0.01 0.10±0.01 32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01 1.00±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.05±0.01 0.17±0.01 35.91±0.92
ACI	0.49±0.04 0.17±0.02 0.82±0.09	9.02±1.17 15.26±0.53 0.05±0.17	98.2±0.0 0.60±0.00 33.4±8.6	0.01±0.01 0.14±0.01 34.00±0.87
QT	0.52±0.04 0.17±0.02 0.86±0.09	8.34±0.97 14.38±0.32 0.02±0.10	99.0±0.8 0.10±0.00 49.7±8.0	0.01±0.01 0.14±0.01 34.17±0.89

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
δ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
default setting: V-C (α_0, α, K) = (0.15, 0.10, 50)					
0.025	0.84±0.05 0.17±0.01 0.93±0.09	1.88±1.07 2.99±1.44 0.00±0.00	76.5±6.9 7.52±2.58 5.3±2.0	21.3±7.1 2.2±0.7 8.66±2.12	0.03±0.01 0.16±0.01 35.21±0.91
0.05	0.83±0.05 0.17±0.01 0.93±0.09	2.12±1.18 3.37±1.63 0.00±0.00	79.2±6.2 6.45±2.37 5.7±2.2	18.3±6.6 2.5±0.7 9.39±2.38	0.03±0.01 0.16±0.01 35.14±0.91
0.10	0.81±0.06 0.17±0.01 0.92±0.09	2.37±1.24 3.82±1.69 0.00±0.00	82.0±5.8 5.38±2.25 6.2±2.3	15.3±6.3 2.7±0.7 10.18±2.40	0.03±0.01 0.16±0.01 35.06±0.91
0.15	0.79±0.06 0.17±0.01 0.92±0.09	2.62±1.39 4.15±1.86 0.00±0.00	83.3±5.5 4.87±2.14 6.6±2.4	13.9±5.9 2.8±0.7 10.64±2.41	0.03±0.01 0.16±0.01 35.01±0.91
0.20	0.78±0.06 0.17±0.01 0.91±0.09	2.86±1.52 4.56±2.07 0.00±0.00	84.7±5.2 4.39±2.02 7.1±2.6	12.5±5.6 2.9±0.7 11.04±2.39	0.03±0.01 0.16±0.01 34.96±0.92

Table 9 (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04 0.00±0.00	30.69±13.51 N-A 22.32±10.04	N-A N-A N-A	−0.27±0.05 −0.02±0.01 70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02 0.66±0.07	28.00±2.99 49.65±2.89 15.27±4.17	100.0±0.0 0.00±0.00 78.8±8.6	−0.03±0.01 0.10±0.01 32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01 1.00±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.05±0.01 0.17±0.01 35.91±0.92
ACI	0.22±0.04 0.17±0.02 0.73±0.08	17.68±1.59 30.21±0.72 2.74±1.43	99.1±0.0 0.30±0.00 57.4±10.2	−0.01±0.01 0.12±0.01 33.13±0.81
QT	0.30±0.03 0.17±0.02 0.78±0.09	15.19±1.31 26.54±1.03 0.61±0.52	99.7±0.1 0.10±0.00 60.1±11.5	−0.00±0.00 0.13±0.01 33.44±0.83

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
δ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S1 and **S2**: V-C (α_0, α, K) = (0.30, 0.25, 50)					
0.025	0.57±0.05 0.17±0.02 0.83±0.09	7.83±1.74 12.14±2.30 0.05±0.19	88.9±1.4 2.74±0.47 18.3±2.9	8.0±1.3 3.1±0.7 26.99±3.13	0.01±0.01 0.14±0.01 34.13±0.89
0.05	0.55±0.05 0.17±0.02 0.82±0.09	8.41±1.80 13.02±2.40 0.08±0.27	90.1±1.5 2.31±0.46 19.1±3.0	6.8±1.3 3.1±0.7 27.95±3.16	0.01±0.01 0.14±0.01 34.03±0.90
0.10	0.53±0.06 0.17±0.02 0.81±0.09	9.00±1.93 13.98±2.76 0.13±0.37	91.0±1.4 1.99±0.40 19.9±3.3	5.8±1.2 3.2±0.8 29.05±3.19	0.01±0.01 0.13±0.01 33.96±0.88
0.15	0.52±0.06 0.17±0.02 0.81±0.09	9.35±1.96 14.52±2.74 0.14±0.36	91.7±1.2 1.74±0.32 20.3±3.3	5.1±0.9 3.2±0.8 29.80±3.27	0.01±0.01 0.13±0.01 33.92±0.88
0.20	0.50±0.06 0.17±0.02 0.80±0.09	9.74±2.12 15.10±2.94 0.18±0.41	92.3±1.1 1.53±0.31 21.1±3.3	4.5±0.9 3.2±0.8 30.46±3.36	0.01±0.01 0.13±0.01 33.86±0.89
S3: V-L (α_0, α, K) = (0.30, 0.20, 50)					
0.025	0.68±0.05 0.17±0.02 0.87±0.09	4.95±1.33 7.74±1.76 0.01±0.04	87.1±1.9 3.39±0.70 12.9±2.9	9.8±2.0 3.1±0.7 21.51±2.91	0.02±0.01 0.15±0.01 34.53±0.87
0.05	0.67±0.05 0.17±0.02 0.86±0.09	5.31±1.34 8.29±1.74 0.01±0.11	88.3±1.9 2.95±0.67 13.4±3.1	8.5±1.9 3.2±0.7 22.33±3.14	0.02±0.01 0.15±0.01 34.47±0.88
0.10	0.64±0.05 0.17±0.02 0.85±0.09	5.71±1.45 9.01±1.95 0.01±0.12	89.7±1.8 2.48±0.60 14.3±3.2	7.2±1.7 3.2±0.7 23.36±3.13	0.02±0.01 0.14±0.01 34.38±0.89

Table 9 (continued).

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
δ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S3: V-L (α_0, α, K) = (0.30, 0.20, 50)					
0.15	0.63±0.06 0.17±0.02 0.85±0.09	6.06±1.56 9.54±2.09 0.03±0.21	90.5±1.8 2.19±0.61 14.9±3.4	6.3±1.7 3.2±0.8 23.89±3.08	0.01±0.01 0.14±0.01 34.33±0.91
0.20	0.62±0.06 0.17±0.02 0.84±0.09	6.38±1.57 10.03±2.14 0.04±0.21	90.9±1.7 2.04±0.54 15.3±3.6	5.9±1.6 3.2±0.8 24.33±3.09	0.01±0.01 0.14±0.01 34.28±0.90
S4: V-Δ (α_0, α, K) = (0.30, 0.25, 100)					
0.025	0.92±0.02 0.17±0.01 0.97±0.08	0.75±0.52 1.44±0.71 0.00±0.00	29.8±3.3 2.74±0.47 11.2±3.4	7.7±1.3 62.5±3.0 16.38±1.82	0.03±0.01 0.17±0.01 35.62±0.88
0.05	0.91±0.02 0.17±0.01 0.97±0.08	0.98±0.61 1.94±0.86 0.00±0.00	33.7±3.1 2.31±0.46 12.3±3.5	6.5±1.3 59.8±2.8 17.01±1.90	0.03±0.01 0.16±0.01 35.57±0.88
0.10	0.89±0.02 0.17±0.01 0.96±0.08	1.24±0.71 2.40±0.94 0.01±0.15	37.6±3.2 1.99±0.40 13.6±3.9	5.6±1.1 56.8±3.0 17.82±1.97	0.03±0.01 0.16±0.01 35.49±0.86
0.15	0.88±0.02 0.17±0.01 0.96±0.08	1.44±0.78 2.82±1.11 0.01±0.15	40.0±3.0 1.74±0.32 14.8±4.1	4.9±0.9 55.1±2.8 18.26±2.06	0.03±0.01 0.16±0.01 35.44±0.86
0.20	0.86±0.02 0.17±0.01 0.95±0.08	1.62±0.82 3.15±1.21 0.01±0.15	41.7±2.9 1.53±0.31 15.6±4.4	4.3±0.9 54.0±2.7 18.69±1.99	0.03±0.01 0.16±0.01 35.40±0.85
S5: V-C (α_0, α, K) = (0.30, 0.25, 10)					
0.025	0.57±0.05 0.17±0.02 0.83±0.09	7.83±1.77 12.15±2.36 0.05±0.19	88.9±1.4 2.74±0.47 18.2±2.9	8.0±1.3 3.1±0.7 112.51±13.86	0.01±0.01 0.14±0.01 34.12±0.89
0.05	0.55±0.06 0.17±0.02 0.82±0.09	8.33±1.84 12.89±2.48 0.08±0.26	90.1±1.5 2.31±0.46 18.8±3.0	6.8±1.3 3.1±0.7 115.70±14.07	0.01±0.01 0.14±0.01 34.04±0.90
0.10	0.53±0.06 0.17±0.02 0.81±0.09	8.94±1.88 13.84±2.67 0.12±0.36	91.0±1.4 1.99±0.40 19.6±3.2	5.8±1.2 3.2±0.8 119.71±13.69	0.01±0.01 0.13±0.01 33.97±0.88
0.15	0.52±0.06 0.17±0.02 0.81±0.09	9.31±1.97 14.40±2.71 0.14±0.36	91.7±1.2 1.74±0.32 20.1±3.2	5.1±0.9 3.2±0.8 122.03±13.45	0.01±0.01 0.13±0.01 33.93±0.88
0.20	0.50±0.06 0.17±0.01 0.80±0.09	9.67±2.09 14.97±2.94 0.18±0.41	92.3±1.1 1.53±0.31 20.8±3.2	4.5±0.9 3.2±0.8 123.83±13.91	0.01±0.01 0.13±0.01 33.88±0.89

Table 10: Maze, as in Table 9.

Maze **Baselines** $\alpha_0 = 0.15,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02 0.00±0.00	55.68±1.58 N-A 53.27±0.95	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02 0.32±0.02	24.93±2.57 49.79±2.97 3.86±2.16	100.0±0.0 0.00±0.00 66.8±12.7	−0.01±0.00 0.21±0.01 48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.22±0.01 54.44±0.32
ACI	0.49±0.05 0.17±0.02 0.50±0.03	7.45±1.47 14.94±1.34 0.01±0.06	97.7±0.0 1.20±0.00 27.6±7.3	0.01±0.00 0.21±0.01 51.16±0.40
QT	0.72±0.02 0.17±0.02 0.61±0.02	2.78±0.88 6.40±0.81 0.00±0.00	99.6±0.1 0.20±0.00 15.3±3.7	0.02±0.00 0.21±0.01 52.52±0.33

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
δ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
default setting: V-C (α_0, α, K) = (0.15, 0.10, 50)					
0.025	0.88±0.04 0.17±0.02 0.71±0.04	0.65±0.63 2.05±1.19 0.00±0.00	68.3±10.3 15.71±5.77 4.4±1.7	29.3±10.6 2.4±0.8 5.93±1.92	0.02±0.00 0.21±0.01 53.52±0.47
0.05	0.86±0.05 0.17±0.02 0.70±0.04	0.77±0.73 2.36±1.25 0.00±0.00	72.6±8.8 13.28±4.84 4.7±1.9	24.8±9.0 2.6±0.7 6.49±1.88	0.02±0.00 0.21±0.01 53.40±0.44
0.10	0.84±0.05 0.17±0.02 0.69±0.04	1.07±0.92 2.92±1.63 0.00±0.00	76.7±7.8 11.01±4.29 5.3±2.1	20.6±7.9 2.7±0.8 7.03±1.88	0.02±0.00 0.21±0.01 53.26±0.50
0.15	0.82±0.06 0.17±0.02 0.68±0.04	1.21±1.03 3.30±1.77 0.00±0.00	79.0±7.2 9.78±3.89 5.7±2.3	18.4±7.2 2.6±0.8 7.70±1.95	0.02±0.00 0.21±0.01 53.13±0.53
0.20	0.81±0.07 0.17±0.02 0.66±0.05	1.51±1.31 3.80±2.20 0.00±0.00	81.1±6.6 8.63±3.59 6.2±2.5	16.3±6.7 2.7±0.8 8.09±2.12	0.02±0.00 0.21±0.01 53.03±0.57

Table 10 (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02 0.00±0.00	55.68±1.58 N-A 53.27±0.95	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02 0.32±0.02	24.93±2.57 49.79±2.97 3.86±2.16	100.0±0.0 0.00±0.00 66.8±12.7	−0.01±0.00 0.21±0.01 48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.22±0.01 54.44±0.32
ACI	0.22±0.04 0.17±0.02 0.39±0.02	15.84±1.93 29.86±1.85 0.48±0.65	98.8±0.0 0.60±0.00 47.1±8.6	−0.00±0.00 0.21±0.01 49.71±0.32
QT	0.50±0.03 0.17±0.02 0.50±0.02	8.04±1.31 15.68±1.10 0.05±0.17	99.6±0.0 0.20±0.00 33.8±7.6	0.01±0.00 0.21±0.01 51.18±0.32

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
δ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S1 and **S2**: V-C (α_0, α, K) = (0.30, 0.25, 50)					
0.025	0.60±0.06 0.17±0.02 0.54±0.04	5.77±1.86 11.20±2.48 0.00±0.04	86.6±1.8 5.45±0.85 17.2±3.2	10.5±1.6 2.9±0.8 20.19±2.36	0.01±0.00 0.21±0.01 51.67±0.48
0.05	0.58±0.06 0.17±0.02 0.53±0.04	6.26±1.92 12.11±2.68 0.01±0.06	88.2±1.7 4.56±0.79 17.8±3.3	8.8±1.5 3.0±0.7 20.93±2.32	0.01±0.00 0.21±0.01 51.57±0.49
0.10	0.55±0.07 0.17±0.02 0.52±0.04	6.78±2.18 13.05±3.18 0.01±0.09	89.3±1.6 3.93±0.77 18.7±3.5	7.6±1.5 3.1±0.8 21.78±2.47	0.01±0.00 0.21±0.01 51.44±0.54
0.15	0.53±0.08 0.17±0.02 0.51±0.04	7.28±2.37 13.89±3.46 0.01±0.07	90.3±1.5 3.45±0.62 19.2±3.7	6.7±1.2 2.9±0.8 22.37±2.50	0.01±0.00 0.21±0.01 51.33±0.53
0.20	0.52±0.07 0.17±0.02 0.51±0.04	7.40±2.27 14.52±3.57 0.01±0.06	91.1±1.4 3.04±0.63 19.7±3.5	5.9±1.2 2.9±0.8 22.83±2.54	0.01±0.00 0.21±0.01 51.25±0.52

Table 10 (continued).

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
δ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S3: V-L (α_0, α, K) = (0.30, 0.20, 50)					
0.025	0.70±0.05 0.17±0.02 0.59±0.04	3.32±1.33 7.18±2.03 0.00±0.00	84.1±2.6 6.75±1.36 11.8±2.8	12.9±2.6 3.0±0.8 15.76±2.14	0.02±0.00 0.21±0.01 52.28±0.46
0.05	0.68±0.06 0.17±0.02 0.58±0.04	3.75±1.51 7.90±2.23 0.00±0.00	85.7±2.4 5.87±1.27 12.6±3.0	11.3±2.4 3.0±0.8 16.57±2.27	0.01±0.00 0.21±0.01 52.16±0.48
0.10	0.65±0.06 0.17±0.02 0.57±0.04	4.26±1.75 8.91±2.56 0.00±0.06	87.6±2.1 4.93±1.09 13.4±3.4	9.4±2.0 3.0±0.8 17.47±2.15	0.01±0.00 0.21±0.01 52.01±0.49
0.15	0.63±0.07 0.17±0.02 0.56±0.04	4.68±1.90 9.54±2.83 0.00±0.06	88.6±2.2 4.31±1.08 14.2±3.7	8.3±2.1 3.1±0.8 17.84±2.27	0.01±0.00 0.21±0.01 51.91±0.53
0.20	0.62±0.07 0.17±0.02 0.55±0.04	5.04±1.93 9.98±2.87 0.00±0.06	89.1±2.0 4.01±0.97 14.5±3.8	7.7±1.9 3.2±0.9 18.19±2.35	0.01±0.00 0.21±0.01 51.86±0.54
S4: V-Δ (α_0, α, K) = (0.30, 0.25, 100)					
0.025	1.00±0.00 0.17±0.01 0.81±0.02	0.00±0.00 0.05±0.13 0.00±0.00	1.4±0.8 7.85±7.25 3.9±3.1	10.0±1.6 88.5±1.0 12.86±1.58	0.03±0.00 0.22±0.01 54.43±0.32
0.05	0.99±0.00 0.17±0.02 0.81±0.02	0.01±0.07 0.16±0.25 0.00±0.00	3.7±1.3 5.29±4.69 6.4±4.1	8.4±1.5 87.9±0.9 13.23±1.61	0.03±0.00 0.22±0.01 54.42±0.33
0.10	0.98±0.01 0.17±0.02 0.80±0.02	0.03±0.10 0.30±0.34 0.00±0.00	6.5±1.7 4.65±4.66 7.7±4.5	7.2±1.4 86.2±1.4 13.84±1.77	0.03±0.00 0.22±0.01 54.41±0.34
0.15	0.98±0.01 0.17±0.02 0.81±0.02	0.05±0.16 0.44±0.44 0.00±0.00	8.6±1.8 3.55±1.60 9.4±4.9	6.4±1.1 85.1±1.6 14.31±1.72	0.03±0.00 0.22±0.01 54.38±0.35
0.20	0.97±0.01 0.17±0.02 0.80±0.02	0.09±0.18 0.59±0.53 0.00±0.00	10.0±1.9 3.04±0.63 11.0±5.1	5.6±1.2 84.4±1.5 14.57±1.69	0.03±0.00 0.22±0.01 54.37±0.34

Table 10 (continued).

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
δ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S5: V-C (α_0, α, K) = (0.30, 0.25, 10)					
0.025	0.60±0.06 0.17±0.02 0.54±0.04	5.73±1.88 11.21±2.53 0.00±0.04	86.6±1.8 5.45±0.85 17.1±3.2	10.5±1.6 2.9±0.8 77.50±9.16	0.01±0.00 0.21±0.01 51.67±0.47
0.05	0.58±0.06 0.17±0.02 0.53±0.04	6.27±1.93 12.09±2.71 0.01±0.06	88.2±1.6 4.56±0.79 17.7±3.2	8.8±1.5 3.0±0.7 80.70±9.27	0.01±0.00 0.21±0.01 51.58±0.50
0.10	0.55±0.07 0.17±0.02 0.52±0.04	6.76±2.17 13.03±3.13 0.01±0.09	89.3±1.7 3.93±0.77 18.5±3.3	7.6±1.5 3.0±0.8 83.58±9.71	0.01±0.00 0.21±0.01 51.43±0.52
0.15	0.53±0.07 0.17±0.02 0.51±0.04	7.22±2.34 13.83±3.42 0.01±0.07	90.3±1.4 3.45±0.62 19.1±3.6	6.7±1.2 2.9±0.8 86.17±10.05	0.01±0.00 0.21±0.01 51.34±0.54
0.20	0.52±0.07 0.17±0.02 0.51±0.04	7.38±2.26 14.39±3.53 0.01±0.06	91.1±1.4 3.04±0.63 19.4±3.5	5.9±1.2 3.0±0.8 87.29±10.22	0.01±0.00 0.21±0.01 51.26±0.52

2.5 Test Period

Tables 11 and 12 vary the test period K over the values of the main grid at the fixed $(\alpha_0, \alpha, \delta)$ of the default setting and of S1–S5. Figs. 42–44 compare the event frequencies with the bounds of Theorems A.1–A.3 for S3 with V-L in the detailed checks (Section 2.1) and for the default setting with V-C at the same values of K and L_μ. Over the 48 series, the smallest unrounded mean $\widehat{m}_c/c_{\max}$ occurs at $K = 50$ in 27, at $K = 10$ in 17, and at $K = 100$ in two. The other two, V-U and V-Δ in the Maze at the default setting, remain equal to one. At two decimals, 30 of them tie at their minimum. $K = \infty$ gives the largest $\widehat{m}_c/c_{\max}$, or ties for it, in all 48 series. That of V-C varies by at most 0.025, and its mean mission time by at most 0.2% of its smallest value over K.

Deployment time For V-U, V-Δ, and V-L with $L_\mu > 0$, the drift term limits deployment to a finite time after the test starts. Fix a certificate (c, r) with starting time $\sigma = \sigma_r$ and monitoring period $[\sigma, \tau)$, and a decision time $t \geq \tau$. For $\mathrm{V} \in \{\mathrm{U}, \Delta, \mathrm{L}\}$, the certificate bound $d_t^{\mathrm{V}}(c; \sigma, \tau)$ defined in (A.47) of Appendix A.D-B equals 1 if a required check fails, and otherwise the minimum of 1 and the maximum over $\sigma \leq s < \tau$ in (A.69), (A.70) or (A.71). Each term of this maximum is the drift term Δ_t defined in (A.46c) of Appendix A.D-A, taken as $\min\{1, L_\mu(t-s)\}$, plus nonnegative terms. The term at $s = \sigma$ therefore gives

$$d_t^{\mathrm{V}}(c; \sigma, \tau) \geq \min\{1, L_\mu(t-\sigma)\}, \qquad \mathrm{V} \in \{\mathrm{U}, \Delta, \mathrm{L}\}. \tag{2.1}$$

The certificate supports deployment at t in (A.25) or (A.31) only if its current risk-increase bound $d_t^{(r)}(c)$ in (A.30), which equals $d_t^{\mathrm{V}}(c; \sigma, \tau)$, is at most $\alpha_0 - \alpha$. Since $\alpha_0 - \alpha < 1$, (2.1) shows that this holds only if

$$L_\mu(t-\sigma) \leq \alpha_0 - \alpha. \tag{2.2}$$

New tests start every K time steps. A smaller K therefore provides more opportunities to issue certificates with recent starting times. This does not guarantee certification or deployment. For V-C, the term at $s = \sigma$ in (A.72) is $(1 - \Delta_t)B_{t,\sigma}(c) + \Delta_t \bar{P}(c) \geq \Delta_t \bar{P}(c)$, since $B_{t,\sigma}(c) \geq 0$ and $\Delta_t \leq 1$. Hence

$$d_t^{\mathrm{C}}(c; \sigma, \tau) \geq \bar{P}(c) \min\{1, L_\mu(t-\sigma)\}, \tag{2.3}$$

with the fraction bound $\bar{P}(c) \leq 1/2$ in (A.54). The right-hand side of (2.3) is at most $\alpha_0 - \alpha$ whenever $L_\mu(t-\sigma) \leq 2(\alpha_0 - \alpha)$. Hence (2.3) does not exclude the deployment of a V-C certificate for $t - \sigma$ up to twice the value allowed by (2.2) when $L_\mu > 0$. If $\bar{P}(c) \leq \alpha_0 - \alpha$, this lower bound does not limit $t - \sigma$. Deployment still requires the certificate bound $d_t^{\mathrm{C}}(c; \sigma, \tau)$ to be at most $\alpha_0 - \alpha$.

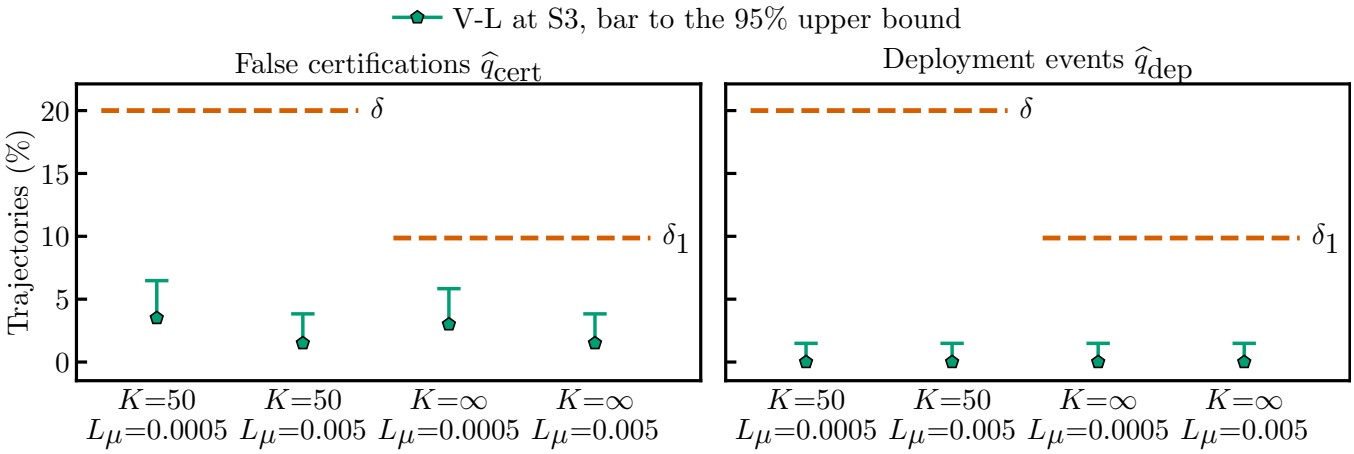


Fig. 42: Corridor, S3 with V-L: $\widehat{q}_{\mathrm{cert}}$ and $\widehat{q}_{\mathrm{dep}}$ (A.78) with upper confidence bounds (A.79), with the bounds δ and δ_1 of Theorems A.1–A.3 (Section A.V-B).

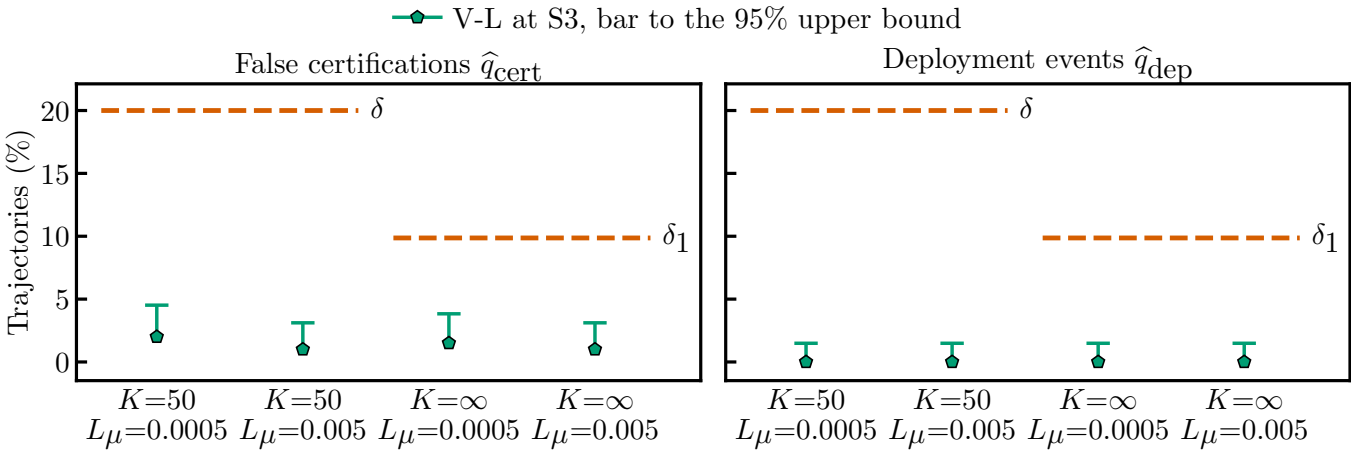


Fig. 43: Maze, as in Fig. 42.

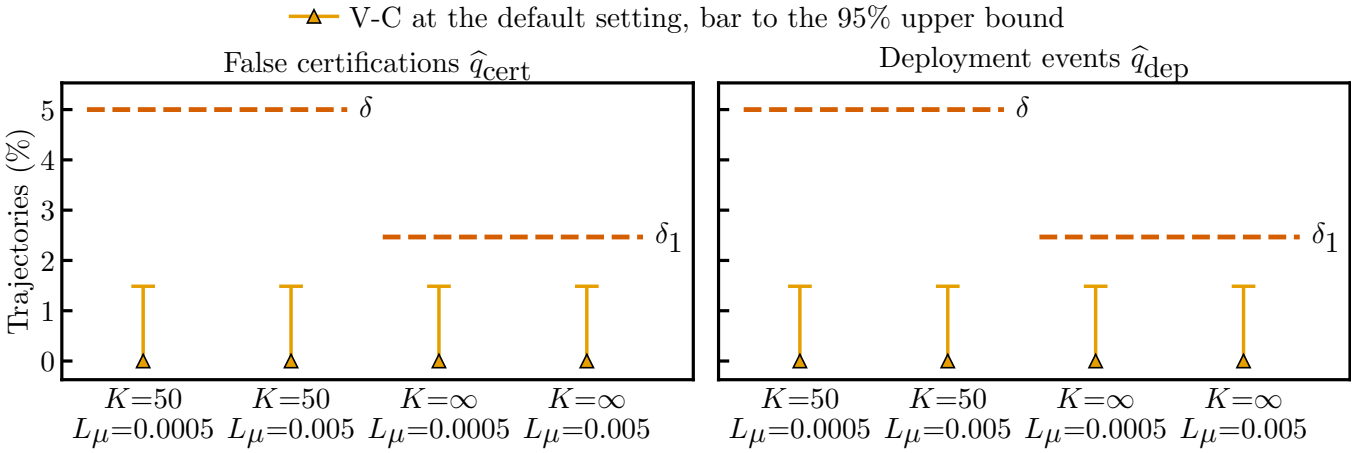


Fig. 44: Default setting with V-C, identical in the Corridor and the Maze (no event in either scenario), as in Fig. 42.

Table 11: Corridor, test period K, default setting and S1–S5.

Corridor Baselines $\alpha_0 = 0.15,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17 ± 0.04 0.00 ± 0.00	30.69 ± 13.51 N-A 22.32 ± 10.04	N-A N-A N-A	-0.27 ± 0.05 -0.02 ± 0.01 70.00 ± 0.00
$c=0$	0.00 ± 0.00 0.17 ± 0.02 0.66 ± 0.07	28.00 ± 2.99 49.65 ± 2.89 15.27 ± 4.17	100.0 ± 0.0 0.00 ± 0.00 78.8 ± 8.6	-0.03 ± 0.01 0.10 ± 0.01 32.45 ± 0.77
$c_{\max}$	1.00 ± 0.00 0.17 ± 0.01 1.00 ± 0.08	0.00 ± 0.00 0.00 ± 0.00 0.00 ± 0.00	0.0 ± 0.0 N-A N-A	0.05 ± 0.01 0.17 ± 0.01 35.91 ± 0.92
ACI	0.49 ± 0.04 0.17 ± 0.02 0.82 ± 0.09	9.02 ± 1.17 15.26 ± 0.53 0.05 ± 0.17	98.2 ± 0.0 0.60 ± 0.00 33.4 ± 8.6	0.01 ± 0.01 0.14 ± 0.01 34.00 ± 0.87
QT	0.52 ± 0.04 0.17 ± 0.02 0.86 ± 0.09	8.34 ± 0.97 14.38 ± 0.32 0.02 ± 0.10	99.0 ± 0.8 0.10 ± 0.00 49.7 ± 8.0	0.01 ± 0.01 0.14 ± 0.01 34.17 ± 0.89

Table 11 (continued).

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\text{m},\ T_s = 0.1\,\text{s},\ N_{\text{MC}} = 200$					
K	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\text{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\text{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\text{nodep}}\downarrow(\%)$ $\widehat{m}_{\text{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$ $\widehat{T}_{\text{mis}}\downarrow(\text{s})$
default setting: V-C $(\alpha_0, \alpha, \delta)$ = (0.15, 0.10, 0.05)					
10	0.83±0.05 0.17±0.01 0.93±0.09	2.09±1.14 3.33±1.59 0.00±0.00	79.2±6.2 6.45±2.36 5.6±2.1	18.3±6.6 2.5±0.7 31.73±8.83	0.03±0.01 0.16±0.01 35.14±0.91
50	0.83±0.05 0.17±0.01 0.93±0.09	2.12±1.18 3.37±1.63 0.00±0.00	79.2±6.2 6.45±2.37 5.7±2.2	18.3±6.6 2.5±0.7 9.39±2.38	0.03±0.01 0.16±0.01 35.14±0.91
100	0.83±0.05 0.17±0.01 0.93±0.09	2.10±1.17 3.36±1.64 0.00±0.00	79.2±6.2 6.46±2.38 5.7±2.1	18.3±6.6 2.5±0.7 5.73±1.36	0.03±0.01 0.16±0.01 35.14±0.91
200	0.83±0.05 0.17±0.01 0.93±0.09	2.09±1.14 3.33±1.58 0.00±0.00	79.2±6.2 6.46±2.38 5.6±2.1	18.3±6.6 2.5±0.7 3.95±0.97	0.03±0.01 0.16±0.01 35.14±0.91
∞	0.83±0.05 0.17±0.01 0.93±0.09	2.08±1.14 3.32±1.59 0.00±0.00	79.2±6.2 6.46±2.38 5.6±2.1	18.3±6.6 2.5±0.7 2.54±0.59	0.03±0.01 0.16±0.01 35.14±0.91

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\text{m},\ T_s = 0.1\,\text{s},\ N_{\text{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\text{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$ $\widehat{T}_{\text{mis}}\downarrow(\text{s})$
Unfiltered	N-A 0.17±0.04 0.00±0.00	30.69±13.51 N-A 22.32±10.04	N-A N-A N-A	−0.27±0.05 −0.02±0.01 70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02 0.66±0.07	28.00±2.99 49.65±2.89 15.27±4.17	100.0±0.0 0.00±0.00 78.8±8.6	−0.03±0.01 0.10±0.01 32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01 1.00±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.05±0.01 0.17±0.01 35.91±0.92
ACI	0.22±0.04 0.17±0.02 0.73±0.08	17.68±1.59 30.21±0.72 2.74±1.43	99.1±0.0 0.30±0.00 57.4±10.2	−0.01±0.01 0.12±0.01 33.13±0.81
QT	0.30±0.03 0.17±0.02 0.78±0.09	15.19±1.31 26.54±1.03 0.61±0.52	99.7±0.1 0.10±0.00 60.1±11.5	−0.00±0.00 0.13±0.01 33.44±0.83

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\text{m},\ T_s = 0.1\,\text{s},\ N_{\text{MC}} = 200$					
K	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\text{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\text{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\text{nodep}}\downarrow(\%)$ $\widehat{m}_{\text{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$ $\widehat{T}_{\text{mis}}\downarrow(\text{s})$
S1: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.20)					
10	0.50±0.06 0.17±0.01 0.80±0.09	9.67±2.09 14.97±2.94 0.18±0.41	92.3±1.1 1.53±0.31 20.8±3.2	4.5±0.9 3.2±0.8 123.83±13.91	0.01±0.01 0.13±0.01 33.88±0.89

Table 11 (continued).

K	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Corridor $\phi=1,\ L_\mu=0.0005,\ c_{\max}=0.02\,\mathrm{m},\ T_s=0.1\,\mathrm{s},\ N_{\mathrm{MC}}=200$					
S1: V-C (α_0,α,δ) = (0.30, 0.25, 0.20)					
50	0.50±0.06 0.17±0.02 0.80±0.09	9.74±2.12 15.10±2.94 0.18±0.41	92.3±1.1 1.53±0.31 21.1±3.3	4.5±0.9 3.2±0.8 30.46±3.36	0.01±0.01 0.13±0.01 33.86±0.89
100	0.50±0.06 0.17±0.01 0.80±0.09	9.61±2.08 14.92±2.83 0.19±0.41	92.3±1.1 1.53±0.31 21.0±3.3	4.5±0.9 3.2±0.8 17.72±1.96	0.00±0.01 0.13±0.01 33.88±0.87
200	0.52±0.06 0.17±0.01 0.81±0.09	9.32±2.05 14.39±2.80 0.17±0.40	92.3±1.1 1.53±0.31 20.6±3.3	4.5±0.9 3.2±0.8 10.63±1.06	0.01±0.01 0.13±0.01 33.91±0.89
∞	0.52±0.06 0.17±0.01 0.81±0.09	9.15±2.05 13.94±2.92 0.15±0.39	92.3±1.1 1.53±0.31 20.3±3.3	4.5±0.9 3.2±0.8 5.98±0.66	0.01±0.01 0.13±0.01 33.93±0.90
S2: V-C (α_0,α,δ) = (0.30, 0.25, 0.15)					
10	0.52±0.06 0.17±0.02 0.81±0.09	9.31±1.97 14.40±2.71 0.14±0.36	91.7±1.2 1.74±0.32 20.1±3.2	5.1±0.9 3.2±0.8 122.03±13.45	0.01±0.01 0.13±0.01 33.93±0.88
50	0.52±0.06 0.17±0.02 0.81±0.09	9.35±1.96 14.52±2.74 0.14±0.36	91.7±1.2 1.74±0.32 20.3±3.3	5.1±0.9 3.2±0.8 29.80±3.27	0.01±0.01 0.13±0.01 33.92±0.88
100	0.52±0.06 0.17±0.02 0.81±0.09	9.23±1.94 14.33±2.65 0.14±0.37	91.7±1.2 1.74±0.32 20.3±3.3	5.1±0.9 3.2±0.8 17.24±1.96	0.01±0.01 0.13±0.01 33.94±0.87
200	0.53±0.06 0.17±0.02 0.81±0.09	8.94±1.93 13.83±2.57 0.12±0.35	91.7±1.2 1.74±0.32 20.0±3.2	5.1±0.9 3.2±0.8 10.41±1.09	0.01±0.01 0.13±0.01 33.96±0.88
∞	0.54±0.06 0.17±0.02 0.81±0.09	8.80±1.93 13.45±2.67 0.11±0.33	91.7±1.2 1.74±0.32 19.6±3.2	5.1±0.9 3.2±0.8 5.86±0.65	0.01±0.01 0.13±0.01 33.98±0.89
S3: V-L (α_0,α,δ) = (0.30, 0.20, 0.20)					
10	0.62±0.06 0.17±0.02 0.85±0.09	6.14±1.63 9.66±2.22 0.03±0.21	90.9±1.7 2.04±0.54 15.0±3.7	5.9±1.6 3.2±0.8 97.10±13.54	0.01±0.01 0.14±0.01 34.32±0.91
50	0.62±0.06 0.17±0.02 0.84±0.09	6.38±1.57 10.03±2.14 0.04±0.21	90.9±1.7 2.04±0.54 15.3±3.6	5.9±1.6 3.2±0.8 24.33±3.09	0.01±0.01 0.14±0.01 34.28±0.90
100	0.63±0.06 0.17±0.02 0.85±0.09	6.18±1.60 9.72±2.17 0.03±0.21	90.9±1.7 2.04±0.54 15.3±3.6	5.9±1.6 3.2±0.8 14.21±1.91	0.01±0.01 0.14±0.01 34.33±0.91

Table 11 (continued).

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
K	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S3: V-L $(\alpha_0, \alpha, \delta)$ = (0.30, 0.20, 0.20)					
200	0.68±0.05 0.17±0.02 0.87±0.09	4.94±1.58 7.95±2.15 0.03±0.21	82.8±2.5 2.04±0.54 15.1±3.7	5.9±1.5 11.3±2.1 8.83±1.14	0.02±0.01 0.15±0.01 34.56±0.95
∞	0.79±0.05 0.17±0.02 0.91±0.09	3.40±1.39 5.29±1.96 0.03±0.20	49.6±2.3 2.04±0.54 14.2±4.0	5.8±1.5 44.6±1.7 5.02±0.73	0.02±0.01 0.15±0.01 34.92±0.98
S4: V-Δ $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.025)					
10	0.83±0.03 0.17±0.01 0.93±0.08	1.90±0.67 3.23±0.74 0.00±0.02	81.2±7.1 2.74±0.47 11.1±3.4	7.8±1.3 11.0±7.0 115.62±12.63	0.03±0.01 0.16±0.01 35.15±0.90
50	0.89±0.02 0.17±0.01 0.96±0.08	1.18±0.59 2.27±0.77 0.00±0.00	47.8±4.0 2.74±0.47 11.7±3.3	7.7±1.3 44.5±3.9 28.37±2.96	0.03±0.01 0.16±0.01 35.43±0.85
100	0.92±0.02 0.17±0.01 0.97±0.08	0.75±0.52 1.44±0.71 0.00±0.00	29.8±3.3 2.74±0.47 11.2±3.4	7.7±1.3 62.5±3.0 16.38±1.82	0.03±0.01 0.17±0.01 35.62±0.88
200	0.95±0.01 0.17±0.01 0.98±0.08	0.43±0.40 0.93±0.62 0.00±0.00	17.7±2.1 2.74±0.47 10.4±3.5	7.6±1.3 74.6±1.8 9.75±0.93	0.04±0.01 0.17±0.01 35.72±0.88
∞	0.97±0.01 0.17±0.01 1.00±0.08	0.13±0.20 0.47±0.42 0.00±0.00	8.5±1.4 2.74±0.47 8.8±4.0	7.6±1.3 83.9±0.8 5.48±0.61	0.05±0.01 0.17±0.01 35.85±0.88
S5: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.10)					
10	0.53±0.06 0.17±0.02 0.81±0.09	8.94±1.88 13.84±2.67 0.12±0.36	91.0±1.4 1.99±0.40 19.6±3.2	5.8±1.2 3.2±0.8 119.71±13.69	0.01±0.01 0.13±0.01 33.97±0.88
50	0.53±0.06 0.17±0.02 0.81±0.09	9.00±1.93 13.98±2.76 0.13±0.37	91.0±1.4 1.99±0.40 19.9±3.3	5.8±1.2 3.2±0.8 29.05±3.19	0.01±0.01 0.13±0.01 33.96±0.88
100	0.53±0.06 0.17±0.02 0.81±0.09	8.90±1.88 13.83±2.58 0.12±0.36	91.0±1.4 1.99±0.40 19.8±3.3	5.8±1.2 3.2±0.8 16.77±1.98	0.01±0.01 0.13±0.01 33.98±0.87
200	0.54±0.06 0.17±0.02 0.81±0.09	8.60±1.87 13.31±2.59 0.11±0.35	91.0±1.4 1.99±0.40 19.5±3.2	5.8±1.2 3.2±0.8 10.25±1.01	0.01±0.01 0.13±0.01 34.00±0.88
∞	0.55±0.06 0.17±0.02 0.82±0.09	8.49±1.87 12.98±2.65 0.10±0.34	91.0±1.4 1.99±0.40 19.2±3.2	5.8±1.2 3.2±0.8 5.78±0.65	0.01±0.01 0.14±0.01 34.01±0.89

Table 12: Maze, as in Table 11.

Maze **Baselines** $\alpha_0 = 0.15,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02 0.00±0.00	55.68±1.58 N-A 53.27±0.95	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02 0.32±0.02	24.93±2.57 49.79±2.97 3.86±2.16	100.0±0.0 0.00±0.00 66.8±12.7	−0.01±0.00 0.21±0.01 48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.22±0.01 54.44±0.32
ACI	0.49±0.05 0.17±0.02 0.50±0.03	7.45±1.47 14.94±1.34 0.01±0.06	97.7±0.0 1.20±0.00 27.6±7.3	0.01±0.00 0.21±0.01 51.16±0.40
QT	0.72±0.02 0.17±0.02 0.61±0.02	2.78±0.88 6.40±0.81 0.00±0.00	99.6±0.1 0.20±0.00 15.3±3.7	0.02±0.00 0.21±0.01 52.52±0.33

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
K	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
default setting: V-C $(\alpha_0, \alpha, \delta)$ = (0.15, 0.10, 0.05)					
10	0.86±0.05 0.17±0.02 0.70±0.04	0.75±0.73 2.34±1.24 0.00±0.00	72.6±8.9 13.27±4.88 4.7±1.9	24.8±9.0 2.6±0.8 20.37±7.43	0.02±0.00 0.21±0.01 53.41±0.45
50	0.86±0.05 0.17±0.02 0.70±0.04	0.77±0.73 2.36±1.25 0.00±0.00	72.6±8.8 13.28±4.84 4.7±1.9	24.8±9.0 2.6±0.7 6.49±1.88	0.02±0.00 0.21±0.01 53.40±0.44
100	0.86±0.05 0.17±0.02 0.70±0.04	0.76±0.71 2.35±1.22 0.00±0.00	72.5±9.0 13.33±4.97 4.7±1.9	24.9±9.2 2.6±0.8 3.95±1.02	0.02±0.00 0.21±0.01 53.41±0.44
200	0.86±0.05 0.17±0.02 0.70±0.04	0.75±0.72 2.33±1.24 0.00±0.00	72.5±9.0 13.33±4.97 4.6±1.9	24.9±9.2 2.6±0.7 2.90±0.98	0.02±0.00 0.21±0.01 53.41±0.44
∞	0.86±0.05 0.17±0.02 0.70±0.04	0.75±0.72 2.33±1.24 0.00±0.00	72.5±9.0 13.33±4.97 4.6±1.8	24.9±9.2 2.6±0.7 2.32±0.59	0.02±0.00 0.21±0.01 53.41±0.44

Table 12 (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02 0.00±0.00	55.68±1.58 N-A 53.27±0.95	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02 0.32±0.02	24.93±2.57 49.79±2.97 3.86±2.16	100.0±0.0 0.00±0.00 66.8±12.7	−0.01±0.00 0.21±0.01 48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.22±0.01 54.44±0.32
ACI	0.22±0.04 0.17±0.02 0.39±0.02	15.84±1.93 29.86±1.85 0.48±0.65	98.8±0.0 0.60±0.00 47.1±8.6	−0.00±0.00 0.21±0.01 49.71±0.32
QT	0.50±0.03 0.17±0.02 0.50±0.02	8.04±1.31 15.68±1.10 0.05±0.17	99.6±0.0 0.20±0.00 33.8±7.6	0.01±0.00 0.21±0.01 51.18±0.32

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
K	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S1: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.20)					
10	0.52±0.07 0.17±0.02 0.51±0.04	7.38±2.26 14.39±3.53 0.01±0.06	91.1±1.4 3.04±0.63 19.4±3.5	5.9±1.2 3.0±0.8 87.29±10.22	0.01±0.00 0.21±0.01 51.26±0.52
50	0.52±0.07 0.17±0.02 0.51±0.04	7.40±2.27 14.52±3.57 0.01±0.06	91.1±1.4 3.04±0.63 19.7±3.5	5.9±1.2 2.9±0.8 22.83±2.54	0.01±0.00 0.21±0.01 51.25±0.52
100	0.52±0.07 0.17±0.02 0.51±0.04	7.42±2.27 14.45±3.56 0.01±0.06	91.1±1.4 3.04±0.63 19.8±3.6	5.9±1.2 3.0±0.8 14.03±1.74	0.01±0.00 0.21±0.01 51.25±0.52
200	0.52±0.08 0.17±0.02 0.51±0.04	7.32±2.30 14.31±3.61 0.01±0.06	91.1±1.4 3.04±0.63 19.4±3.5	5.9±1.2 2.9±0.8 9.35±1.29	0.01±0.00 0.21±0.01 51.27±0.52
∞	0.52±0.08 0.17±0.02 0.51±0.04	7.32±2.30 14.31±3.62 0.01±0.06	91.1±1.4 3.04±0.63 19.4±3.5	5.9±1.2 2.9±0.8 5.77±0.69	0.01±0.00 0.21±0.01 51.27±0.52

Table 12 (continued).

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
K	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S2: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.15)					
10	0.53±0.07 0.17±0.02 0.51±0.04	7.22±2.34 13.83±3.42 0.01±0.07	90.3±1.4 3.45±0.62 19.1±3.6	6.7±1.2 2.9±0.8 86.17±10.05	0.01±0.00 0.21±0.01 51.34±0.54
50	0.53±0.08 0.17±0.02 0.51±0.04	7.28±2.37 13.89±3.46 0.01±0.07	90.3±1.5 3.45±0.62 19.2±3.7	6.7±1.2 2.9±0.8 22.37±2.50	0.01±0.00 0.21±0.01 51.33±0.53
100	0.53±0.08 0.17±0.02 0.51±0.04	7.21±2.33 13.77±3.45 0.01±0.08	90.3±1.5 3.45±0.62 19.3±3.7	6.7±1.2 2.9±0.8 13.73±1.73	0.01±0.00 0.21±0.01 51.33±0.54
200	0.53±0.08 0.17±0.02 0.51±0.04	7.15±2.36 13.69±3.46 0.01±0.07	90.4±1.5 3.45±0.62 19.0±3.6	6.7±1.2 2.9±0.8 9.22±1.25	0.01±0.00 0.21±0.01 51.35±0.53
∞	0.53±0.08 0.17±0.02 0.51±0.04	7.15±2.36 13.68±3.47 0.01±0.07	90.4±1.5 3.45±0.62 18.9±3.6	6.7±1.2 2.9±0.8 5.68±0.72	0.01±0.00 0.21±0.01 51.35±0.53
S3: V-L $(\alpha_0, \alpha, \delta)$ = (0.30, 0.20, 0.20)					
10	0.62±0.07 0.17±0.02 0.56±0.04	4.88±1.86 9.85±2.79 0.00±0.06	89.2±2.0 4.01±0.97 14.2±3.6	7.7±1.9 3.1±0.9 67.48±8.99	0.01±0.00 0.21±0.01 51.87±0.54
50	0.62±0.07 0.17±0.02 0.55±0.04	5.04±1.93 9.98±2.87 0.00±0.06	89.1±2.0 4.01±0.97 14.5±3.8	7.7±1.9 3.2±0.9 18.19±2.35	0.01±0.00 0.21±0.01 51.86±0.54
100	0.63±0.07 0.17±0.02 0.56±0.04	4.83±1.81 9.71±2.76 0.00±0.06	89.1±2.0 4.01±0.97 14.4±3.8	7.7±1.8 3.2±0.9 11.26±1.62	0.01±0.00 0.21±0.01 51.88±0.53
200	0.69±0.07 0.17±0.02 0.58±0.04	4.01±1.75 7.98±2.75 0.00±0.06	78.8±2.6 4.01±0.97 14.0±3.7	7.7±1.8 13.5±2.1 7.68±1.19	0.01±0.00 0.21±0.01 52.20±0.52
∞	0.72±0.07 0.17±0.02 0.60±0.04	3.78±1.73 7.36±2.65 0.00±0.06	67.1±2.1 4.01±0.97 13.8±3.7	7.7±1.8 25.2±1.0 4.86±0.77	0.01±0.00 0.21±0.01 52.35±0.53

Table 12 (continued).

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
K	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S4: V-Δ $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.025)					
10	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.04±0.13 0.00±0.00	1.3±0.8 5.00±0.00 3.6±3.2	10.0±1.6 88.7±1.0 83.68±11.09	0.03±0.00 0.22±0.01 54.44±0.32
50	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.04 0.05±0.15 0.00±0.00	1.7±0.9 7.09±4.38 3.9±3.1	10.0±1.6 88.3±1.0 21.00±2.61	0.03±0.00 0.21±0.01 54.46±0.31
100	1.00±0.00 0.17±0.01 0.81±0.02	0.00±0.00 0.05±0.13 0.00±0.00	1.4±0.8 7.85±7.25 3.9±3.1	10.0±1.6 88.5±1.0 12.86±1.58	0.03±0.00 0.22±0.01 54.43±0.32
200	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.03 0.04±0.13 0.00±0.00	1.5±0.8 10.33±13.87 3.9±3.3	10.0±1.6 88.5±1.0 8.55±1.22	0.03±0.00 0.22±0.01 54.44±0.32
∞	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.04±0.13 0.00±0.00	1.3±0.8 5.00±0.00 3.6±3.2	10.0±1.6 88.7±1.0 5.29±0.59	0.03±0.00 0.22±0.01 54.44±0.32
S5: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.10)					
10	0.55±0.07 0.17±0.02 0.52±0.04	6.76±2.17 13.03±3.13 0.01±0.09	89.3±1.7 3.93±0.77 18.5±3.3	7.6±1.5 3.0±0.8 83.58±9.71	0.01±0.00 0.21±0.01 51.43±0.52
50	0.55±0.07 0.17±0.02 0.52±0.04	6.78±2.18 13.05±3.18 0.01±0.09	89.3±1.6 3.93±0.77 18.7±3.5	7.6±1.5 3.1±0.8 21.78±2.47	0.01±0.00 0.21±0.01 51.44±0.54
100	0.55±0.07 0.17±0.02 0.52±0.04	6.74±2.17 12.94±3.20 0.01±0.09	89.3±1.6 3.93±0.77 18.6±3.4	7.6±1.5 3.0±0.8 13.27±1.57	0.01±0.00 0.21±0.01 51.46±0.54
200	0.56±0.07 0.17±0.02 0.52±0.04	6.71±2.18 12.88±3.20 0.01±0.09	89.3±1.6 3.93±0.77 18.4±3.4	7.6±1.5 3.0±0.8 8.82±1.15	0.01±0.00 0.21±0.01 51.47±0.54
∞	0.56±0.07 0.17±0.02 0.52±0.04	6.71±2.18 12.88±3.20 0.01±0.09	89.3±1.6 3.93±0.77 18.4±3.4	7.6±1.5 3.0±0.8 5.58±0.67	0.01±0.00 0.21±0.01 51.47±0.54

2.6 Budget Normalizer

Tables 13 and 14 vary the budget normalizer b at the default setting and at S1–S5. In all 48 series, $\widehat{m}_c/c_{\max}$ is nondecreasing in b. From $b = \kappa$ to $b = 4$, it increases in 46, by at most 0.035, and the mean mission time increases in 44. No case has a deployment event.

Table 13: Corridor, budget normalizer b, default setting and S1–S5.

Corridor **Baselines** $\alpha_0 = 0.15,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04 0.00±0.00	30.69±13.51 N-A 22.32±10.04	N-A N-A N-A	−0.27±0.05 −0.02±0.01 70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02 0.66±0.07	28.00±2.99 49.65±2.89 15.27±4.17	100.0±0.0 0.00±0.00 78.8±8.6	−0.03±0.01 0.10±0.01 32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01 1.00±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.05±0.01 0.17±0.01 35.91±0.92
ACI	0.49±0.04 0.17±0.02 0.82±0.09	9.02±1.17 15.26±0.53 0.05±0.17	98.2±0.0 0.60±0.00 33.4±8.6	0.01±0.01 0.14±0.01 34.00±0.87
QT	0.52±0.04 0.17±0.02 0.86±0.09	8.34±0.97 14.38±0.32 0.02±0.10	99.0±0.8 0.10±0.00 49.7±8.0	0.01±0.01 0.14±0.01 34.17±0.89

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
b	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
default setting: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.15, 0.10, 0.05, 50)					
κ	0.83±0.05 0.17±0.01 0.93±0.09	2.12±1.18 3.37±1.63 0.00±0.00	79.2±6.2 6.45±2.37 5.7±2.2	18.3±6.6 2.5±0.7 9.39±2.38	0.03±0.01 0.16±0.01 35.14±0.91
2.5	0.83±0.05 0.17±0.01 0.93±0.09	2.08±1.16 3.28±1.58 0.00±0.00	78.6±6.3 6.68±2.38 5.6±2.2	18.9±6.6 2.4±0.7 9.17±2.27	0.03±0.01 0.16±0.01 35.16±0.92
3	0.83±0.05 0.17±0.01 0.93±0.09	2.00±1.11 3.16±1.52 0.00±0.00	77.8±6.5 6.99±2.48 5.5±2.1	19.8±6.9 2.4±0.7 9.03±2.20	0.03±0.01 0.16±0.01 35.17±0.91
4	0.84±0.05 0.17±0.01 0.93±0.08	1.88±1.07 2.98±1.42 0.00±0.00	76.7±6.7 7.44±2.50 5.3±2.0	21.0±6.9 2.3±0.7 8.67±2.13	0.03±0.01 0.16±0.01 35.20±0.91

Table 13 (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04 0.00±0.00	30.69±13.51 N-A 22.32±10.04	N-A N-A N-A	−0.27±0.05 −0.02±0.01 70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02 0.66±0.07	28.00±2.99 49.65±2.89 15.27±4.17	100.0±0.0 0.00±0.00 78.8±8.6	−0.03±0.01 0.10±0.01 32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01 1.00±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.05±0.01 0.17±0.01 35.91±0.92
ACI	0.22±0.04 0.17±0.02 0.73±0.08	17.68±1.59 30.21±0.72 2.74±1.43	99.1±0.0 0.30±0.00 57.4±10.2	−0.01±0.01 0.12±0.01 33.13±0.81
QT	0.30±0.03 0.17±0.02 0.78±0.09	15.19±1.31 26.54±1.03 0.61±0.52	99.7±0.1 0.10±0.00 60.1±11.5	−0.00±0.00 0.13±0.01 33.44±0.83

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
b	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S1: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.20, 50)					
κ	0.50±0.06 0.17±0.02 0.80±0.09	9.74±2.12 15.10±2.94 0.18±0.41	92.3±1.1 1.53±0.31 21.1±3.3	4.5±0.9 3.2±0.8 30.46±3.36	0.01±0.01 0.13±0.01 33.86±0.89
2.5	0.51±0.06 0.17±0.02 0.80±0.09	9.54±2.07 14.79±2.89 0.14±0.35	92.0±1.2 1.64±0.33 20.7±3.3	4.8±0.9 3.2±0.8 30.18±3.34	0.01±0.01 0.13±0.01 33.89±0.88
3	0.52±0.06 0.17±0.02 0.81±0.09	9.31±1.93 14.46±2.73 0.14±0.36	91.7±1.2 1.75±0.35 20.3±3.2	5.1±1.0 3.2±0.8 29.67±3.21	0.01±0.01 0.13±0.01 33.93±0.89
4	0.53±0.06 0.17±0.02 0.81±0.09	9.02±1.91 14.01±2.73 0.13±0.37	91.0±1.3 1.98±0.40 19.9±3.2	5.8±1.1 3.2±0.8 29.12±3.23	0.01±0.01 0.13±0.01 33.95±0.88
S2: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.15, 50)					
κ	0.52±0.06 0.17±0.02 0.81±0.09	9.35±1.96 14.52±2.74 0.14±0.36	91.7±1.2 1.74±0.32 20.3±3.3	5.1±0.9 3.2±0.8 29.80±3.27	0.01±0.01 0.13±0.01 33.92±0.88
2.5	0.52±0.06 0.17±0.02 0.81±0.09	9.22±1.92 14.27±2.72 0.13±0.35	91.4±1.3 1.85±0.37 20.2±3.2	5.4±1.1 3.1±0.8 29.44±3.17	0.01±0.01 0.13±0.01 33.94±0.88
3	0.53±0.06 0.17±0.02 0.81±0.09	9.02±1.91 14.01±2.73 0.13±0.37	91.0±1.3 1.98±0.40 19.9±3.2	5.8±1.1 3.2±0.8 29.12±3.23	0.01±0.01 0.13±0.01 33.95±0.88
4	0.54±0.06 0.17±0.02 0.81±0.09	8.72±1.83 13.47±2.63 0.11±0.35	90.7±1.5 2.11±0.45 19.6±3.1	6.2±1.3 3.1±0.7 28.67±3.28	0.01±0.01 0.13±0.01 34.00±0.88

Table 13 (continued).

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
b	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S3: V-L $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.20, 0.20, 50)					
κ	0.62±0.06 0.17±0.02 0.84±0.09	6.38±1.57 10.03±2.14 0.04±0.21	90.9±1.7 2.04±0.54 15.3±3.6	5.9±1.6 3.2±0.8 24.33±3.09	0.01±0.01 0.14±0.01 34.28±0.90
2.5	0.62±0.06 0.17±0.02 0.85±0.09	6.15±1.58 9.70±2.09 0.03±0.21	90.6±1.7 2.16±0.56 15.1±3.5	6.3±1.6 3.2±0.8 24.02±3.18	0.01±0.01 0.14±0.01 34.32±0.91
3	0.63±0.06 0.17±0.02 0.85±0.09	5.98±1.57 9.42±2.06 0.03±0.19	90.2±1.7 2.27±0.58 14.8±3.5	6.6±1.7 3.2±0.8 23.74±3.07	0.01±0.01 0.14±0.01 34.34±0.91
4	0.64±0.05 0.17±0.02 0.85±0.09	5.76±1.47 9.06±1.94 0.02±0.12	89.7±1.8 2.48±0.60 14.4±3.2	7.2±1.7 3.2±0.7 23.44±3.13	0.02±0.01 0.14±0.01 34.37±0.90
S4: V-Δ $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.025, 100)					
κ	0.92±0.02 0.17±0.01 0.97±0.08	0.75±0.52 1.44±0.71 0.00±0.00	29.8±3.3 2.74±0.47 11.2±3.4	7.7±1.3 62.5±3.0 16.38±1.82	0.03±0.01 0.17±0.01 35.62±0.88
2.5	0.93±0.02 0.17±0.01 0.98±0.08	0.69±0.49 1.35±0.69 0.00±0.00	29.2±3.3 2.77±0.51 10.7±3.3	7.7±1.4 63.1±3.0 16.26±1.74	0.04±0.01 0.17±0.01 35.63±0.89
3	0.93±0.01 0.17±0.01 0.98±0.08	0.62±0.47 1.23±0.65 0.00±0.00	28.3±3.3 2.88±0.52 10.3±3.2	8.0±1.4 63.7±3.0 16.08±1.81	0.04±0.01 0.17±0.01 35.64±0.89
4	0.94±0.01 0.17±0.01 0.98±0.08	0.56±0.42 1.10±0.57 0.00±0.00	26.7±3.3 3.06±0.49 9.9±3.1	8.5±1.3 64.7±3.0 15.82±1.72	0.04±0.01 0.17±0.01 35.67±0.89
S5: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.10, 10)					
κ	0.53±0.06 0.17±0.02 0.81±0.09	8.94±1.88 13.84±2.67 0.12±0.36	91.0±1.4 1.99±0.40 19.6±3.2	5.8±1.2 3.2±0.8 119.71±13.69	0.01±0.01 0.13±0.01 33.97±0.88
2.5	0.54±0.06 0.17±0.02 0.81±0.09	8.78±1.88 13.59±2.63 0.11±0.34	90.7±1.4 2.11±0.43 19.5±3.0	6.1±1.2 3.1±0.8 118.62±13.67	0.01±0.01 0.13±0.01 33.99±0.88
3	0.54±0.06 0.17±0.02 0.82±0.09	8.61±1.83 13.33±2.53 0.10±0.32	90.4±1.5 2.22±0.45 19.3±3.0	6.5±1.3 3.1±0.7 117.37±13.53	0.01±0.01 0.13±0.01 34.01±0.89
4	0.55±0.05 0.17±0.02 0.82±0.09	8.35±1.83 12.92±2.47 0.08±0.26	90.1±1.5 2.31±0.46 18.9±3.0	6.8±1.3 3.1±0.7 115.90±14.00	0.01±0.01 0.14±0.01 34.04±0.91

Table 14: Maze, as in Table 13.

Maze **Baselines** $\alpha_0 = 0.15,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02 0.00±0.00	55.68±1.58 N-A 53.27±0.95	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02 0.32±0.02	24.93±2.57 49.79±2.97 3.86±2.16	100.0±0.0 0.00±0.00 66.8±12.7	−0.01±0.00 0.21±0.01 48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.22±0.01 54.44±0.32
ACI	0.49±0.05 0.17±0.02 0.50±0.03	7.45±1.47 14.94±1.34 0.01±0.06	97.7±0.0 1.20±0.00 27.6±7.3	0.01±0.00 0.21±0.01 51.16±0.40
QT	0.72±0.02 0.17±0.02 0.61±0.02	2.78±0.88 6.40±0.81 0.00±0.00	99.6±0.1 0.20±0.00 15.3±3.7	0.02±0.00 0.21±0.01 52.52±0.33

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
b	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
default setting: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.15, 0.10, 0.05, 50)					
κ	0.86±0.05 0.17±0.02 0.70±0.04	0.77±0.73 2.36±1.25 0.00±0.00	72.6±8.8 13.28±4.84 4.7±1.9	24.8±9.0 2.6±0.7 6.49±1.88	0.02±0.00 0.21±0.01 53.40±0.44
2.5	0.86±0.05 0.17±0.02 0.70±0.04	0.72±0.65 2.29±1.26 0.00±0.00	71.6±9.3 13.90±5.13 4.5±1.8	26.0±9.5 2.5±0.7 6.36±1.99	0.02±0.00 0.21±0.01 53.44±0.47
3	0.87±0.05 0.17±0.02 0.71±0.04	0.72±0.67 2.20±1.27 0.00±0.00	70.3±9.8 14.57±5.42 4.5±1.8	27.2±10.0 2.4±0.7 6.18±1.93	0.02±0.00 0.21±0.01 53.45±0.45
4	0.87±0.04 0.17±0.02 0.71±0.04	0.70±0.64 2.09±1.17 0.00±0.00	68.4±10.1 15.60±5.68 4.3±1.8	29.1±10.5 2.5±0.8 5.95±1.93	0.02±0.00 0.21±0.01 53.53±0.47

Table 14 (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\text{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$ $\widehat{T}_{\text{mis}}\downarrow(\text{s})$
Unfiltered	N-A 0.17±0.02 0.00±0.00	55.68±1.58 N-A 53.27±0.95	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02 0.32±0.02	24.93±2.57 49.79±2.97 3.86±2.16	100.0±0.0 0.00±0.00 66.8±12.7	−0.01±0.00 0.21±0.01 48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.22±0.01 54.44±0.32
ACI	0.22±0.04 0.17±0.02 0.39±0.02	15.84±1.93 29.86±1.85 0.48±0.65	98.8±0.0 0.60±0.00 47.1±8.6	−0.00±0.00 0.21±0.01 49.71±0.32
QT	0.50±0.03 0.17±0.02 0.50±0.02	8.04±1.31 15.68±1.10 0.05±0.17	99.6±0.0 0.20±0.00 33.8±7.6	0.01±0.00 0.21±0.01 51.18±0.32

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$					
b	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\text{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\text{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\text{nodep}}\downarrow(\%)$ $\widehat{m}_{\text{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$ $\widehat{T}_{\text{mis}}\downarrow(\text{s})$
S1: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.20, 50)					
κ	0.52±0.07 0.17±0.02 0.51±0.04	7.40±2.27 14.52±3.57 0.01±0.06	91.1±1.4 3.04±0.63 19.7±3.5	5.9±1.2 2.9±0.8 22.83±2.54	0.01±0.00 0.21±0.01 51.25±0.52
2.5	0.53±0.07 0.17±0.02 0.51±0.04	7.32±2.32 14.12±3.48 0.01±0.08	90.6±1.4 3.25±0.64 19.3±3.6	6.3±1.2 3.0±0.8 22.49±2.47	0.01±0.00 0.21±0.01 51.31±0.50
3	0.53±0.08 0.17±0.02 0.51±0.04	7.18±2.36 13.81±3.46 0.01±0.08	90.3±1.5 3.46±0.64 19.1±3.7	6.7±1.2 3.0±0.8 22.28±2.52	0.01±0.00 0.21±0.01 51.34±0.53
4	0.55±0.07 0.17±0.02 0.52±0.04	6.75±2.20 13.07±3.19 0.01±0.09	89.3±1.6 3.92±0.75 18.5±3.5	7.6±1.4 3.1±0.9 21.86±2.45	0.01±0.00 0.21±0.01 51.44±0.51
S2: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.15, 50)					
κ	0.53±0.08 0.17±0.02 0.51±0.04	7.28±2.37 13.89±3.46 0.01±0.07	90.3±1.5 3.45±0.62 19.2±3.7	6.7±1.2 2.9±0.8 22.37±2.50	0.01±0.00 0.21±0.01 51.33±0.53
2.5	0.54±0.07 0.17±0.02 0.51±0.04	7.09±2.34 13.57±3.52 0.01±0.08	89.8±1.5 3.67±0.65 18.9±3.6	7.1±1.3 3.0±0.8 22.12±2.39	0.01±0.00 0.21±0.01 51.37±0.51
3	0.55±0.07 0.17±0.02 0.52±0.04	6.75±2.20 13.07±3.19 0.01±0.09	89.3±1.6 3.92±0.75 18.5±3.5	7.6±1.4 3.1±0.9 21.86±2.45	0.01±0.00 0.21±0.01 51.44±0.51
4	0.56±0.07 0.17±0.02 0.52±0.04	6.70±2.05 12.83±2.95 0.01±0.07	89.0±1.7 4.16±0.77 18.3±3.4	8.1±1.5 3.0±0.8 21.39±2.45	0.01±0.00 0.21±0.01 51.44±0.52

Table 14 (continued).

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
b	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S3: V-L $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.20, 0.20, 50)					
κ	0.62±0.07 0.17±0.02 0.55±0.04	5.04±1.93 9.98±2.87 0.00±0.06	89.1±2.0 4.01±0.97 14.5±3.8	7.7±1.9 3.2±0.9 18.19±2.35	0.01±0.00 0.21±0.01 51.86±0.54
2.5	0.63±0.07 0.17±0.02 0.56±0.04	4.85±2.01 9.78±2.80 0.00±0.06	88.7±2.0 4.26±0.99 14.3±3.8	8.2±1.9 3.2±0.8 18.08±2.32	0.01±0.00 0.21±0.01 51.88±0.53
3	0.64±0.07 0.17±0.02 0.56±0.04	4.63±1.89 9.39±2.69 0.01±0.06	88.3±2.1 4.50±1.01 14.1±3.6	8.6±2.0 3.1±0.8 17.83±2.29	0.01±0.00 0.21±0.01 51.91±0.53
4	0.65±0.06 0.17±0.02 0.57±0.04	4.31±1.74 9.00±2.54 0.00±0.06	87.5±2.2 4.93±1.09 13.5±3.5	9.4±2.0 3.0±0.8 17.46±2.15	0.01±0.00 0.21±0.01 52.01±0.50
S4: V-Δ $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.025, 100)					
κ	1.00±0.00 0.17±0.01 0.81±0.02	0.00±0.00 0.05±0.13 0.00±0.00	1.4±0.8 7.85±7.25 3.9±3.1	10.0±1.6 88.5±1.0 12.86±1.58	0.03±0.00 0.22±0.01 54.43±0.32
2.5	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.04±0.13 0.00±0.00	1.3±0.8 5.00±0.00 3.6±3.2	10.1±1.7 88.6±1.1 12.75±1.59	0.03±0.00 0.22±0.01 54.43±0.32
3	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.03±0.11 0.00±0.00	1.0±0.6 5.20±0.00 3.4±3.0	10.5±1.7 88.5±1.2 12.61±1.61	0.03±0.00 0.22±0.01 54.43±0.32
4	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.01±0.07 0.00±0.00	0.5±0.3 5.60±0.00 3.1±2.7	11.2±1.7 88.4±1.5 12.38±1.63	0.03±0.00 0.22±0.01 54.44±0.32
S5: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.10, 10)					
κ	0.55±0.07 0.17±0.02 0.52±0.04	6.76±2.17 13.03±3.13 0.01±0.09	89.3±1.7 3.93±0.77 18.5±3.3	7.6±1.5 3.0±0.8 83.58±9.71	0.01±0.00 0.21±0.01 51.43±0.52
2.5	0.56±0.07 0.17±0.02 0.52±0.04	6.66±2.09 12.78±2.95 0.01±0.07	88.9±1.7 4.16±0.77 18.3±3.4	8.0±1.5 3.0±0.8 82.98±9.69	0.01±0.00 0.21±0.01 51.47±0.51
3	0.57±0.06 0.17±0.02 0.53±0.04	6.40±1.95 12.53±2.83 0.01±0.07	88.5±1.6 4.39±0.77 18.0±3.1	8.4±1.5 3.0±0.8 81.93±9.55	0.01±0.00 0.21±0.01 51.49±0.50
4	0.58±0.06 0.17±0.02 0.53±0.04	6.27±1.92 12.11±2.75 0.01±0.06	88.2±1.7 4.56±0.79 17.6±3.3	8.8±1.5 3.0±0.7 80.99±9.01	0.01±0.00 0.21±0.01 51.57±0.49

2.7 Bound Factor and Distribution-Drift Bound

Tables 15 and 16 vary the bound factor ϕ (Appendix A.F-B) and the distribution-drift bound L_μ (A.46a) for the default setting with V-C and S1–S5 with their variants. Values $\phi < 1$ need not satisfy Assumption A.2 and carry no guarantee. The ϕ columns of the sweeps (Section 2.2.3) show all four variants. Of the 96 combinations of a setting (default setting or S1–S5), a variant, a scenario, and a value of L_μ, an excess $e_{\mathrm{A2},t}$ (A.83) above $\varepsilon_{\mathrm{chk}}$ occurs in 16, 22, 52, and 96 at $\phi = 0.75$, 0.5, 0.25, and 0, and a deployment event in 3, 6, 15, and 92. At $\phi = 1$, neither occurs. Reducing ϕ from one to zero lowers $\widehat{m}_c/c_{\max}$ in all 96 series, raises the CBF-violation fraction $\widehat{q}^{+}_{\mathrm{CBF}}$ in 95, and shortens the mean mission time in 93, with at most 106/200 (59.0%) trajectories with a deployment event per case.

With $\phi = 1$ and $L_\mu = 0.005$, V-U, V-Δ, and V-L stay at $\widehat{m}_c/c_{\max} \geq 0.998$, while at the selected settings that of V-C is 0.81–0.96 and its mean mission time, paired by initial state and disturbance realization, is 0.4–2.5% below that of the $c_{\max}$ baseline. Tables 17 and 18 give the event counts and the bound-excess counts of S1–S5 with their variants.

Table 15: Corridor, bound factor ϕ and distribution-drift bound L_μ, default setting and S1–S5.

Corridor Baselines $\alpha_0 = 0.15,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
$L_\mu = 0.0005$				
Unfiltered	N-A 0.17±0.04 0.00±0.00	30.69±13.51 N-A 22.32±10.04	N-A N-A N-A	−0.27±0.05 −0.02±0.01 70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02 0.66±0.07	28.00±2.99 49.65±2.89 15.27±4.17	100.0±0.0 0.00±0.00 78.8±8.6	−0.03±0.01 0.10±0.01 32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01 1.00±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.05±0.01 0.17±0.01 35.91±0.92
ACI	0.49±0.04 0.17±0.02 0.82±0.09	9.02±1.17 15.26±0.53 0.05±0.17	98.2±0.0 0.60±0.00 33.4±8.6	0.01±0.01 0.14±0.01 34.00±0.87
QT	0.52±0.04 0.17±0.02 0.86±0.09	8.34±0.97 14.38±0.32 0.02±0.10	99.0±0.8 0.10±0.00 49.7±8.0	0.01±0.01 0.14±0.01 34.17±0.89
$L_\mu = 0.005$				
Unfiltered	N-A 0.18±0.05 0.00±0.00	30.50±13.38 N-A 22.41±9.99	N-A N-A N-A	−0.27±0.05 −0.02±0.01 70.00±0.00
$c = 0$	0.00±0.00 0.21±0.02 0.67±0.08	28.69±2.97 49.74±2.87 16.43±4.06	100.0±0.0 0.00±0.00 82.7±8.6	−0.03±0.01 0.09±0.01 32.51±0.82
$c_{\max}$	1.00±0.00 0.20±0.01 1.01±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.05±0.01 0.17±0.01 35.95±0.98
ACI	0.59±0.04 0.20±0.02 0.87±0.09	8.44±1.11 14.21±0.59 0.02±0.13	98.3±0.0 0.60±0.00 33.9±8.6	0.01±0.01 0.14±0.01 34.47±0.91
QT	0.59±0.03 0.20±0.02 0.90±0.09	8.36±0.96 14.38±0.24 0.01±0.05	98.2±1.0 0.10±0.00 48.1±7.5	0.01±0.01 0.14±0.01 34.57±0.92

Table 15 (continued).

Corridor $c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
default setting: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.15, 0.10, 0.05, 50), $L_\mu = 0.0005$					
0	0.82±0.06 0.17±0.01 0.93±0.09	2.12±1.18 3.46±1.68 0.00±0.00	81.7±6.6 6.43±2.38 6.7±3.1	18.3±6.6 0.0±0.0 9.39±2.38	0.03±0.01 0.16±0.01 35.14±0.91
0.25	0.83±0.05 0.17±0.01 0.93±0.09	2.12±1.18 3.38±1.64 0.00±0.00	79.2±6.2 6.45±2.37 5.7±2.2	18.3±6.6 2.5±0.7 9.39±2.38	0.03±0.01 0.16±0.01 35.14±0.91
0.5	0.83±0.05 0.17±0.01 0.93±0.09	2.12±1.18 3.38±1.64 0.00±0.00	79.2±6.2 6.45±2.37 5.7±2.2	18.3±6.6 2.5±0.7 9.39±2.38	0.03±0.01 0.16±0.01 35.14±0.91
0.75	0.83±0.05 0.17±0.01 0.93±0.09	2.12±1.18 3.38±1.64 0.00±0.00	79.2±6.2 6.45±2.37 5.7±2.2	18.3±6.6 2.5±0.7 9.39±2.38	0.03±0.01 0.16±0.01 35.14±0.91
1	0.83±0.05 0.17±0.01 0.93±0.09	2.12±1.18 3.37±1.63 0.00±0.00	79.2±6.2 6.45±2.37 5.7±2.2	18.3±6.6 2.5±0.7 9.39±2.38	0.03±0.01 0.16±0.01 35.14±0.91
default setting: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.15, 0.10, 0.05, 50), $L_\mu = 0.005$					
0	0.87±0.05 0.20±0.02 0.95±0.09	1.72±1.17 2.76±1.63 0.00±0.00	79.9±8.6 7.12±3.12 5.6±3.0	20.1±8.6 0.0±0.0 7.53±2.02	0.03±0.01 0.16±0.01 35.38±0.98
0.25	0.90±0.03 0.20±0.02 0.97±0.09	1.19±0.72 1.86±0.98 0.00±0.00	77.4±8.1 7.14±3.11 4.5±2.5	20.0±8.6 2.6±0.9 7.58±2.07	0.04±0.01 0.16±0.01 35.51±0.98
0.5	0.95±0.02 0.20±0.01 0.99±0.09	0.54±0.36 0.86±0.43 0.00±0.00	47.8±17.2 7.34±3.79 3.3±2.0	19.9±8.6 32.3±15.5 7.64±2.02	0.04±0.01 0.16±0.01 35.72±0.98
0.75	0.99±0.01 0.20±0.01 1.00±0.09	0.12±0.18 0.22±0.23 0.00±0.00	14.2±8.0 7.87±5.52 2.4±1.4	19.8±8.5 66.0±7.1 7.64±2.02	0.04±0.01 0.17±0.01 35.88±0.99
1	1.00±0.00 0.20±0.01 1.01±0.09	0.03±0.08 0.05±0.12 0.00±0.00	3.4±2.8 6.42±2.74 1.7±0.1	19.8±8.5 76.9±6.6 7.70±2.09	0.05±0.01 0.17±0.01 35.94±0.98

Table 15 (continued).

Corridor Baselines $\alpha_0 = 0.30,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
$L_\mu = 0.0005$				
Unfiltered	N-A 0.17±0.04 0.00±0.00	30.69±13.51 N-A 22.32±10.04	N-A N-A N-A	−0.27±0.05 −0.02±0.01 70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02 0.66±0.07	28.00±2.99 49.65±2.89 15.27±4.17	100.0±0.0 0.00±0.00 78.8±8.6	−0.03±0.01 0.10±0.01 32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01 1.00±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.05±0.01 0.17±0.01 35.91±0.92
ACI	0.22±0.04 0.17±0.02 0.73±0.08	17.68±1.59 30.21±0.72 2.74±1.43	99.1±0.0 0.30±0.00 57.4±10.2	−0.01±0.01 0.12±0.01 33.13±0.81
QT	0.30±0.03 0.17±0.02 0.78±0.09	15.19±1.31 26.54±1.03 0.61±0.52	99.7±0.1 0.10±0.00 60.1±11.5	−0.00±0.00 0.13±0.01 33.44±0.83
$L_\mu = 0.005$				
Unfiltered	N-A 0.18±0.05 0.00±0.00	30.50±13.38 N-A 22.41±9.99	N-A N-A N-A	−0.27±0.05 −0.02±0.01 70.00±0.00
$c = 0$	0.00±0.00 0.21±0.02 0.67±0.08	28.69±2.97 49.74±2.87 16.43±4.06	100.0±0.0 0.00±0.00 82.7±8.6	−0.03±0.01 0.09±0.01 32.51±0.82
$c_{\max}$	1.00±0.00 0.20±0.01 1.01±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.05±0.01 0.17±0.01 35.95±0.98
ACI	0.33±0.04 0.20±0.02 0.79±0.09	17.07±1.68 29.20±0.64 1.13±0.99	99.1±0.0 0.30±0.00 58.2±11.3	−0.01±0.01 0.12±0.01 33.62±0.87
QT	0.36±0.03 0.20±0.02 0.83±0.09	15.96±1.32 27.34±0.84 0.41±0.45	99.6±0.2 0.10±0.00 63.5±12.0	−0.00±0.00 0.13±0.01 33.78±0.90

Table 15 (continued).

Corridor $c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S1: V-C $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.20, 50)$, $L_\mu = 0.0005$					
0	0.47±0.07 0.17±0.02 0.80±0.09	10.02±2.38 16.14±3.49 0.22±0.50	95.5±0.9 1.51±0.30 29.7±8.8	4.5±0.9 0.0±0.0 30.40±3.39	0.00±0.01 0.13±0.01 33.84±0.89
0.25	0.49±0.07 0.17±0.02 0.80±0.09	10.02±2.38 15.61±3.40 0.21±0.45	92.3±1.1 1.53±0.31 21.6±3.5	4.5±0.9 3.2±0.8 30.38±3.39	0.00±0.01 0.13±0.01 33.84±0.89
0.5	0.49±0.07 0.17±0.02 0.80±0.09	10.02±2.38 15.61±3.40 0.21±0.45	92.3±1.1 1.53±0.31 21.6±3.5	4.5±0.9 3.2±0.8 30.38±3.39	0.00±0.01 0.13±0.01 33.84±0.89
0.75	0.49±0.07 0.17±0.02 0.80±0.09	10.02±2.38 15.60±3.40 0.21±0.45	92.3±1.1 1.53±0.31 21.5±3.5	4.5±0.9 3.2±0.8 30.38±3.39	0.00±0.01 0.13±0.01 33.84±0.89
1	0.50±0.06 0.17±0.02 0.80±0.09	9.74±2.12 15.10±2.94 0.18±0.41	92.3±1.1 1.53±0.31 21.1±3.3	4.5±0.9 3.2±0.8 30.46±3.36	0.01±0.01 0.13±0.01 33.86±0.89
S1: V-C $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.20, 50)$, $L_\mu = 0.005$					
0	0.55±0.09 0.20±0.02 0.84±0.10	10.74±3.52 17.11±5.23 0.21±0.53	95.5±1.0 1.53±0.33 32.2±12.7	4.5±1.0 0.0±0.0 26.61±3.41	0.01±0.01 0.13±0.01 34.19±0.98
0.25	0.66±0.04 0.20±0.02 0.88±0.09	7.23±1.50 11.45±2.03 0.04±0.24	92.1±1.2 1.54±0.34 22.5±5.9	4.4±1.0 3.4±0.9 27.23±3.36	0.01±0.01 0.14±0.01 34.59±0.93
0.5	0.75±0.03 0.20±0.02 0.92±0.09	4.10±0.96 6.78±1.29 0.01±0.15	92.1±1.2 1.54±0.34 18.2±5.0	4.4±1.0 3.5±0.9 27.83±3.42	0.02±0.01 0.15±0.01 34.99±0.93
0.75	0.83±0.02 0.20±0.02 0.95±0.08	2.28±0.85 4.02±1.06 0.01±0.15	91.9±1.4 1.54±0.34 15.4±4.8	4.3±1.0 3.8±1.2 28.33±3.29	0.03±0.01 0.15±0.01 35.28±0.92
1	0.89±0.01 0.20±0.02 0.97±0.08	1.35±0.68 2.36±0.84 0.00±0.02	74.4±3.7 1.54±0.34 12.9±4.7	4.3±0.9 21.3±3.6 28.60±3.39	0.03±0.01 0.16±0.01 35.50±0.96
S2: V-C $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.15, 50)$, $L_\mu = 0.0005$					
0	0.49±0.07 0.17±0.02 0.80±0.09	9.58±2.17 15.50±3.18 0.17±0.44	94.9±0.9 1.72±0.32 28.0±8.4	5.1±0.9 0.0±0.0 29.82±3.24	0.01±0.01 0.13±0.01 33.90±0.88
0.25	0.51±0.07 0.17±0.02 0.80±0.09	9.59±2.17 14.98±3.09 0.16±0.38	91.7±1.2 1.74±0.32 21.0±3.4	5.1±0.9 3.2±0.8 29.80±3.25	0.01±0.01 0.13±0.01 33.90±0.88
0.5	0.51±0.07 0.17±0.02 0.80±0.09	9.59±2.17 14.98±3.09 0.16±0.38	91.7±1.2 1.74±0.32 21.0±3.4	5.1±0.9 3.2±0.8 29.80±3.25	0.01±0.01 0.13±0.01 33.90±0.88
0.75	0.51±0.07 0.17±0.02 0.80±0.09	9.59±2.17 14.97±3.09 0.16±0.38	91.7±1.2 1.74±0.32 20.9±3.4	5.1±0.9 3.2±0.8 29.80±3.25	0.01±0.01 0.13±0.01 33.90±0.88
1	0.52±0.06 0.17±0.02 0.81±0.09	9.35±1.96 14.52±2.74 0.14±0.36	91.7±1.2 1.74±0.32 20.3±3.3	5.1±0.9 3.2±0.8 29.80±3.27	0.01±0.01 0.13±0.01 33.92±0.88

Table 15 (continued).

Corridor $c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S2: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.15, 50), $L_\mu = 0.005$					
0	0.57±0.09 0.20±0.02 0.84±0.10	10.07±3.37 16.03±5.05 0.16±0.47	94.9±1.0 1.74±0.35 29.7±11.8	5.1±1.0 0.0±0.0 26.05±3.40	0.01±0.01 0.13±0.01 34.25±0.97
0.25	0.67±0.04 0.20±0.02 0.88±0.09	6.86±1.45 10.83±1.92 0.03±0.22	91.7±1.3 1.75±0.35 21.2±5.9	5.0±1.0 3.3±0.9 26.68±3.27	0.02±0.01 0.14±0.01 34.62±0.93
0.5	0.76±0.03 0.20±0.02 0.92±0.09	3.86±0.92 6.37±1.27 0.01±0.15	91.6±1.3 1.75±0.35 17.2±4.9	5.0±1.0 3.4±0.9 27.34±3.41	0.02±0.01 0.15±0.01 35.03±0.93
0.75	0.84±0.02 0.20±0.02 0.95±0.08	2.07±0.73 3.70±0.95 0.01±0.15	91.1±1.5 1.75±0.35 14.5±4.7	4.9±1.0 3.9±1.3 27.66±3.34	0.03±0.01 0.16±0.01 35.30±0.93
1	0.90±0.01 0.20±0.02 0.97±0.08	1.20±0.62 2.14±0.80 0.00±0.02	71.5±3.9 1.75±0.35 12.2±4.6	4.9±1.0 23.6±3.8 28.06±3.45	0.03±0.01 0.16±0.01 35.54±0.95
S3: V-L $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.20, 0.20, 50), $L_\mu = 0.0005$					
0	0.57±0.08 0.17±0.02 0.83±0.09	7.44±2.15 12.14±3.27 0.07±0.32	94.1±1.6 2.03±0.55 21.9±7.9	5.9±1.6 0.0±0.0 24.29±3.09	0.01±0.01 0.14±0.01 34.18±0.91
0.25	0.58±0.07 0.17±0.02 0.83±0.09	7.44±2.16 11.73±3.19 0.06±0.27	90.9±1.7 2.04±0.54 16.6±3.8	5.9±1.6 3.2±0.8 24.29±3.09	0.01±0.01 0.14±0.01 34.18±0.91
0.5	0.58±0.07 0.17±0.02 0.83±0.09	7.44±2.16 11.73±3.19 0.06±0.27	90.9±1.7 2.04±0.54 16.6±3.8	5.9±1.6 3.2±0.8 24.29±3.09	0.01±0.01 0.14±0.01 34.18±0.91
0.75	0.59±0.07 0.17±0.02 0.83±0.09	7.15±1.96 11.18±2.78 0.06±0.26	90.9±1.7 2.04±0.54 16.1±3.6	5.9±1.6 3.2±0.8 24.31±3.05	0.01±0.01 0.14±0.01 34.20±0.91
1	0.62±0.06 0.17±0.02 0.84±0.09	6.38±1.57 10.03±2.14 0.04±0.21	90.9±1.7 2.04±0.54 15.3±3.6	5.9±1.6 3.2±0.8 24.33±3.09	0.01±0.01 0.14±0.01 34.28±0.90
S3: V-L $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.20, 0.20, 50), $L_\mu = 0.005$					
0	0.64±0.09 0.20±0.02 0.87±0.10	7.79±3.15 12.33±4.75 0.06±0.31	94.1±1.6 2.05±0.57 22.3±10.0	5.9±1.6 0.0±0.0 20.96±2.99	0.01±0.01 0.14±0.01 34.52±0.99
0.25	0.83±0.03 0.20±0.02 0.95±0.09	2.61±0.88 4.30±1.32 0.01±0.17	76.5±5.6 2.06±0.57 14.4±5.5	5.8±1.6 17.7±5.4 22.21±3.01	0.03±0.01 0.15±0.01 35.31±0.97
0.5	0.97±0.01 0.20±0.01 1.00±0.09	0.36±0.43 0.68±0.61 0.01±0.12	13.0±3.2 2.16±1.16 10.2±5.2	5.7±1.6 81.3±2.7 22.74±3.02	0.04±0.01 0.17±0.01 35.83±0.97
0.75	1.00±0.00 0.20±0.01 1.01±0.08	0.08±0.18 0.11±0.21 0.00±0.00	2.1±1.0 2.49±1.86 5.8±4.9	5.7±1.6 92.2±0.9 22.74±2.99	0.04±0.01 0.17±0.01 35.94±0.99
1	1.00±0.00 0.20±0.01 1.01±0.09	0.02±0.07 0.03±0.09 0.00±0.00	0.6±0.3 1.81±0.04 5.1±4.9	5.7±1.6 93.7±1.3 22.75±2.94	0.05±0.01 0.17±0.01 35.95±0.99

Table 15 (continued).

Corridor $c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S4: V-Δ $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.025, 100), $L_\mu = 0.0005$					
0	0.55±0.06 0.17±0.02 0.83±0.09	8.03±1.85 12.97±2.61 0.06±0.21	92.0±1.3 2.73±0.47 24.5±7.7	8.0±1.3 0.0±0.0 15.14±1.83	0.01±0.01 0.14±0.01 34.11±0.89
0.25	0.58±0.05 0.17±0.02 0.83±0.09	7.51±1.62 11.74±2.11 0.05±0.19	88.9±1.4 2.74±0.47 18.1±3.1	8.0±1.3 3.1±0.7 15.15±1.86	0.01±0.01 0.14±0.01 34.16±0.87
0.5	0.70±0.04 0.17±0.02 0.88±0.09	4.66±0.98 7.10±1.27 0.01±0.05	84.5±2.7 2.74±0.47 15.2±3.2	7.9±1.3 7.7±2.4 15.57±1.90	0.02±0.01 0.15±0.01 34.65±0.90
0.75	0.86±0.02 0.17±0.01 0.94±0.08	1.97±0.82 3.10±1.04 0.00±0.00	48.6±3.0 2.74±0.47 13.1±3.3	7.7±1.3 43.6±2.8 16.09±1.97	0.03±0.01 0.16±0.01 35.30±0.92
1	0.92±0.02 0.17±0.01 0.97±0.08	0.75±0.52 1.44±0.71 0.00±0.00	29.8±3.3 2.74±0.47 11.2±3.4	7.7±1.3 62.5±3.0 16.38±1.82	0.03±0.01 0.17±0.01 35.62±0.88
S4: V-Δ $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.025, 100), $L_\mu = 0.005$					
0	0.64±0.06 0.20±0.02 0.87±0.10	7.78±2.43 12.29±3.57 0.02±0.11	92.0±1.4 2.76±0.50 22.8±9.2	8.0±1.4 0.0±0.0 13.49±2.03	0.01±0.01 0.14±0.01 34.51±0.94
0.25	0.99±0.00 0.20±0.01 1.00±0.09	0.10±0.18 0.18±0.25 0.00±0.00	4.6±1.6 4.02±5.26 6.4±3.8	7.7±1.4 87.7±1.3 15.02±2.07	0.04±0.01 0.17±0.01 35.91±0.97
0.5	1.00±0.00 0.20±0.01 1.01±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	7.7±1.4 92.3±1.4 15.03±2.07	0.05±0.01 0.17±0.01 35.95±0.98
0.75	1.00±0.00 0.20±0.01 1.01±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	7.7±1.4 92.3±1.4 15.03±2.07	0.05±0.01 0.17±0.01 35.95±0.98
1	1.00±0.00 0.20±0.01 1.01±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	7.7±1.4 92.3±1.4 15.03±2.07	0.05±0.01 0.17±0.01 35.95±0.98

Table 15 (continued).

Corridor $c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S5: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.10, 10), $L_\mu = 0.0005$					
0	0.51±0.07 0.17±0.02 0.81±0.09	9.20±2.07 14.89±3.11 0.15±0.44	94.2±1.2 1.97±0.40 27.5±8.6	5.8±1.2 0.0±0.0 119.56±13.82	0.01±0.01 0.13±0.01 33.95±0.88
0.25	0.52±0.07 0.17±0.02 0.81±0.09	9.20±2.08 14.40±3.06 0.14±0.37	91.0±1.4 1.99±0.40 20.4±3.4	5.8±1.2 3.2±0.8 119.63±13.83	0.01±0.01 0.13±0.01 33.95±0.88
0.5	0.52±0.07 0.17±0.02 0.81±0.09	9.20±2.08 14.40±3.06 0.14±0.37	91.0±1.4 1.99±0.40 20.4±3.4	5.8±1.2 3.2±0.8 119.63±13.83	0.01±0.01 0.13±0.01 33.95±0.88
0.75	0.52±0.07 0.17±0.02 0.81±0.09	9.20±2.08 14.39±3.06 0.14±0.37	91.0±1.4 1.99±0.40 20.3±3.3	5.8±1.2 3.2±0.8 119.63±13.83	0.01±0.01 0.13±0.01 33.95±0.88
1	0.53±0.06 0.17±0.02 0.81±0.09	8.94±1.88 13.84±2.67 0.12±0.36	91.0±1.4 1.99±0.40 19.6±3.2	5.8±1.2 3.2±0.8 119.71±13.69	0.01±0.01 0.13±0.01 33.97±0.88
S5: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.10, 10), $L_\mu = 0.005$					
0	0.59±0.08 0.20±0.02 0.85±0.10	9.46±3.08 15.01±4.61 0.11±0.39	94.2±1.2 1.99±0.43 28.4±11.0	5.8±1.2 0.0±0.0 103.12±14.06	0.01±0.01 0.14±0.01 34.33±0.96
0.25	0.69±0.04 0.20±0.02 0.89±0.09	6.34±1.28 9.95±1.77 0.02±0.21	91.0±1.4 2.00±0.43 19.8±5.7	5.7±1.2 3.3±0.9 105.33±13.81	0.02±0.01 0.14±0.01 34.69±0.93
0.5	0.77±0.03 0.20±0.02 0.92±0.09	3.66±0.82 6.07±1.05 0.01±0.15	90.9±1.5 2.00±0.43 15.7±4.8	5.7±1.2 3.4±0.9 107.94±13.88	0.02±0.01 0.15±0.01 35.04±0.94
0.75	0.83±0.02 0.20±0.02 0.94±0.09	2.24±0.71 3.78±0.87 0.01±0.15	90.9±1.4 2.00±0.43 13.0±4.5	5.6±1.2 3.4±0.9 109.88±13.10	0.03±0.01 0.15±0.01 35.28±0.95
1	0.88±0.01 0.20±0.02 0.96±0.09	1.32±0.58 2.27±0.69 0.00±0.02	89.2±3.6 2.00±0.43 10.4±3.9	5.6±1.2 5.2±3.4 111.21±14.56	0.03±0.01 0.16±0.01 35.48±0.96

Table 16: Maze, as in Table 15.

Maze Baselines $\alpha_0 = 0.15,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
$L_\mu = 0.0005$				
Unfiltered	N-A 0.17±0.02 0.00±0.00	55.68±1.58 N-A 53.27±0.95	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02 0.32±0.02	24.93±2.57 49.79±2.97 3.86±2.16	100.0±0.0 0.00±0.00 66.8±12.7	−0.01±0.00 0.21±0.01 48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.22±0.01 54.44±0.32
ACI	0.49±0.05 0.17±0.02 0.50±0.03	7.45±1.47 14.94±1.34 0.01±0.06	97.7±0.0 1.20±0.00 27.6±7.3	0.01±0.00 0.21±0.01 51.16±0.40
QT	0.72±0.02 0.17±0.02 0.61±0.02	2.78±0.88 6.40±0.81 0.00±0.00	99.6±0.1 0.20±0.00 15.3±3.7	0.02±0.00 0.21±0.01 52.52±0.33
$L_\mu = 0.005$				
Unfiltered	N-A 0.22±0.02 0.00±0.00	54.91±1.61 N-A 53.15±1.09	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.17±0.16
$c = 0$	0.00±0.00 0.22±0.02 0.32±0.02	28.79±2.72 49.92±3.15 5.43±2.54	100.0±0.0 0.00±0.00 72.2±14.1	−0.01±0.00 0.20±0.01 48.69±0.28
$c_{\max}$	1.00±0.00 0.21±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.21±0.01 54.50±0.34
ACI	0.59±0.04 0.22±0.02 0.56±0.03	7.22±1.42 15.57±1.34 0.00±0.06	97.7±0.0 1.20±0.00 28.8±7.7	0.01±0.00 0.21±0.01 51.85±0.38
QT	0.75±0.02 0.22±0.02 0.64±0.02	3.11±0.91 7.89±0.74 0.00±0.00	99.6±0.1 0.20±0.00 15.7±3.0	0.02±0.00 0.21±0.01 52.74±0.33

Table 16 (continued).

Maze $c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
default setting: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.15, 0.10, 0.05, 50), $L_\mu = 0.0005$					
0	0.86±0.05 0.17±0.02 0.70±0.04	0.81±0.72 2.48±1.31 0.00±0.00	75.2±9.0 13.27±4.84 5.2±2.3	24.8±9.0 0.0±0.0 6.49±1.94	0.02±0.00 0.21±0.01 53.40±0.45
0.25	0.86±0.05 0.17±0.02 0.70±0.04	0.77±0.73 2.36±1.25 0.00±0.00	72.6±8.8 13.28±4.84 4.7±1.9	24.8±9.0 2.6±0.7 6.49±1.88	0.02±0.00 0.21±0.01 53.40±0.44
0.5	0.86±0.05 0.17±0.02 0.70±0.04	0.77±0.73 2.36±1.25 0.00±0.00	72.6±8.8 13.28±4.84 4.7±1.9	24.8±9.0 2.6±0.7 6.49±1.88	0.02±0.00 0.21±0.01 53.40±0.44
0.75	0.86±0.05 0.17±0.02 0.70±0.04	0.77±0.73 2.36±1.25 0.00±0.00	72.6±8.8 13.28±4.84 4.7±1.9	24.8±9.0 2.6±0.7 6.49±1.88	0.02±0.00 0.21±0.01 53.40±0.44
1	0.86±0.05 0.17±0.02 0.70±0.04	0.77±0.73 2.36±1.25 0.00±0.00	72.6±8.8 13.28±4.84 4.7±1.9	24.8±9.0 2.6±0.7 6.49±1.88	0.02±0.00 0.21±0.01 53.40±0.44
default setting: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.15, 0.10, 0.05, 50), $L_\mu = 0.005$					
0	0.89±0.05 0.22±0.02 0.72±0.04	0.71±0.76 2.64±1.77 0.00±0.00	73.1±11.8 14.47±6.43 5.2±3.0	26.9±11.8 0.0±0.0 4.44±1.32	0.02±0.00 0.21±0.01 53.63±0.50
0.25	0.90±0.04 0.21±0.02 0.73±0.04	0.54±0.58 2.08±1.22 0.00±0.00	70.7±11.6 14.48±6.42 4.3±2.4	26.9±11.8 2.4±0.7 4.43±1.32	0.02±0.00 0.21±0.01 53.76±0.44
0.5	0.95±0.02 0.22±0.02 0.76±0.03	0.19±0.26 1.04±0.61 0.00±0.00	49.6±19.2 14.18±6.40 3.4±2.1	26.8±11.8 23.6±14.3 4.47±1.37	0.02±0.00 0.21±0.01 54.05±0.39
0.75	0.98±0.01 0.22±0.02 0.79±0.02	0.05±0.14 0.31±0.33 0.00±0.00	17.0±10.5 14.79±8.71 2.6±1.5	26.6±11.8 56.4±8.1 4.40±1.31	0.03±0.00 0.21±0.01 54.34±0.35
1	1.00±0.00 0.21±0.02 0.80±0.02	0.01±0.06 0.07±0.17 0.00±0.00	4.8±4.0 12.29±5.13 1.7±0.1	26.6±11.8 68.6±8.8 4.48±1.27	0.03±0.00 0.21±0.01 54.41±0.34

Table 16 (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
$L_\mu = 0.0005$				
Unfiltered	N-A 0.17±0.02 0.00±0.00	55.68±1.58 N-A 53.27±0.95	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02 0.32±0.02	24.93±2.57 49.79±2.97 3.86±2.16	100.0±0.0 0.00±0.00 66.8±12.7	−0.01±0.00 0.21±0.01 48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.22±0.01 54.44±0.32
ACI	0.22±0.04 0.17±0.02 0.39±0.02	15.84±1.93 29.86±1.85 0.48±0.65	98.8±0.0 0.60±0.00 47.1±8.6	−0.00±0.00 0.21±0.01 49.71±0.32
QT	0.50±0.03 0.17±0.02 0.50±0.02	8.04±1.31 15.68±1.10 0.05±0.17	99.6±0.0 0.20±0.00 33.8±7.6	0.01±0.00 0.21±0.01 51.18±0.32
$L_\mu = 0.005$				
Unfiltered	N-A 0.22±0.02 0.00±0.00	54.91±1.61 N-A 53.15±1.09	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.17±0.16
$c = 0$	0.00±0.00 0.22±0.02 0.32±0.02	28.79±2.72 49.92±3.15 5.43±2.54	100.0±0.0 0.00±0.00 72.2±14.1	−0.01±0.00 0.20±0.01 48.69±0.28
$c_{\max}$	1.00±0.00 0.21±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.21±0.01 54.50±0.34
ACI	0.33±0.05 0.22±0.02 0.44±0.03	17.20±1.96 31.05±1.81 0.25±0.45	98.8±0.0 0.60±0.00 47.6±8.1	0.00±0.00 0.21±0.01 50.35±0.39
QT	0.57±0.03 0.22±0.02 0.54±0.02	9.29±1.32 18.33±0.96 0.01±0.09	99.6±0.0 0.20±0.00 34.9±8.7	0.01±0.00 0.21±0.01 51.63±0.30

Table 16 (continued).

Maze $c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S1: V-C $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.20, 50)$, $L_\mu = 0.0005$					
0	0.50±0.08 0.17±0.02 0.50±0.04	7.53±2.28 15.01±3.62 0.01±0.09	94.1±1.2 3.04±0.63 24.1±7.3	5.9±1.2 0.0±0.0 22.68±2.53	0.01±0.00 0.21±0.01 51.21±0.52
0.25	0.52±0.07 0.17±0.02 0.51±0.04	7.40±2.27 14.53±3.57 0.01±0.06	91.1±1.4 3.04±0.63 19.8±3.6	5.9±1.2 2.9±0.8 22.83±2.54	0.01±0.00 0.21±0.01 51.25±0.52
0.5	0.52±0.07 0.17±0.02 0.51±0.04	7.40±2.27 14.53±3.57 0.01±0.06	91.1±1.4 3.04±0.63 19.8±3.6	5.9±1.2 2.9±0.8 22.83±2.54	0.01±0.00 0.21±0.01 51.25±0.52
0.75	0.52±0.07 0.17±0.02 0.51±0.04	7.40±2.27 14.53±3.57 0.01±0.06	91.1±1.4 3.04±0.63 19.8±3.6	5.9±1.2 2.9±0.8 22.83±2.54	0.01±0.00 0.21±0.01 51.25±0.52
1	0.52±0.07 0.17±0.02 0.51±0.04	7.40±2.27 14.52±3.57 0.01±0.06	91.1±1.4 3.04±0.63 19.7±3.5	5.9±1.2 2.9±0.8 22.83±2.54	0.01±0.00 0.21±0.01 51.25±0.52
S1: V-C $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.20, 50)$, $L_\mu = 0.005$					
0	0.56±0.10 0.22±0.02 0.54±0.05	9.24±3.91 18.31±6.00 0.03±0.21	94.1±1.2 3.05±0.63 27.6±9.7	5.9±1.2 0.0±0.0 17.42±1.89	0.01±0.00 0.21±0.01 51.63±0.67
0.25	0.65±0.05 0.22±0.02 0.58±0.03	6.13±1.79 12.92±2.86 0.00±0.03	91.1±1.3 3.05±0.63 22.5±6.1	5.8±1.2 3.0±0.8 17.61±1.98	0.01±0.00 0.21±0.01 52.16±0.43
0.5	0.75±0.03 0.22±0.02 0.64±0.03	2.98±1.02 7.77±1.68 0.00±0.00	91.1±1.4 3.05±0.63 18.3±5.2	5.8±1.2 3.1±0.8 17.79±2.03	0.02±0.00 0.21±0.01 52.82±0.38
0.75	0.82±0.02 0.22±0.02 0.68±0.03	1.52±0.71 4.57±1.32 0.00±0.00	90.9±1.7 3.05±0.63 15.3±5.0	5.7±1.2 3.4±1.3 17.96±1.94	0.02±0.00 0.21±0.01 53.30±0.35
1	0.88±0.01 0.21±0.02 0.72±0.02	0.71±0.57 2.76±0.99 0.00±0.00	77.1±4.4 3.05±0.63 12.9±5.0	5.7±1.2 17.2±4.3 18.03±1.93	0.02±0.00 0.21±0.01 53.68±0.37
S2: V-C $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.15, 50)$, $L_\mu = 0.0005$					
0	0.52±0.08 0.17±0.02 0.51±0.04	7.30±2.38 14.44±3.51 0.01±0.09	93.3±1.2 3.45±0.62 23.3±7.1	6.7±1.2 0.0±0.0 22.33±2.57	0.01±0.00 0.21±0.01 51.31±0.53
0.25	0.53±0.08 0.17±0.02 0.51±0.04	7.28±2.37 13.90±3.47 0.01±0.07	90.3±1.5 3.45±0.62 19.2±3.7	6.7±1.2 2.9±0.8 22.37±2.50	0.01±0.00 0.21±0.01 51.33±0.53
0.5	0.53±0.08 0.17±0.02 0.51±0.04	7.28±2.37 13.90±3.47 0.01±0.07	90.3±1.5 3.45±0.62 19.2±3.7	6.7±1.2 2.9±0.8 22.37±2.50	0.01±0.00 0.21±0.01 51.33±0.53
0.75	0.53±0.08 0.17±0.02 0.51±0.04	7.28±2.37 13.90±3.47 0.01±0.07	90.3±1.5 3.45±0.62 19.2±3.7	6.7±1.2 2.9±0.8 22.37±2.50	0.01±0.00 0.21±0.01 51.33±0.53
1	0.53±0.08 0.17±0.02 0.51±0.04	7.28±2.37 13.89±3.46 0.01±0.07	90.3±1.5 3.45±0.62 19.2±3.7	6.7±1.2 2.9±0.8 22.37±2.50	0.01±0.00 0.21±0.01 51.33±0.53

Table 16 (continued).

Maze $c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S2: V-C $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.15, 50)$, $L_\mu = 0.005$					
0	0.58±0.09 0.22±0.02 0.54±0.05	8.72±3.75 17.40±5.70 0.02±0.13	93.3±1.2 3.46±0.63 27.0±10.1	6.7±1.2 0.0±0.0 16.89±1.92	0.01±0.00 0.21±0.01 51.69±0.63
0.25	0.67±0.05 0.22±0.02 0.58±0.03	5.75±1.81 12.28±2.73 0.00±0.00	90.2±1.3 3.46±0.63 21.4±6.0	6.6±1.2 3.2±0.8 17.24±2.04	0.01±0.00 0.21±0.01 52.24±0.41
0.5	0.76±0.03 0.22±0.02 0.64±0.03	2.80±1.08 7.33±1.51 0.00±0.00	90.4±1.4 3.46±0.63 17.5±5.0	6.5±1.2 3.1±0.8 17.45±1.99	0.02±0.00 0.21±0.01 52.88±0.36
0.75	0.83±0.02 0.22±0.02 0.69±0.02	1.36±0.72 4.20±1.14 0.00±0.00	89.9±1.9 3.46±0.63 14.6±4.8	6.5±1.2 3.6±1.6 17.64±1.92	0.02±0.00 0.21±0.01 53.32±0.36
1	0.89±0.01 0.21±0.02 0.72±0.02	0.65±0.53 2.47±0.90 0.00±0.00	74.4±4.8 3.46±0.63 11.7±4.6	6.4±1.2 19.2±4.7 17.67±1.92	0.02±0.00 0.21±0.01 53.73±0.37
S3: V-L $(\alpha_0, \alpha, \delta, K) = (0.30, 0.20, 0.20, 50)$, $L_\mu = 0.0005$					
0	0.59±0.08 0.17±0.02 0.55±0.04	5.42±2.13 11.07±3.41 0.01±0.08	92.2±1.8 4.01±0.97 18.0±6.3	7.8±1.8 0.0±0.0 18.22±2.34	0.01±0.00 0.21±0.01 51.76±0.57
0.25	0.61±0.08 0.17±0.02 0.55±0.04	5.36±2.18 10.62±3.32 0.00±0.06	89.0±2.0 4.01±0.97 15.2±3.8	7.7±1.8 3.2±0.9 18.23±2.33	0.01±0.00 0.21±0.01 51.79±0.58
0.5	0.61±0.08 0.17±0.02 0.55±0.04	5.36±2.18 10.62±3.32 0.00±0.06	89.0±2.0 4.01±0.97 15.2±3.8	7.7±1.8 3.2±0.9 18.23±2.33	0.01±0.00 0.21±0.01 51.79±0.58
0.75	0.61±0.08 0.17±0.02 0.55±0.04	5.36±2.18 10.62±3.32 0.00±0.06	89.0±2.0 4.01±0.97 15.2±3.8	7.7±1.8 3.2±0.9 18.23±2.33	0.01±0.00 0.21±0.01 51.79±0.58
1	0.62±0.07 0.17±0.02 0.55±0.04	5.04±1.93 9.98±2.87 0.00±0.06	89.1±2.0 4.01±0.97 14.5±3.8	7.7±1.9 3.2±0.9 18.19±2.35	0.01±0.00 0.21±0.01 51.86±0.54
S3: V-L $(\alpha_0, \alpha, \delta, K) = (0.30, 0.20, 0.20, 50)$, $L_\mu = 0.005$					
0	0.66±0.09 0.22±0.02 0.58±0.05	6.07±3.41 12.95±5.24 0.01±0.12	92.3±1.9 4.04±1.01 20.1±8.6	7.7±1.9 0.0±0.0 13.80±1.88	0.01±0.00 0.21±0.01 52.14±0.60
0.25	0.83±0.04 0.22±0.02 0.70±0.03	1.84±1.05 4.93±1.71 0.00±0.00	77.9±6.8 4.04±1.01 14.2±5.6	7.6±1.9 14.5±6.8 14.11±1.81	0.02±0.00 0.21±0.01 53.30±0.40
0.5	0.97±0.01 0.22±0.02 0.79±0.02	0.26±0.31 0.82±0.64 0.00±0.00	15.7±3.5 4.10±1.40 9.7±5.1	7.4±1.9 76.8±3.2 14.34±2.02	0.03±0.00 0.21±0.01 54.26±0.36
0.75	0.99±0.01 0.21±0.02 0.81±0.02	0.00±0.03 0.16±0.27 0.00±0.00	2.7±1.3 5.05±3.85 6.4±5.0	7.4±1.9 89.8±1.0 14.34±2.03	0.03±0.00 0.21±0.01 54.49±0.34
1	1.00±0.00 0.21±0.02 0.81±0.02	0.00±0.00 0.04±0.13 0.00±0.00	0.6±0.3 3.60±0.00 5.4±4.8	7.4±1.9 92.0±1.6 14.36±2.07	0.03±0.00 0.21±0.01 54.50±0.34

Table 16 (continued).

Maze $c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S4: V-Δ $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.025, 100), $L_\mu = 0.0005$					
0	0.59±0.06 0.17±0.02 0.54±0.04	5.82±1.84 11.47±2.58 0.00±0.04	89.5±1.6 5.45±0.84 19.9±6.0	10.5±1.6 0.0±0.0 12.22±1.49	0.01±0.00 0.21±0.01 51.64±0.50
0.25	0.61±0.06 0.17±0.02 0.55±0.04	5.43±1.74 10.61±2.25 0.01±0.05	86.7±1.8 5.45±0.85 16.6±3.3	10.5±1.6 2.8±0.7 12.25±1.43	0.01±0.00 0.21±0.01 51.73±0.45
0.5	0.91±0.02 0.17±0.02 0.73±0.03	0.92±0.69 1.96±0.99 0.00±0.00	35.5±2.8 5.45±0.85 11.1±3.6	10.1±1.6 54.4±2.4 12.73±1.70	0.02±0.00 0.21±0.01 53.70±0.34
0.75	0.98±0.01 0.17±0.02 0.80±0.02	0.14±0.24 0.38±0.37 0.00±0.00	9.6±2.2 5.53±1.66 7.5±3.7	10.0±1.6 80.4±1.9 12.84±1.66	0.03±0.00 0.21±0.01 54.34±0.31
1	1.00±0.00 0.17±0.01 0.81±0.02	0.00±0.00 0.05±0.13 0.00±0.00	1.4±0.8 7.85±7.25 3.9±3.1	10.0±1.6 88.5±1.0 12.86±1.58	0.03±0.00 0.22±0.01 54.43±0.32
S4: V-Δ $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.025, 100), $L_\mu = 0.005$					
0	0.66±0.07 0.22±0.02 0.58±0.04	6.04±2.71 12.95±4.13 0.00±0.06	89.5±1.7 5.50±0.91 21.7±7.6	10.5±1.7 0.0±0.0 9.31±1.17	0.01±0.00 0.21±0.01 52.14±0.53
0.25	1.00±0.00 0.21±0.02 0.81±0.02	0.01±0.04 0.08±0.19 0.00±0.00	2.5±1.2 6.48±4.88 6.1±4.3	10.1±1.7 87.4±0.9 9.83±1.35	0.03±0.00 0.21±0.01 54.50±0.32
0.5	1.00±0.00 0.21±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	10.1±1.7 89.9±1.7 9.86±1.33	0.03±0.00 0.21±0.01 54.50±0.34
0.75	1.00±0.00 0.21±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	10.1±1.7 89.9±1.7 9.86±1.33	0.03±0.00 0.21±0.01 54.50±0.34
1	1.00±0.00 0.21±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	10.1±1.7 89.9±1.7 9.86±1.33	0.03±0.00 0.21±0.01 54.50±0.34

Table 16 (continued).

Maze $c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S5: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.10, 10), $L_\mu = 0.0005$					
0	0.54±0.07 0.17±0.02 0.52±0.04	6.88±2.10 13.56±3.19 0.02±0.11	92.4±1.5 3.93±0.77 22.3±6.6	7.6±1.5 0.0±0.0 83.81±9.69	0.01±0.00 0.21±0.01 51.40±0.49
0.25	0.55±0.07 0.17±0.02 0.52±0.04	6.76±2.17 13.04±3.14 0.01±0.09	89.3±1.7 3.93±0.77 18.5±3.4	7.6±1.5 3.0±0.8 83.58±9.71	0.01±0.00 0.21±0.01 51.43±0.52
0.5	0.55±0.07 0.17±0.02 0.52±0.04	6.76±2.17 13.04±3.14 0.01±0.09	89.3±1.7 3.93±0.77 18.5±3.4	7.6±1.5 3.0±0.8 83.58±9.71	0.01±0.00 0.21±0.01 51.43±0.52
0.75	0.55±0.07 0.17±0.02 0.52±0.04	6.76±2.17 13.04±3.14 0.01±0.09	89.3±1.7 3.93±0.77 18.5±3.4	7.6±1.5 3.0±0.8 83.58±9.71	0.01±0.00 0.21±0.01 51.43±0.52
1	0.55±0.07 0.17±0.02 0.52±0.04	6.76±2.17 13.03±3.13 0.01±0.09	89.3±1.7 3.93±0.77 18.5±3.3	7.6±1.5 3.0±0.8 83.58±9.71	0.01±0.00 0.21±0.01 51.43±0.52
S5: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.10, 10), $L_\mu = 0.005$					
0	0.60±0.08 0.22±0.02 0.55±0.05	7.99±3.50 16.17±5.05 0.02±0.13	92.4±1.5 3.94±0.78 25.1±8.8	7.6±1.5 0.0±0.0 61.57±7.92	0.01±0.00 0.21±0.01 51.80±0.60
0.25	0.69±0.05 0.22±0.02 0.59±0.03	5.25±1.85 11.34±2.51 0.00±0.00	89.3±1.7 3.94±0.78 20.1±5.8	7.5±1.5 3.1±0.8 62.80±7.58	0.01±0.00 0.21±0.01 52.29±0.42
0.5	0.77±0.03 0.21±0.02 0.64±0.03	2.76±1.00 6.96±1.38 0.00±0.00	89.4±1.6 3.94±0.78 15.8±4.7	7.5±1.5 3.2±0.8 64.77±7.77	0.02±0.00 0.21±0.01 52.85±0.36
0.75	0.83±0.02 0.21±0.02 0.69±0.03	1.46±0.62 4.38±1.03 0.00±0.00	89.4±1.7 3.94±0.78 12.9±4.3	7.4±1.5 3.2±1.0 65.62±7.79	0.02±0.00 0.21±0.01 53.26±0.37
1	0.88±0.02 0.22±0.02 0.72±0.02	0.71±0.49 2.76±0.80 0.00±0.00	87.1±5.6 3.94±0.78 10.3±3.9	7.3±1.5 5.5±5.5 65.88±8.11	0.02±0.00 0.21±0.01 53.64±0.31

Table 17: Corridor, S1–S5 with their variants, bound factor ϕ and distribution-drift bound L_μ: events and bound excesses.

Corridor $c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$ †: $d_t^{\mathrm{V}}(c;\sigma,\tau)$ scaled by $\phi < 1$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{A2}}$	$\widehat{q}_{\mathrm{pw}}$
S1: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.20, 50), $L_\mu = 0.0005$					
$0^\dagger$	0.47±0.07	10.02±2.38	91 (51.6)	200 (100.0)	200 (100.0)
$0.25^\dagger$	0.49±0.07	10.02±2.38	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.5^\dagger$	0.49±0.07	10.02±2.38	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.75^\dagger$	0.49±0.07	10.02±2.38	0 (1.5) ✓	32 (20.9)	123 (67.3)
1	0.50±0.06	9.74±2.12	0 (1.5) ✓	0 (1.5)	0 (1.5)
S1: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.20, 50), $L_\mu = 0.005$					
$0^\dagger$	0.55±0.09	10.74±3.52	106 (59.0)	200 (100.0)	200 (100.0)
$0.25^\dagger$	0.66±0.04	7.23±1.50	16 (11.9) ✓	199 (100.0)	200 (100.0)
$0.5^\dagger$	0.75±0.03	4.10±0.96	1 (2.4) ✓	168 (88.2)	168 (88.2)
$0.75^\dagger$	0.83±0.02	2.28±0.85	1 (2.4) ✓	5 (5.2)	5 (5.2)
1	0.89±0.01	1.35±0.68	0 (1.5) ✓	0 (1.5)	0 (1.5)
S2: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.15, 50), $L_\mu = 0.0005$					
$0^\dagger$	0.49±0.07	9.58±2.17	75 (43.5)	200 (100.0)	200 (100.0)
$0.25^\dagger$	0.51±0.07	9.59±2.17	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.5^\dagger$	0.51±0.07	9.59±2.17	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.75^\dagger$	0.51±0.07	9.59±2.17	0 (1.5) ✓	24 (16.5)	116 (63.9)
1	0.52±0.06	9.35±1.96	0 (1.5) ✓	0 (1.5)	0 (1.5)
S2: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.15, 50), $L_\mu = 0.005$					
$0^\dagger$	0.57±0.09	10.07±3.37	85 (48.6)	200 (100.0)	200 (100.0)
$0.25^\dagger$	0.67±0.04	6.86±1.45	10 (8.4) ✓	199 (100.0)	200 (100.0)
$0.5^\dagger$	0.76±0.03	3.86±0.92	1 (2.4) ✓	161 (85.0)	161 (85.0)
$0.75^\dagger$	0.84±0.02	2.07±0.73	0 (1.5) ✓	5 (5.2)	5 (5.2)
1	0.90±0.01	1.20±0.62	0 (1.5) ✓	0 (1.5)	0 (1.5)
S3: V-L $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.20, 0.20, 50), $L_\mu = 0.0005$					
$0^\dagger$	0.57±0.08	7.44±2.15	32 (20.9)	200 (100.0)	200 (100.0)
$0.25^\dagger$	0.58±0.07	7.44±2.16	0 (1.5) ✓	45 (27.9)	54 (32.7)
$0.5^\dagger$	0.58±0.07	7.44±2.16	0 (1.5) ✓	0 (1.5)	0 (1.5)
$0.75^\dagger$	0.59±0.07	7.15±1.96	0 (1.5) ✓	0 (1.5)	0 (1.5)
1	0.62±0.06	6.38±1.57	0 (1.5) ✓	0 (1.5)	0 (1.5)

Table 17 (continued).

Corridor $c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$ †: $d_t^{\mathrm{V}}(c;\sigma,\tau)$ scaled by $\phi < 1$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{A2}}$	$\widehat{q}_{\mathrm{pw}}$
S3: V-L $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.20, 0.20, 50), $L_\mu = 0.005$					
$0^{\dagger}$	0.64±0.09	7.79±3.15	40 (25.3)	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	0.83±0.03	2.61±0.88	3 (3.9) ✓	11 (9.0)	12 (9.6)
$0.5^{\dagger}$	0.97±0.01	0.36±0.43	0 (1.5) ✓	0 (1.5)	0 (1.5)
$0.75^{\dagger}$	1.00±0.00	0.08±0.18	0 (1.5) ✓	0/181 (1.5)	0/181 (1.5)
1	1.00±0.00	0.02±0.07	0 (1.5) ✓	0/158 (1.5)	0/158 (1.5)
S4: V-Δ $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.025, 100), $L_\mu = 0.0005$					
$0^{\dagger}$	0.55±0.06	8.03±1.85	43 (26.9)	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	0.58±0.05	7.51±1.62	0 (1.5) ✓	0 (1.5)	0 (1.5)
$0.5^{\dagger}$	0.70±0.04	4.66±0.98	0 (1.5) ✓	0 (1.5)	0 (1.5)
$0.75^{\dagger}$	0.86±0.02	1.97±0.82	0 (1.5) ✓	0 (1.5)	0 (1.5)
1	0.92±0.02	0.75±0.52	0 (1.5) ✓	0 (1.5)	0 (1.5)
S4: V-Δ $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.025, 100), $L_\mu = 0.005$					
$0^{\dagger}$	0.64±0.06	7.78±2.43	34 (22.0)	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	0.99±0.00	0.10±0.18	0 (1.5) ✓	0/199 (1.5)	0/199 (1.5)
$0.5^{\dagger}$	1.00±0.00	0.00±0.00	0 (1.5) ✓	N-A	N-A
$0.75^{\dagger}$	1.00±0.00	0.00±0.00	0 (1.5) ✓	N-A	N-A
1	1.00±0.00	0.00±0.00	0 (1.5) ✓	N-A	N-A
S5: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.10, 10), $L_\mu = 0.0005$					
$0^{\dagger}$	0.51±0.07	9.20±2.07	69 (40.5)	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	0.52±0.07	9.20±2.08	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.5^{\dagger}$	0.52±0.07	9.20±2.08	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.75^{\dagger}$	0.52±0.07	9.20±2.08	0 (1.5) ✓	36 (23.1)	108 (60.0)
1	0.53±0.06	8.94±1.88	0 (1.5) ✓	0 (1.5)	0 (1.5)
S5: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.10, 10), $L_\mu = 0.005$					
$0^{\dagger}$	0.59±0.08	9.46±3.08	80 (46.1)	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	0.69±0.04	6.34±1.28	8 (7.2) ✓	200 (100.0)	200 (100.0)
$0.5^{\dagger}$	0.77±0.03	3.66±0.82	0 (1.5) ✓	146 (78.2)	146 (78.2)
$0.75^{\dagger}$	0.83±0.02	2.24±0.71	0 (1.5) ✓	1 (2.4)	1 (2.4)
1	0.88±0.01	1.32±0.58	0 (1.5) ✓	0 (1.5)	0 (1.5)

Table 18: Maze, as in Table 17.

Maze $c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$ †: $d_t^{\mathrm{V}}(c;\sigma,\tau)$ scaled by $\phi < 1$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{A2}}$	$\widehat{q}_{\mathrm{pw}}$
S1: V-C $(\alpha_0,\alpha,\delta,K)$ = (0.30, 0.25, 0.20, 50), $L_\mu = 0.0005$					
$0^\dagger$	0.50±0.08	7.53±2.28	40 (25.3)	200 (100.0)	200 (100.0)
$0.25^\dagger$	0.52±0.07	7.40±2.27	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.5^\dagger$	0.52±0.07	7.40±2.27	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.75^\dagger$	0.52±0.07	7.40±2.27	0 (1.5) ✓	106 (59.0)	121 (66.3)
1	0.52±0.07	7.40±2.27	0 (1.5) ✓	0 (1.5)	0 (1.5)
S1: V-C $(\alpha_0,\alpha,\delta,K)$ = (0.30, 0.25, 0.20, 50), $L_\mu = 0.005$					
$0^\dagger$	0.56±0.10	9.24±3.91	77 (44.6)	200 (100.0)	200 (100.0)
$0.25^\dagger$	0.65±0.05	6.13±1.79	16 (11.9) ✓	200 (100.0)	200 (100.0)
$0.5^\dagger$	0.75±0.03	2.98±1.02	1 (2.4) ✓	166 (87.3)	166 (87.3)
$0.75^\dagger$	0.82±0.02	1.52±0.71	0 (1.5) ✓	10 (8.4)	12 (9.6)
1	0.88±0.01	0.71±0.57	0 (1.5) ✓	0 (1.5)	0 (1.5)
S2: V-C $(\alpha_0,\alpha,\delta,K)$ = (0.30, 0.25, 0.15, 50), $L_\mu = 0.0005$					
$0^\dagger$	0.52±0.08	7.30±2.38	35 (22.6)	200 (100.0)	200 (100.0)
$0.25^\dagger$	0.53±0.08	7.28±2.37	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.5^\dagger$	0.53±0.08	7.28±2.37	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.75^\dagger$	0.53±0.08	7.28±2.37	0 (1.5) ✓	99 (55.6)	112 (62.0)
1	0.53±0.08	7.28±2.37	0 (1.5) ✓	0 (1.5)	0 (1.5)
S2: V-C $(\alpha_0,\alpha,\delta,K)$ = (0.30, 0.25, 0.15, 50), $L_\mu = 0.005$					
$0^\dagger$	0.58±0.09	8.72±3.75	63 (37.4)	200 (100.0)	200 (100.0)
$0.25^\dagger$	0.67±0.05	5.75±1.81	9 (7.8) ✓	200 (100.0)	200 (100.0)
$0.5^\dagger$	0.76±0.03	2.80±1.08	1 (2.4) ✓	157 (83.2)	157 (83.2)
$0.75^\dagger$	0.83±0.02	1.36±0.72	0 (1.5) ✓	6 (5.9)	8 (7.2)
1	0.89±0.01	0.65±0.53	0 (1.5) ✓	0 (1.5)	0 (1.5)
S3: V-L $(\alpha_0,\alpha,\delta,K)$ = (0.30, 0.20, 0.20, 50), $L_\mu = 0.0005$					
$0^\dagger$	0.59±0.08	5.42±2.13	11 (9.0) ✓	200 (100.0)	200 (100.0)
$0.25^\dagger$	0.61±0.08	5.36±2.18	0 (1.5) ✓	44 (27.4)	45 (27.9)
$0.5^\dagger$	0.61±0.08	5.36±2.18	0 (1.5) ✓	0 (1.5)	0 (1.5)
$0.75^\dagger$	0.61±0.08	5.36±2.18	0 (1.5) ✓	0 (1.5)	0 (1.5)
1	0.62±0.07	5.04±1.93	0 (1.5) ✓	0 (1.5)	0 (1.5)

Table 18 (continued).

Maze $c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$ †: $d_t^{\mathrm{V}}(c;\sigma,\tau)$ scaled by $\phi < 1$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{A2}}$	$\widehat{q}_{\mathrm{pw}}$
S3: V-L $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.20, 0.20, 50), $L_\mu = 0.005$					
$0^{\dagger}$	0.66±0.09	6.07±3.41	24 (16.5) ✓	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	0.83±0.04	1.84±1.05	3 (3.9) ✓	10 (8.4)	11 (9.0)
$0.5^{\dagger}$	0.97±0.01	0.26±0.31	0 (1.5) ✓	0 (1.5)	0 (1.5)
$0.75^{\dagger}$	0.99±0.01	0.00±0.03	0 (1.5) ✓	0/184 (1.5)	0/184 (1.5)
1	1.00±0.00	0.00±0.00	0 (1.5) ✓	0/161 (1.5)	0/161 (1.5)
S4: V-Δ $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.025, 100), $L_\mu = 0.0005$					
$0^{\dagger}$	0.59±0.06	5.82±1.84	12 (9.6)	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	0.61±0.06	5.43±1.74	0 (1.5) ✓	0 (1.5)	0 (1.5)
$0.5^{\dagger}$	0.91±0.02	0.92±0.69	0 (1.5) ✓	0 (1.5)	0 (1.5)
$0.75^{\dagger}$	0.98±0.01	0.14±0.24	0 (1.5) ✓	0 (1.5)	0 (1.5)
1	1.00±0.00	0.00±0.00	0 (1.5) ✓	0/171 (1.5)	0/171 (1.5)
S4: V-Δ $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.025, 100), $L_\mu = 0.005$					
$0^{\dagger}$	0.66±0.07	6.04±2.71	29 (19.3)	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	1.00±0.00	0.01±0.04	0 (1.5) ✓	0/194 (1.5)	0/194 (1.5)
$0.5^{\dagger}$	1.00±0.00	0.00±0.00	0 (1.5) ✓	N-A	N-A
$0.75^{\dagger}$	1.00±0.00	0.00±0.00	0 (1.5) ✓	N-A	N-A
1	1.00±0.00	0.00±0.00	0 (1.5) ✓	N-A	N-A
S5: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.10, 10), $L_\mu = 0.0005$					
$0^{\dagger}$	0.54±0.07	6.88±2.10	29 (19.3)	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	0.55±0.07	6.76±2.17	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.5^{\dagger}$	0.55±0.07	6.76±2.17	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.75^{\dagger}$	0.55±0.07	6.76±2.17	0 (1.5) ✓	88 (50.1)	96 (54.1)
1	0.55±0.07	6.76±2.17	0 (1.5) ✓	0 (1.5)	0 (1.5)
S5: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.10, 10), $L_\mu = 0.005$					
$0^{\dagger}$	0.60±0.08	7.99±3.50	50 (30.6)	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	0.69±0.05	5.25±1.85	6 (5.9) ✓	200 (100.0)	200 (100.0)
$0.5^{\dagger}$	0.77±0.03	2.76±1.00	0 (1.5) ✓	145 (77.7)	145 (77.7)
$0.75^{\dagger}$	0.83±0.02	1.46±0.62	0 (1.5) ✓	0 (1.5)	2 (3.2)
1	0.88±0.02	0.71±0.49	0 (1.5) ✓	0 (1.5)	0 (1.5)

2.8 Certification, Deployment, and Assumption Checks

Tables 19–22 give the counts in (A.78), (A.81), and (A.83) for S1–S5 with their variants in the detailed checks (Section 2.1), for the default setting with V-C, and for the baselines. The repository contains the cases with the other three variants. V-U and V-Δ in the Maze at the default setting have no checked decision at either L_μ, while at $L_\mu = 0.005$, 26 of the 40 cases of S1–S5 and 6 of the 8 of the default setting have none at each test period. In the detailed checks, every count is consistent with its guarantee except at S4 with $K = \infty$, where $\delta_1 = 1.23\%$ lies

below the 1.5% upper confidence bound at zero possible events, although none of its 16 cases has an event. Over the main grid, 154 of the 3000 grid cells have a deployment event, all at $\alpha_0 \leq 0.25$. A count has an upper confidence bound above the bound of its guarantee in 9 grid cells with finite K, each a false-certification count of $2/200$ (3.2%) at $\delta = 0.025$, and, with $K = \infty$, in every grid cell with $\delta = 0.025$, where δ_1 lies below the 1.5% upper confidence bound at zero possible events, and in 40 Corridor grid cells with $\alpha = 0.05$, with counts $2/200$ (3.2%) at $\delta = 0.05$ and $5/200$ (5.2%) at $\delta = 0.10$. The rows with $K = \infty$ check Theorems A.1 and A.2, and the rows with the setting's K check Theorem A.3. Appendix A.F-D summarizes the assumption checks.

Table 19: Corridor, detailed checks at $L_\mu = 0.0005$.

Corridor **Baselines** $\alpha_0 = 0.15,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$			
Method	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$
Unfiltered	N-A	200	200
$c = 0$	200 (100.0)	200	200
$c_{\max}$	0 (1.5)	0	0
ACI	200 (100.0)	200	26
QT	200 (100.0)	200	12

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$						
K	$\widehat{q}_{\mathrm{cert}}\downarrow$	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$	$\widehat{q}_{\mathrm{A2}}$	$\widehat{q}_{\mathrm{pw}}$
default setting: V-C $(\alpha_0, \alpha, \delta)$ = (0.15, 0.10, 0.05)						
50	0 (1.5) ✓	0 (1.5) ✓	200	0	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	200	0	0 (1.5)	0 (1.5)

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$			
Method	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$
Unfiltered	N-A	200	200
$c = 0$	200 (100.0)	200	200
$c_{\max}$	0 (1.5)	0	0
ACI	200 (100.0)	200	198
QT	200 (100.0)	200	156

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$						
K	$\widehat{q}_{\mathrm{cert}}\downarrow$	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$	$\widehat{q}_{\mathrm{A2}}$	$\widehat{q}_{\mathrm{pw}}$
S1: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.20)						
50	3 (3.9) ✓	0 (1.5) ✓	200	51	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	200	45	0 (1.5)	0 (1.5)
S2: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.15)						
50	1 (2.4) ✓	0 (1.5) ✓	200	43	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	200	36	0 (1.5)	0 (1.5)
S3: V-L $(\alpha_0, \alpha, \delta)$ = (0.30, 0.20, 0.20)						
50	7 (6.5) ✓	0 (1.5) ✓	200	13	0 (1.5)	0 (1.5)
∞	6 (5.9) ✓	0 (1.5) ✓	200	10	0 (1.5)	0 (1.5)
S4: V-Δ $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.025)						
100	0 (1.5) ✓	0 (1.5) ✓	178	0	0 (1.5)	0 (1.5)
∞	0 (1.5)	0 (1.5)	74	0	0 (1.5)	0 (1.5)
S5: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.10)						
10	0 (1.5) ✓	0 (1.5) ✓	200	36	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	200	30	0 (1.5)	0 (1.5)

Table 20: Corridor, detailed checks at $L_\mu = 0.005$.

Corridor **Baselines** $\alpha_0 = 0.15,\ L_\mu = 0.005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$			
Method	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$
Unfiltered	N-A	200	200
$c = 0$	200 (100.0)	200	200
$c_{\max}$	0 (1.5)	0	0
ACI	200 (100.0)	200	8
QT	200 (100.0)	200	3

Corridor $\phi = 1,\ L_\mu = 0.005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$						
K	$\widehat{q}_{\mathrm{cert}}\downarrow$	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$	$\widehat{q}_{\mathrm{A2}}$	$\widehat{q}_{\mathrm{pw}}$
default setting: V-C $(\alpha_0, \alpha, \delta)$ = (0.15, 0.10, 0.05)						
50	0 (1.5) ✓	0 (1.5) ✓	19	0	0/139 (1.5)	0/139 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	13	0	0/107 (1.5)	0/107 (1.5)

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$			
Method	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$
Unfiltered	N-A	200	200
$c = 0$	200 (100.0)	200	200
$c_{\max}$	0 (1.5)	0	0
ACI	200 (100.0)	200	171
QT	200 (100.0)	200	128

Corridor $\phi = 1,\ L_\mu = 0.005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$						
K	$\widehat{q}_{\mathrm{cert}}\downarrow$	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$	$\widehat{q}_{\mathrm{A2}}$	$\widehat{q}_{\mathrm{pw}}$
S1: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.20)						
50	0 (1.5) ✓	0 (1.5) ✓	198	1	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	119	1	0 (1.5)	0 (1.5)
S2: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.15)						
50	0 (1.5) ✓	0 (1.5) ✓	199	1	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	114	1	0 (1.5)	0 (1.5)
S3: V-L $(\alpha_0, \alpha, \delta)$ = (0.30, 0.20, 0.20)						
50	3 (3.9) ✓	0 (1.5) ✓	14	0	0/158 (1.5)	0/158 (1.5)
∞	3 (3.9) ✓	0 (1.5) ✓	14	0	0/158 (1.5)	0/158 (1.5)
S4: V-Δ $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.025)						
100	0 (1.5) ✓	0 (1.5) ✓	0	0	N-A	N-A
∞	0 (1.5)	0 (1.5)	0	0	N-A	N-A
S5: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.10)						
10	1 (2.4) ✓	0 (1.5) ✓	200	1	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	108	1	0 (1.5)	0 (1.5)

Table 21: Maze, detailed checks at $L_\mu = 0.0005$.

Maze **Baselines** $\alpha_0 = 0.15,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$			
Method	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$
Unfiltered	N-A	200	200
$c = 0$	200 (100.0)	200	195
$c_{\max}$	0 (1.5)	0	0
ACI	200 (100.0)	200	2
QT	85–87 (49.6)	200	0

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$						
K	$\widehat{q}_{\mathrm{cert}}\downarrow$	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$	$\widehat{q}_{\mathrm{A2}}$	$\widehat{q}_{\mathrm{pw}}$
default setting: V-C $(\alpha_0, \alpha, \delta)$ = (0.15, 0.10, 0.05)						
50	0 (1.5) ✓	0 (1.5) ✓	155	0	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	154	0	0 (1.5)	0 (1.5)

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$			
Method	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$
Unfiltered	N-A	200	200
$c = 0$	200 (100.0)	200	195
$c_{\max}$	0 (1.5)	0	0
ACI	199 (100.0)	200	102
QT	131 (71.1)	200	17

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$						
K	$\widehat{q}_{\mathrm{cert}}\downarrow$	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$	$\widehat{q}_{\mathrm{A2}}$	$\widehat{q}_{\mathrm{pw}}$
S1: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.20)						
50	1 (2.4) ✓	0 (1.5) ✓	200	4	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	200	4	0 (1.5)	0 (1.5)
S2: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.15)						
50	0 (1.5) ✓	0 (1.5) ✓	200	4	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	200	3	0 (1.5)	0 (1.5)
S3: V-L $(\alpha_0, \alpha, \delta)$ = (0.30, 0.20, 0.20)						
50	4 (4.6) ✓	0 (1.5) ✓	200	1	0 (1.5)	0 (1.5)
∞	3 (3.9) ✓	0 (1.5) ✓	200	1	0 (1.5)	0 (1.5)
S4: V-Δ $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.025)						
100	0 (1.5) ✓	0 (1.5) ✓	0	0	0/171 (1.5)	0/171 (1.5)
∞	0 (1.5)	0 (1.5)	0	0	0/148 (1.5)	0/148 (1.5)
S5: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.10)						
10	0 (1.5) ✓	0 (1.5) ✓	200	4	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	200	4	0 (1.5)	0 (1.5)

Table 22: Maze, detailed checks at $L_\mu = 0.005$.

Maze **Baselines** $\alpha_0 = 0.15,\ L_\mu = 0.005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$			
Method	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$
Unfiltered	N-A	200	200
$c = 0$	200 (100.0)	200	200
$c_{\max}$	0 (1.5)	0	0
ACI	199 (100.0)	200	1
QT	111 (61.5)	200	0

Maze $\phi = 1,\ L_\mu = 0.005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$						
K	$\widehat{q}_{\mathrm{cert}}\downarrow$	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$	$\widehat{q}_{\mathrm{A2}}$	$\widehat{q}_{\mathrm{pw}}$
default setting: V-C $(\alpha_0, \alpha, \delta)$ = (0.15, 0.10, 0.05)						
50	0 (1.5) ✓	0 (1.5) ✓	6	0	0/136 (1.5)	0/136 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	5	0	0/110 (1.5)	0/110 (1.5)

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$			
Method	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$
Unfiltered	N-A	200	200
$c = 0$	200 (100.0)	200	200
$c_{\max}$	0 (1.5)	0	0
ACI	200 (100.0)	200	65
QT	133 (72.1)	200	4

Maze $\phi = 1,\ L_\mu = 0.005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$						
K	$\widehat{q}_{\mathrm{cert}}\downarrow$	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$	$\widehat{q}_{\mathrm{A2}}$	$\widehat{q}_{\mathrm{pw}}$
S1: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.20)						
50	1 (2.4) ✓	0 (1.5) ✓	166	0	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	66	0	0 (1.5)	0 (1.5)
S2: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.15)						
50	1 (2.4) ✓	0 (1.5) ✓	166	0	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	58	0	0 (1.5)	0 (1.5)
S3: V-L $(\alpha_0, \alpha, \delta)$ = (0.30, 0.20, 0.20)						
50	2 (3.2) ✓	0 (1.5) ✓	0	0	0/161 (1.5)	0/161 (1.5)
∞	2 (3.2) ✓	0 (1.5) ✓	0	0	0/161 (1.5)	0/161 (1.5)
S4: V-Δ $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.025)						
100	0 (1.5) ✓	0 (1.5) ✓	0	0	N-A	N-A
∞	0 (1.5)	0 (1.5)	0	0	N-A	N-A
S5: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.10)						
10	0 (1.5) ✓	0 (1.5) ✓	167	0	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	57	0	0 (1.5)	0 (1.5)

2.9 Variation in Robustness Margin and Estimated Exceedance Risk

Tables 23 and 24 report the temporal variance $\widehat{V}_v$, root-mean-square change $\widehat{J}_v$, and population size $|\mathcal{K}_v|$ defined in (1.26) of Section 1.4.

Table 23: Corridor, $\widehat{V}_v$ and $\widehat{J}_v$, default setting and S1–S5.

Corridor **Baselines** $L_\mu = 0.0005,\ c_{\max} = 0.02\,\text{m},\ T_s = 0.1\,\text{s},\ N_{\text{MC}} = 200$					
Method	$10^3\widehat{V}_{c/c_{\max}}\downarrow$	$100\widehat{J}_{c/c_{\max}}\downarrow$	$10^3\widehat{V}_{\widehat{p}}\downarrow$	$100\widehat{J}_{\widehat{p}}\downarrow$	$\lvert\mathcal{K}_{c/c_{\max}}\rvert, \lvert\mathcal{K}_{\widehat{p}}\rvert$
$\alpha_0 = 0.15$					
Unfiltered	N-A	N-A	N-A	N-A	0, 0
$c = 0$	0.000±0.000	0.000±0.000	0.633±0.582	3.108±1.339	200, 200
$c_{\max}$	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000	200, 200
ACI	21.158±7.176	5.841±1.733	3.553±1.710	2.694±0.958	200, 200
QT	41.791±7.940	10.394±0.125	8.745±2.405	5.152±0.606	200, 200
$\alpha_0 = 0.30$					
Unfiltered	N-A	N-A	N-A	N-A	0, 0
$c = 0$	0.000±0.000	0.000±0.000	0.633±0.582	3.108±1.339	200, 200
$c_{\max}$	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000	200, 200
ACI	19.851±6.237	6.803±1.683	6.433±2.338	4.360±1.196	200, 200
QT	41.165±6.218	13.125±0.171	15.176±1.820	9.347±0.644	200, 200

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\text{m},\ T_s = 0.1\,\text{s},\ N_{\text{MC}} = 200$					
Method	$10^3\widehat{V}_{c/c_{\max}}\downarrow$	$100\widehat{J}_{c/c_{\max}}\downarrow$	$10^3\widehat{V}_{\widehat{p}}\downarrow$	$100\widehat{J}_{\widehat{p}}\downarrow$	$\lvert\mathcal{K}_{c/c_{\max}}\rvert, \lvert\mathcal{K}_{\widehat{p}}\rvert$
default setting: $(\alpha_0, \alpha, \delta, K)$ = (0.15, 0.10, 0.05, 50)					
V-U	0.883±1.430	1.404±1.255	0.023±0.059	0.194±0.236	200, 200
V-Δ	0.883±1.430	1.404±1.255	0.023±0.059	0.194±0.236	200, 200
V-L	5.719±5.367	2.950±1.337	0.168±0.262	0.450±0.306	200, 200
V-C	9.966±6.831	3.754±1.522	0.373±0.393	0.650±0.383	200, 200
S1: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.20, 50)					
V-U	38.489±13.559	10.377±2.082	3.039±1.783	2.687±0.805	200, 200
V-Δ	38.486±13.556	10.364±2.066	3.039±1.782	2.683±0.798	200, 200
V-L	31.330±10.482	10.693±2.072	3.182±1.717	2.975±0.841	200, 200
V-C	35.985±11.153	11.789±2.294	3.936±1.849	3.444±0.976	200, 200
S2: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.15, 50)					
V-U	36.427±13.399	9.859±2.081	2.718±1.673	2.491±0.779	200, 200
V-Δ	36.424±13.395	9.846±2.077	2.718±1.672	2.487±0.777	200, 200
V-L	30.562±10.395	10.304±1.967	2.908±1.635	2.806±0.788	200, 200
V-C	36.287±10.834	11.407±2.181	3.801±1.740	3.276±0.907	200, 200
S3: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.20, 0.20, 50)					
V-U	26.713±11.116	9.101±2.042	2.252±1.615	2.311±0.782	200, 200
V-Δ	26.675±11.017	9.064±1.965	2.249±1.606	2.301±0.754	200, 200
V-L	29.294±10.894	9.777±2.240	2.584±1.555	2.574±0.883	200, 200
V-C	29.239±10.806	9.733±2.145	2.579±1.546	2.560±0.848	200, 200
S4: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.025, 100)					
V-U	21.183±9.515	6.521±1.713	1.132±0.798	1.487±0.557	200, 200
V-Δ	21.188±9.513	6.523±1.713	1.133±0.798	1.488±0.558	200, 200
V-L	35.672±11.899	8.240±1.720	2.505±1.346	2.013±0.612	200, 200
V-C	36.754±11.519	9.761±1.907	3.367±1.675	2.622±0.737	200, 200
S5: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.10, 10)					
V-U	22.998±10.232	8.394±1.770	1.790±1.306	2.084±0.679	200, 200
V-Δ	22.991±10.221	8.385±1.755	1.790±1.304	2.081±0.673	200, 200
V-L	29.026±10.255	9.674±1.906	2.558±1.534	2.539±0.740	200, 200
V-C	36.889±10.943	11.005±2.135	3.755±1.746	3.119±0.869	200, 200

Table 24: Maze, as in Table 23.

Maze **Baselines** $L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$					
Method	$10^3\widehat{V}_{c/c_{\max}}\downarrow$	$100\widehat{J}_{c/c_{\max}}\downarrow$	$10^3\widehat{V}_{\widehat{p}}\downarrow$	$100\widehat{J}_{\widehat{p}}\downarrow$	$\lvert\mathcal{K}_{c/c_{\max}}\rvert, \lvert\mathcal{K}_{\widehat{p}}\rvert$
$\alpha_0 = 0.15$					
Unfiltered	N-A	N-A	N-A	N-A	0, 0
$c = 0$	0.000±0.000	0.000±0.000	0.189±0.277	1.416±1.322	200, 200
$c_{\max}$	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000	200, 200
ACI	16.729±5.454	4.931±1.900	2.431±1.098	1.996±0.933	200, 200
QT	14.664±3.332	0.472±0.032	1.147±0.393	0.203±0.119	200, 200
$\alpha_0 = 0.30$					
Unfiltered	N-A	N-A	N-A	N-A	0, 0
$c = 0$	0.000±0.000	0.000±0.000	0.189±0.277	1.416±1.322	200, 200
$c_{\max}$	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000	200, 200
ACI	17.309±6.033	6.280±1.812	4.700±2.066	3.382±1.162	200, 200
QT	42.553±7.890	0.749±0.015	6.713±1.861	0.482±0.278	200, 200

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$					
Method	$10^3\widehat{V}_{c/c_{\max}}\downarrow$	$100\widehat{J}_{c/c_{\max}}\downarrow$	$10^3\widehat{V}_{\widehat{p}}\downarrow$	$100\widehat{J}_{\widehat{p}}\downarrow$	$\lvert\mathcal{K}_{c/c_{\max}}\rvert, \lvert\mathcal{K}_{\widehat{p}}\rvert$
default setting: $(\alpha_0, \alpha, \delta, K)$ = (0.15, 0.10, 0.05, 50)					
V-U	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000	200, 200
V-Δ	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000	200, 200
V-L	4.908±4.252	2.717±1.385	0.131±0.165	0.404±0.278	200, 200
V-C	8.343±5.253	3.627±1.401	0.284±0.268	0.607±0.322	200, 200
S1: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.20, 50)					
V-U	12.962±8.699	6.149±1.909	0.774±0.797	1.390±0.653	200, 200
V-Δ	13.241±8.720	5.833±1.868	0.782±0.798	1.335±0.644	200, 200
V-L	28.807±10.025	10.428±2.437	3.003±1.674	2.908±0.896	200, 200
V-C	32.238±8.987	11.641±2.981	3.538±1.465	3.432±1.132	200, 200
S2: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.15, 50)					
V-U	9.916±7.184	5.593±1.762	0.561±0.646	1.209±0.590	200, 200
V-Δ	10.208±7.168	5.318±1.698	0.572±0.645	1.164±0.574	200, 200
V-L	29.416±10.534	10.069±2.565	2.969±1.730	2.771±0.915	200, 200
V-C	33.493±9.698	11.186±2.772	3.590±1.569	3.246±1.044	200, 200
S3: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.20, 0.20, 50)					
V-U	26.345±12.501	7.669±2.054	1.681±1.388	1.741±0.696	200, 200
V-Δ	26.011±12.344	7.296±1.974	1.666±1.376	1.666±0.674	200, 200
V-L	26.642±9.971	9.821±2.738	2.358±1.401	2.593±0.986	200, 200
V-C	26.537±9.947	9.635±2.782	2.353±1.398	2.552±0.995	200, 200
S4: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.025, 100)					
V-U	0.956±1.324	2.009±1.405	0.039±0.084	0.348±0.334	200, 200
V-Δ	1.010±1.364	1.959±1.355	0.040±0.085	0.340±0.325	200, 200
V-L	34.866±12.512	8.010±1.804	2.396±1.406	1.978±0.599	200, 200
V-C	34.436±10.441	9.665±2.205	3.198±1.522	2.635±0.807	200, 200
S5: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.10, 10)					
V-U	5.706±5.234	4.568±1.920	0.290±0.430	0.913±0.564	200, 200
V-Δ	5.791±5.249	4.486±1.877	0.293±0.430	0.898±0.555	200, 200
V-L	26.952±9.363	9.377±2.368	2.401±1.375	2.453±0.806	200, 200
V-C	33.277±9.583	10.767±2.728	3.356±1.457	3.051±0.996	200, 200

References

[1] P. S. Trakas, O. Mirzaeedodangeh, and L. Lindemann, "Anytime-valid certification of robustness margins for safe control via e-values," arXiv preprint, 2026.